## Illustrative Examples

36. Find the torque on a loop of wire area $12cm^2$ at 30° to magnetic field of strength 50mT with a current of 4A flowing through it.

***Solution***: $\tau = IAB \sin\theta = (4)(12 \times 10^{-4})(50\times10^{-3}) \sin 30°$

$\tau = 1.2 \times 10^{-4}$ N.m

37. A rectangular coil with 200 turns is 5 cm high and 4cm wide. When the coil is placed in a magnetic field of 0.5T, its maximum torque is 0.2N.m. What is the current in the coil?

***Solution***: $\tau = NIAB \Rightarrow I = \dfrac{\tau}{NAB} = \dfrac{0.2}{(200)(0.05)(0.04)(0.5)}$

$I = 1A$

**Note**: The direction of the current will determine whether the torque on the coil is clockwise or counterclockwise.

**Magnetic dipole moment**

A current loop creates a magnetic dipole moment. If we consider a coil of wire consisting of N turns, carrying current I, and having area A, then the magnetic moment of such a coil is given by:

$$\boxed{\mu = NIA}$$

Magnetic moment is a vector quantity and its direction is determined using the right hand rule.

When a magnetic dipole moment is placed in a magnetic field, it experiences a torque. This torque is given by

$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

This can be written as

$$\boxed{\tau = \mu B \sin\theta}$$

## Illustrative Example

38. A current of 17mA is maintained in a single circular loop with a circumference of 2m. A magnetic field of 0.8T is directed parallel to the plane of the loop.

a) Calculate the magnetic moment of the loop.

b) What is the magnitude of the torque exerted on the loop by the magnetic field?

***Solution***:

a) $\mu = NIA$, $c = 2\pi r \Rightarrow 2 = 2\pi r$, $\pi r = 1m$, $A = \pi r^2$ $\pi\left(\dfrac{1}{\pi}\right)^2 = \dfrac{1}{\pi}$

Generated by CamScanner from intsig.com

$$\mu = (1)(17 \times 10^{-3})\left(\frac{1}{3.14}\right)$$

$$\mu = 5.4 \times 10^{-3}\ A.m^2$$

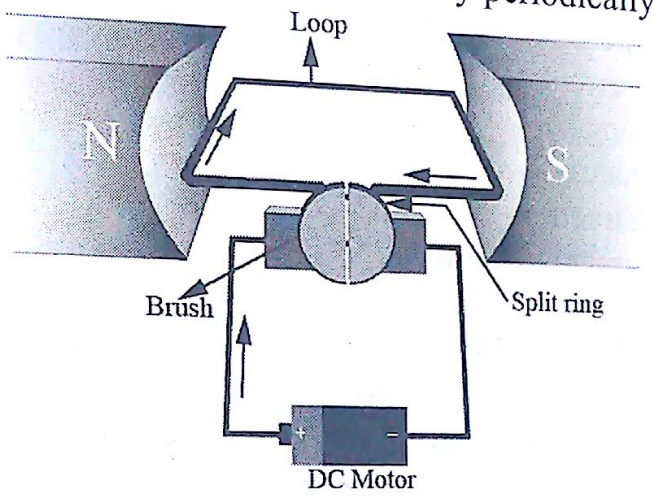
b) $\tau = \mu B = (5.4 \times 10^{-3})(0.8)$

$$\tau = 4.3 \times 10^{-3}\ N.m$$

### A Direct Current Motor

It's hard to imagine life in the 21 century without electric motors. Computer disk drives, CD players, DVD players, food processors, automobiles, furnaces, washing machines, air conditioners etc all contain motors. The motor convert electrical energy into rotational kinetic energy that is mechanical energy.

A simple DC motor consist current carrying loop called *armatures*, split - ring commutator, brushes and DC source. The split – ring commutator provides continuous rotation of the loop in one direction by periodically reversing the current direction. The brushes, which are usually made from graphite are used to reduce sparking and heat.

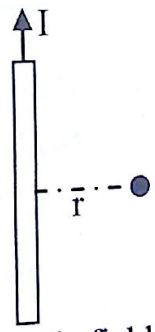



The working principle of electric motor or the galvanometer is the rotation of a current carrying coil placed in a magnetic field. Thus, these devices use the concept of torque for their operation.

## 14.6 Magnetic Field due to a Wire

During a lecture demonstration in 1819, the Danish scientist Hans Christain Oersted found that an electric current in a wire deflected a nearby compass needle. This momentous discovery lead Oersted in 1820 to experimentally confirm that a current – carrying conductor produces a magnetic field.

The magnitude of the magnetic field produced by the current – carrying wire is found to be directly




Magnetic field at distance r

Generated by CamScanner from intsig.com

proportional to the current and inversely proportional to the distance from the wire.

$$\Rightarrow B \sim \frac{I}{r}$$

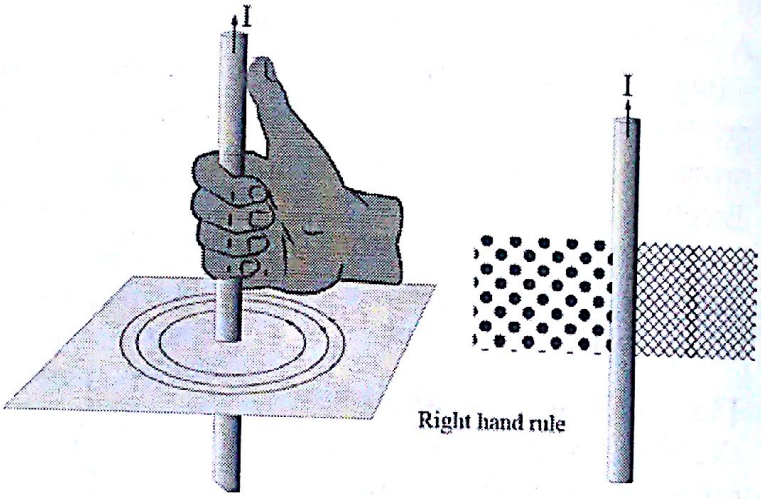
$$\boxed{B = \frac{\mu o I}{2\pi r}}$$

Where: $\mu_o$ is called *permeability of free space* and has the value: $\mu_o = 4\pi \times 10^{-7} \frac{T.m}{A}$.

The direction of the magnetic field produced by the current – carrying wire is determined by the right hand rule.

**Right Hand Rule**

"Point the thumb of your right hand in the direction of the current. Your fingers then naturally curl in the direction of the magnetic field."



Right hand rule

☛ **Note that:** The magnetic field lines are represented by concentric circles.

- **The dot (•)** represents the field lines are out of the page. And,
- **The cross (×)** represents, the field lines are into the page.

## Conceptual Examples

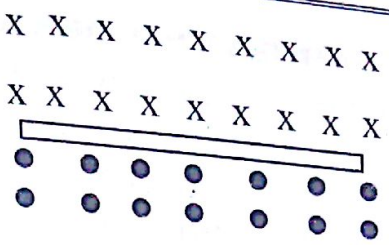
39. What will be the direction of compass needles that are placed around a straight wire when
    a. There is no current in the wire.
    b. There is current in the wire.

    *Explanation*:

    a) The compass needles all will point in the same direction.
    b) The compass needles will defect in the direction of the magnetic field.

DIGITAL LIBRARY FOR ETHIOPIAN STUDENTS
Digital Library for Ethiopian students
Link: https://telegram.me//digitallibrary4ethiopianstudents or digitallibrary4ethiopianstudents
YouTube Digital[0,1] Ethiopia
Generated by CamScanner from intsig.com

40. The magnetic field shown below is due to a horizontal current carrying wire. Does the current in the wire flow to the left or to the right?



*Explanation*: If you grasp the wire in such a way that your right hand fingers are into the page from above and out of the page below the wire the thumb will point to the left. Thus, the current in the wire flows to the left.

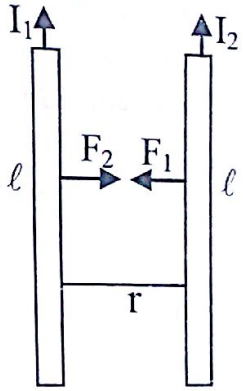
**Illustrative Example**

41. Find the magnitude of the magnetic field 1m from a long, straight wire carrying a current of 1A.

*Solution*: $B = \frac{\mu_o I}{2\pi r} = \frac{(4\pi \times 10^{-7})(1)}{(2\pi)(1)} = 2 \times 10^{-7}\,T$

## 14.7 Force Between Current Carrying Wire

We know that a current – carrying wire in a magnetic field experiences a force. We also know that a current – carrying wire produces a magnetic field. It follows, then, that one current – carrying wire will exert a force on another.



Force between wires

Now, consider two long, straight, parallel wires separated by distance r and carrying currents $I_1$ and $I_2$ in the same direction. The magnetic force on one wire due to a magnetic field set up by the other wire can be calculated as follows. For example, let us see the force on wire 2 due to the magnetic field of wire 1. The current $I_1$, sets up magnetic field, $B_1$ at wire 2. Which is given by,

$$B_1 = \frac{\mu_o I_1}{2\pi r}$$

And according to the relation, $F = I\ell B$,
The magnitude of the magnetic force on wire 2 due to the magnetic field of wire 1 is then $F_2 = B_1 I_2 \ell$, substitute for $B_1$

$$F_2 = \left(\frac{\mu_o I_1}{2\pi r}\right) I_2 \ell \text{, } F_1 = F_2\text{; From Newton's 3}^{rd}\text{ law } \boxed{F = \frac{\mu_o I_1 I_2 \ell}{2\pi r}}$$

The force per unit length is given by $\boxed{\frac{F}{\ell} = \frac{\mu_o I_1 I_2}{2\pi r}}$

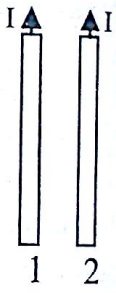
Generated by CamScanner from intsig.com

**Important points to remember**

- The force that wire 1 exerts on wire 2 is equal in magnitude and opposite in direction to the force that wire 2 exerts on wire 1 due to Newton's 3rd law.
- Wires carrying – currents in the same direction attract each other while wires carrying – currents in opposite direction repel each other.
- The force between two parallel wires carrying a current is used to define the SI unit of current, the ampere.
  "The ampere is that current which if maintained in two straight parallel wires of infinite length of negligible cross – section and placed one meter apart in vacuum, would produce between these two wires a force equal to $2 \times 10^{-7}$ Newton per meter of length."

## Conceptual Examples

42. Parallel wires exert magnetic force on each other. What about perpendicular wires? Explain




*Explanation*: There is no net force on the wires, but there is a torque.

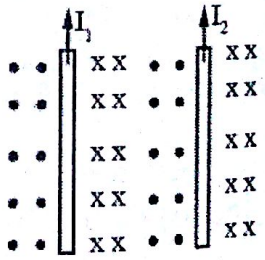
43. In the figure shown, $I_1 = 2A$ and $I_2 = 6A$. Which of the following is true?

a) $F_1 = -3F_2$ b) $F_1 = -F_2$ c) $F_1 = \frac{F_2}{3}$

*Explanation*: We know, $F_1 - F_2 = \frac{\mu_o I_1 I_2 \ell}{2\pi r}$ hence, $F_1 = F_2$

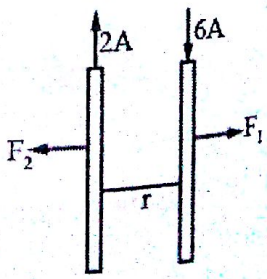
**Answer: B**

44. Why do two straight parallel wires carrying currents in the same direction attract each other?




*Explanation*: It is because, when the wires carry currents in the same direction, the magnetic field between them is opposite and hence, the force is attractive. Right hand rule can verify this fact.

45. If two parallel wires carry currents of 2A and 6A respectively, in opposite directions, which wire exert a larger force on the other? Explain




*Explanation*: Even if the wires carry currents

Generated by CamScanner from intsig.com

of different magnitude, the force one exerts on the other in equal in magnitude and opposite in direction according to Newton's 3rd law.

$$F_1 = F_2 = \frac{\mu_o I_1 I_2 \ell}{2\pi r}$$

## Illustrative Examples

46. Two wires separated by a distance of 12cm carry currents in the same direction. The current in one wire is 2.5A and the current in the other wire is 4.5A. Calculate:
a) The magnetic field midway between the wires.
b) The force per unit length of wire.

*Solution*: a) The magnetic field due to wire 1 is

$$B_1 = \frac{\mu_o I_1}{2\pi r} = \frac{(4\pi\times10^{-7})(2.5)}{(2\pi)(6\times10^{-2})} = 8.3\times10^{-6} \text{ into the page}$$

The magnetic field due to wire 2 is,

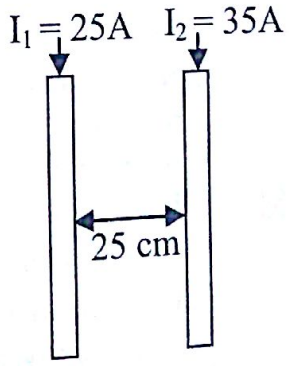
$$B_2 = \frac{\mu_o I_2}{2\pi r} = \frac{(4\pi\times10^{-7})(4.5)}{(2\pi)(6\times10^{-2})} = 15\times10^{-6} \text{ out of the page}$$

The net field midway the wires is
$B = B_2 - B_1 = 15 \times 10^{-6} - 8.3 \times 10^{-6} = 6.7 \times 10^{-6}$T out of the page

b) $$F = \frac{\mu_o I_1 I_2 \ell}{2\pi r} \Rightarrow \frac{F}{\ell} = \frac{\mu_o I_1 I_2}{2\pi r} = \frac{(4\pi\times10^{-7})(2.5)(4.5)}{(2\pi)(12\times10^{-2})}$$

$$\frac{F}{\ell} = 1.9 \times 10^{-5} \text{ N/m attractive}$$

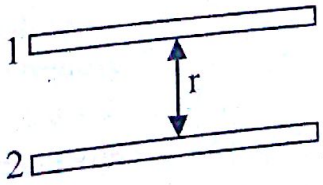
47. What is the force per unit length between the wires shown below?




*Solution*: $$F = \frac{\mu_o I_1 I_2 \ell}{2\pi r} \Rightarrow \frac{F}{\ell} = \frac{\mu_o I_1 I_2}{2\pi r}$$

$$\frac{F}{\ell} = \frac{(4\pi\times10^{-7})(25)(35)}{2\pi\times25\times10^{-2}} = 7 \times 0^{-4} \text{ N/m}$$

48. A wire with a weight per unit length of 0.08N/m is suspended directly above a second wire. The top wire carries a current of 60A. Find the separation between the wires.




Generated by CamScanner from intsig.com

***Solution***: $F = \frac{\mu_o I_1 I_2 \ell}{2\pi r} \Rightarrow \frac{F}{\ell} = \frac{\mu_o I_1 I_2}{2\pi r}$

Since, $\frac{F}{\ell} = 0.08 \Rightarrow r = \frac{\mu_o I_1 I_2}{2\pi \times 0.08}$

$$r = \frac{(4\pi \times 10^{-7})(30)(60)}{2\pi \times 0.08} = 0.0045\text{m} \Rightarrow r = 4.5\text{mm}$$

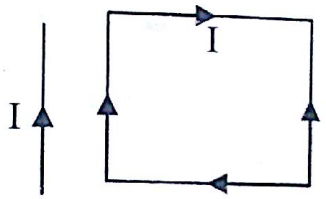
49. Two parallel copper wires are 1 cm apart. Lighting sends a 10,000A pulse of current along each conductor. What is the force per unit length on one conductor? (UEE 2002)

A. 1000N/m B. 200N/m C. 20N/m D. 2000N/m

***Solution***: $\frac{F}{\ell} = \frac{\mu_o I_1 I_2}{2\pi r} = \frac{(4\pi \times 10^{-7})(10,000)(10,000)}{2\pi(0.01)} = 2000\text{N/m}$

**Answer: D**

50. A rectangular loop carrying a current is situated near a long straight wire such that the wire is parallel to one of the sides of the loop and it is in a plane of the loop. If a steady current is established in the wire as shown, the loop will (UEE 2005)




A. Remain stationary.
B. Move toward the wire.
C. Move away from the wire.
D. Rotate about an axis parallel to the wire.

***Explanation***: Since the straight wire and the side of the loop near the wire carry currents in the same direction there is attraction force between the wire and the loop. **Answer: B**

51. Which one of the following statements is correct about the effect and formation of a magnetic field? (UEE 2007)

A. A magnetic field exerts a non zero force on a stationary charged object.
B. A magnetic field exerts a non zero force on a conducting wire placed in the field.
C. A uniform magnetic field can be created at all points around a long current carrying wire.
D. A magnetic field can be created in the space surrounding any charged object that is in motion. **Answer: D**

Generated by CamScanner from intsig.com

52. The torque experienced by a current carrying loop, in an external, magnetic field is (UEE 2008)
   A. Independent of the number of turns of the loop.
   B. Independent of the current in the loop.
   C. Dependent on the orientation of the loop.
   D. Dependent on magnetic field generated according to Lenz's law.

**Answer: C**

53. Two long wires of the same length and cross – sectional area carry the same amount of current. Which one of the following statements is correct about the magnetic field and the magnetic forces on the wires? (UEE 2008)
   A. The magnetic field is zero at the midpoint between the wires when the currents are in the same direction.
   B. The force between the wires is attractive when they carry currents in opposite direction.
   C. The magnetic fields between the wires reinforce each other when the currents are in the same direction.
   D. The magnetic field due to one is proportional to the current through the other wire.

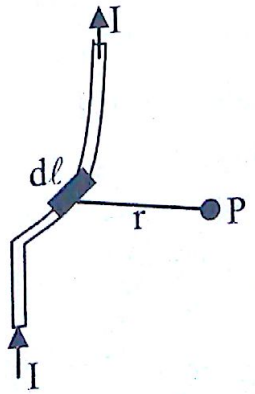
**Answer: A**

## The Biot – Savart Law

In the 19th Century, Jean Biot and Felix Savart, investigate the magnetic field due to a very small element of current that is part of a large current distribution.

**Bilot – Savart Law** states that: "The magnetic field contribution dB at a point by a current element of length $d\ell$ at a point P a distance r from that element is by $\boxed{d\vec{B} = \frac{\mu_o I d\ell}{4\pi r^2} \times \vec{n}r}$"

$$\Rightarrow dB = \frac{\mu_o I d\ell \sin\theta}{4\pi r^2}$$




Generated by CamScanner from intsig.com

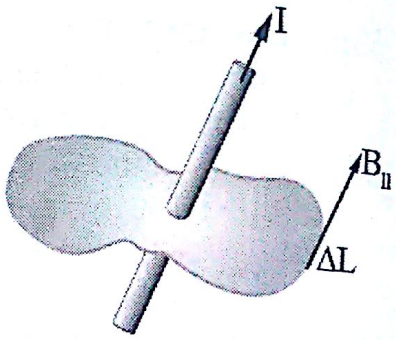
**Illustrative Example**

54. Find the contribution to the magnetic field at a point that is at a perpendicular distance of 10cm from a current element of length 10cm and current 1.5A.

***Solution***: $dB = \frac{\mu_o I d\ell \sin\theta}{4\pi r^2} = \frac{(4\pi \times 10^{-7})(1.5)(0.1)}{4\pi(0.1)^2}$

$$dB = 1.5 \times 10^{-6} T$$

## 14.8 Ampere's Law




Ampere's law relates that magnetic field along a closed path to the electric current enclosed by the path. Now, consider a straight current – carrying wire enclosed by a closed path, which can be divided into many small, straight line segments of length ΔL with the parallel component of the magnetic field, $B_{11}$ in the direction of the length element.

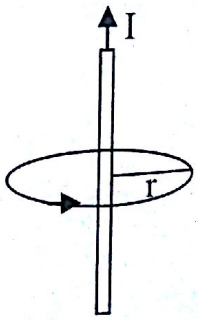
> **Ampere's Law** states that: "In any closed loop path, the sum of the length element the magnetic field in the direction of the length element is equal to the permeability of free space times the electric current enclosed in the loop."

Mathematically, we can write this as:

$$\boxed{\sum B_{11} \Delta L = \mu_o I} \longleftarrow \text{Ampere's law}$$

Where: $\sum$ (sigma) stands for "sum of" $\mu_o$, is called permeability of free space, and is equal to, $\mu_o = 4\pi \times 10^{-7}$T.m/A.

**Ampere's Law and Magnetic Field outside a wire**




The magnetic field lines due to a straight current – carrying wire form circles around the wire. Thus, the length enclosed equals the circumference of the circle.

$\sum B_{11} \Delta\ell = \mu_o I \leftarrow$ Ampere's law

But $\Delta\ell = 2\pi r$

$$\Rightarrow \boxed{B = \frac{\mu_o I}{2\pi r}}$$

Generated by CamScanner from intsig.com

Where, $n = \frac{N}{\ell}$, number of turns per length

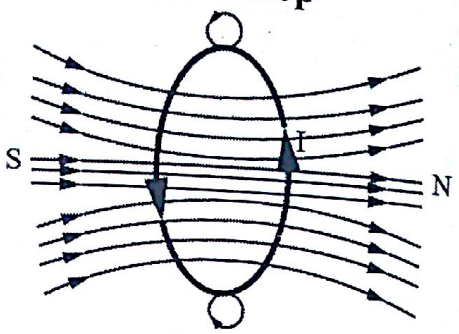
**Illustrative Example**

57. Find the magnitude field inside a solenoid of 500 turns per unit length with a current of 1.5A flowing through it?

***Solution***: $B = \mu_o nI = (4\pi \times 10^{-7})(500)(1.5) = 9.4 \times 10^{-4} T$

**Ampere's Law and the Magnetic Field of a Current Loop**

Because the loop can be thought of as a wire bent into a circle of radius R, the expression for the magnetic field at the centre of the loop can be found as




$$\sum B_{11}\Delta L = \mu_o I$$

Here, $\Delta L = 2R$

$$\Rightarrow \boxed{B = \frac{\mu_o I}{2R}}$$

If N loops of wire are all tightly wound into a single thin coil, then the field at the center is

$$\boxed{B = \frac{\mu_o NI}{2R}}$$

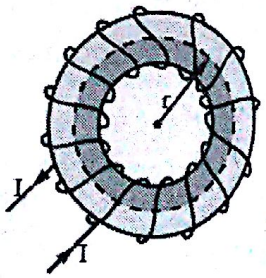
**Illustrative Example**

58. A circular coil of radius 5 cm has 200 turns and carries a current of 0.2A. What is the magnetic field at the center of the coil?

***Solution***: $B = \frac{\mu_o NI}{2R} = \frac{(4\pi \times 10^{-7})(200)(0.2)}{2(5 \times 10^{-2})} = 5 \times 10^{-4} T$

**Ampere's Law and the Magnetic Field of Toroid**

A toroid, is a device consisting of a conducting wire wrapped around a ring, and used to create uniform magnetic field. For a toroid having N turns, the magnetic field inside the toroid at a distance of r from the centre is




$$\sum B_{11}\Delta L = \mu_o I$$

$$B(2\pi r) = N\mu_o I \Rightarrow \boxed{B = \frac{\mu_o NI}{2\pi r}}$$

Generated by CamScanner from intsig.com

**Illustrative Example**

59. Find the magnetic field for a toroid of radius 5 cm with 1000 turns and a current of 3A flowing through it

*Solution*: $B = \frac{\mu_o NI}{2\pi r} = \frac{(4\pi \times 10^{-7})(1000)(3)}{2\pi(0.05)} = 0.012T$

## 14.9 Magnitude and Direction of Earth's Magnetic Field

The magnitude and direction of the earth's field varies with position over the earth's surface. At a given point the earth's field is inclined downward at an angle called *angle of dip*. At the magnetic poles it is vertical, angle of dip = 90°. At the magnetic equator, it is parallel to the earth's surface, angle of dip = 0°. It is thus convenient to resolve the earth's field into horizontal and vertical components.

- The strength of the earth's field ranges from less than $3 \times 10^{-5}$T around America and South Africa to over $6 \times 10^{-5}$T around North Canada and South Australia. Near the poles, the field diminishes with the inverse square of earth radius $\left(1/R^2\right)$. Whereas at greater distances, such as in outer space, it diminishes with the inverse cube of distance $\left(1/R^3\right)$.

**The tangent galvanometer**

A tangent galvanometer (TG) is an instrument for measuring the strength of an electrical current in terms of the magnetic field it produces.

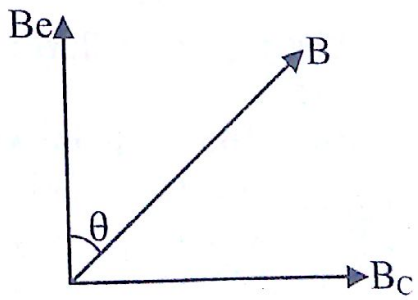
When current flows through a coil of wire, it generates a magnetic field at the center of the coil. The instrument derives its name from the fact that the current is proportional to the tangent of the angle of the needle deflection. When current is passed though the TG a magnetic field is created at its center. This field is given by

$$B_C = \frac{\mu_o NI}{2R}$$

Where, N, is the number of turns of wire in the coil

I, is the current through it and

R, is the radius of the coil




Generated by CamScanner from intsig.com

If $B_C$ is perpendicular to the horizontal component of the earth's field, the needle rests along the resultant.

The horizontal component of the earth's magnetic field is then, expressed as

$$\boxed{B_e = \frac{B_c}{\tan\theta}}$$ Where; θ, is angular deflection.

## Illustrative Example

60. At a particular location, θ = 63°. And a coil of radius 10cm with 1000 turns and a current of 5A is used. What is the horizontal component of the earth's magnetic field at this location?

***Solution***: First of all, find the magnetic field due to coil of wire.

$$B_C = \frac{\mu_o NI}{2R} = \frac{(4\pi\times10^{-7})(1000)(5)}{2(0.1)} = 3\times10^{-2}\,T$$

Next, the horizontal component of the field

$$B_e = \frac{B_C}{\tan\theta} = \frac{3\times10^{-2}}{\tan 63} = \frac{3\times10^{-2}}{2} = 1.5\times10^{-2}\,T$$

## Additional Examples

61. If a compass needle is placed at the magnetic pole.
A. Points south – east
B. Points east – west
C. May stay in any direction
D. Becomes vertical

***Explanation***: At the magnetic pole of earth, the horizontal component of the field is zero, $B_H = 0$. Since a compass needle oscillates in the horizontal plane, no field no deflection.

62. A permanent magnet **Answer: C**
A. Attracts all substances.
B. Attracts only magnetic substances.
C. Attracts magnetic substances and repel all non – magnetic substances.
D. Attracts non – magnetic substances and repels magnetic substances.

63. Which of the following demonstrates that earth has a magnetic field? **Answer: B**
A. The earth is surrounded by ionosphere, i.e., a shell of charged particles.
B. A large quantity of iron ore is found in the earth.

DIGITAL LIBRARY FOR ETHIOPIAN STUDENTS
Digital Library for Ethiopian students
Link: https://telegram.me//digitallibrary4ethiopianstudents or digitallibrary4ethiopianstudents
YouTube Digital[0,1] Ethiopia
Generated by CamScanner from intsig.com

C. The intensity of Cosmic rays, i.e., stream of charged particles coming from outer space is more at the poles than at the equator.

D. There is a huge bar magnet inside the earth.

**Answer: C**

64. The value of angle of dip is zero at the magnetic equator because on it

A. V and H are equal
B. The value of V and H is zero
C. The value of v is zero
D. The value of H is zero

**Answer: C**

65. Magnetic lines of force

A. Cannot intersect at all C) Intersect within the magnet
B. Intersect at infinity D) Intersect at the neutral point

**Answer: A**

66. The core of an electromagnet is made of soft iron, because soft iron has

A. Small susceptibility and small retentivity.
B. Large susceptibility and small retentivity.
C. Large density and large retentivity.
D. Small density and large retentivity.

**Answer: B**

67. The North Pole of earth's magnet is near the geographical.

A. East B. West C. North D. South

**Answer: D**

68. The magnetic susceptibility for a diamagnetic material is

A. Small and negative C. Small and positive
B. Large and Positive D. Large and negative

**Answer: A**

69. A ferromagnetic material is placed in an external magnetic field. The magnetic domains

A. Increase in size C. May increase or decrease in size
B. Decrease in size D. Have no relation with the field

**Answer: C**

70. Of the following the most suitable material for making permanent magnet is

A. Steel B. soft-iron C. copper D. Nickel

**Answer: A**

Generated by CamScanner from intsig.com

71. The magnetism of the magnet is due to
A. The spin of electron C. Big magnet inside earth
B. Earth D. Cosmic rays

**Answer: A**

72. In a bar magnet, lines of magnetic induction
A. Start from the North Pole and end at the South Pole.
B. Run continuously through the bar and outside.
C. Emerge in circular paths from the middle of the bar.
D. Are produced only at north pole like ray of light from a bulb.

**Answer: B**

73. The magnetic fields at two points lying at the same distance from an isolated pole are
A. The same both in magnitude and direction.
B. Different both in magnitude and direction.
C. The same in magnitude and different in direction.
D. Different in magnitude but same in direction.

**Answer: C**

74. Force between two identical bar magnets whose centers are r meter apart is 4.8N. If the separation is increased to 2r, the force between them is reduced to:
A. 2.4N B. 1.2N C. 0.6N D. 0.3N

***Solution***: $F \sim \frac{1}{r^4} \Rightarrow \frac{F^1}{F} = \frac{r^4}{(2r)^4}$

$$F^1 = \frac{1}{2^4} F = \frac{4.8}{16} = 0.3N$$

**Answer: D**

75. The incorrect statement regarding the lines of force of the magnetic field is
A. Magnetic intensity is a measure of lines of force passing through unit area held normal to it.
B. Magnetic lines of force form a close curve.
C. Inside a magnet, its magnetic lines of force move from the north pole of a magnet towards its south pole.
D. Due to a magnet, magnetic lines of force never cut each other.

**Answer: C**

76. The direction of magnet in tan B position of a deflection magnetometer is
A. North – South C. North – West
B. East – West D. South – West

**Answer: B**

Generated by CamScanner from intsig.com

77. When a South pole is placed in the magnetic field of a north pole, the south pole moves towards the north pole and its potential energy
A. Decreases
B. Increases
C. Remains unchanged
D. Gets doubled

**Answer: A**

78. A magnet is placed in iron powder and then taken out. Then, maximum iron powder is at
A. Some away from north pole
B. Some away from south pole
C. The middle of the magnet
D. The end of the magnet

*Explanation*: The field is strong at the end of the magnet.

**Answer: D**

79. The magnet can be completely demagnetized by
A. Heating it slightly.
B. Dropping it into ice cold water.
C. A reverse field of appropriate strength.
D. Breaking the magnet into small pieces.

**Answer: C**

80. The direction of earth's magnetic field at a place is from
A. North to South
B. South to north
C. Magnetic north to south
D. Magnetic south to north

**Answer: D**

81. Out of the following, which is non – magnetic?
A. Zinc B. Steel C. Copper D. Cobalt

**Answer: A**

82. A freely suspended magnet stands vertical at
A. Magnetic equator
B. Geographic south pole
C. Geographic north pole
D. Magnetic South pole

**Answer: D**

**Magnetic Effects of Current**

83. A positively charged particle falls vertically down wards. The horizontal component of the earth's magnetic field will deflect it towards.
A. North B. South C. East D. West

*Explanation*: Apply Fleming's left hand rule.

**Answer: C**

Generated by CamScanner from intsig.com

84. An electron and a proton enter a magnetic field perpendicularly. Both have same kinetic energy. Which of the following is true?
A. Trajectory of electron is less curved.
B. Trajectory of proton is less curved.
C. Both trajectories are equally curved.
D. Both move on straight line path.

*Explanation*: $Bqv = \frac{mv^2}{r} \Rightarrow Bq = \frac{mv}{r}$

$\Rightarrow Bq = \frac{\sqrt{2mk_E}}{r} \Rightarrow r = \frac{\sqrt{2mk_E}}{Bq}$, thus $r \sim \sqrt{m}$

→ As m increases, r increase, hence trajectory is less curved for proton since $m_P > m_e$ and $r_P > r_e$.

**Answer: B**

85. Current is flowing in two long parallel conductors in the same direction. The conductors
A. Attract each other C. Neither attract nor repel
B. Repel each other D. Get perpendicular to each other
*Explanation*: Using right hand rule the magnetic field between conductors carrying currents in the same direction is opposite. Hence, the conductors attract. **Answer: A**

86. A positive charge is moving vertically upward. It enters a magnetic field directed to the north. The force on the charge will be towards
A. North B. South C. East D. West
*Explanation*: Apply Fleming's left hand rule. **Answer: D**

87. A current – carrying power line carries current from west to east. What will be the direction of the magnetic field at a point above the wire?
A. East to West C. North to South
B. West to East D. South to North
*Explanation*: Apply the right hand rule. **Answer: C**

88. An electron is moving along positive x-axis. To get it moving on an anti – clockwise circular path in x – y plane, a magnetic field is applied
A. Along positive y – axis C. Along negative y-axis
B. Along positive z – axis D. Along negative z-axis

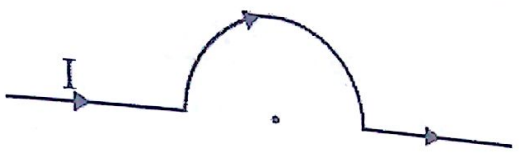
Generated by CamScanner from intsig.com

*Explanation*: For the particle to move along anticlockwise path, force should be along j, velocity is along i.

$\Rightarrow \vec{F} = -e(\vec{v} \times \vec{B}) \Rightarrow j = -(i \times k)$

89. The magnetic field at the center of the semi – circle due to:

A. Each straight portion
B. Semi – circle alone




**Answer: B**

A. Zero B. $\frac{\mu_0 I}{4R}$ C. $\frac{\mu_0 I}{R}$ D. Infinity

*Explanation*: a) The magnetic field at a point on the axis of the current element is zero.

b) $B = \frac{1}{2}\left(\frac{\mu_0 I}{2R}\right) = \frac{\mu_0 I}{4R}$ **Answer: B**

90. A proton is moving with velocity $\vec{v}$ in a direction opposite to the direction of magnetic field $\vec{B}$. The magnetic force experienced by the proton is

A. Bev B. -Bev C. Bv D. zero

*Explanation*: $F = Bqv \sin\theta$, $\theta = 0 \Rightarrow F = 0$ **Answer: D**

91. An electron accelerated through a potential difference v passes through a uniform transverse magnetic field and experiences a force F. If the potential is doubled, the electron in the field experience a force.

A. F B. $F/2$ C. $\sqrt{2}F$ D. 2F

*Explanation*: $F = Bqv$ and $\frac{1}{2}mv^2 = ev \Rightarrow v = \sqrt{\frac{2ev}{m}}$

$$F = Bq\sqrt{\frac{2ev}{m}} \Rightarrow F \sim \sqrt{v} \Rightarrow \frac{F^1}{F} = \frac{\sqrt{2v}}{\sqrt{v}} \Rightarrow F^1 = \sqrt{2F}$$

**Answer: C**

92. A neutron, a proton, an electron and an α-particle enter a region of uniform magnetic field with equal velocities. The magnetic field is directed into the paper. The track followed by

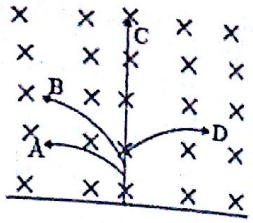
A. The neutron b) the proton

A. A B. B C. C D. D

Generated by CamScanner from intsig.com

*Explanation*:

a) Neutron follows straight path. Since no force on it.




**Answer: C**

b) $Bqv = \frac{mv^2}{r}$

$\Rightarrow r = \frac{mv}{Bq}$, $r \sim m$. The proton has less mass than α **Answer: A**

93. A uniform electric field and a uniform magnetic field are produced, pointed in the same direction. An electron is projected with its velocity pointed in the same direction.

A. The electron will turn to its right.
B. The electron will turn to its left.
C. The electron velocity will increase in magnitude.
D. The electron velocity will decrease in magnitude.

*Explanation*: The force on electron will be opposite to the direction of electric field. **Answer: D**

94. Two beams of protons moving parallel and in the same direction will

A. Attract each other
B. Exert no force.
C. Deflect perpendicular to the beam.
D. Repel each other

*Explanation*: The electrostatic force dominates the magnetic force. **Answer: D**

95. A positively charged particle projected towards east is deflected towards north by a magnetic field. The field may be

A. Towards west
B. Towards South
C. Upward
D. Downward

*Explanation*: Apply Fleming's left hand rule. **Answer: D**

96. An electric charge in uniform motion produces

A. An electric field only
B. A magnetic field only
C. Both electric and magnetic field
D. No such field at all **Answer: C**

97. A current – carrying straight wire is kept along the axis of a circular loop carrying a current. The straight wire

A. Will exert an inward force on the circular loop.
B. Will exert an outward force on the circular loop.

Generated by CamScanner from intsig.com

C. Will not exert a force on the circular loop parallel to itself.
D. Will exert a force on the circular loop parallel to itself.

*Explanation*: Magnetic lines of force are parallel to circular loop.

**Answer: C**

98. A charge moving with velocity $\vec{v}$ in x-direction is subjected to a field of magnetic induction in the negative x-direction. As a result, the charge will
A. Remain unaffected
B. Start moving in a circular path in y – z plane
C. Retard along x – axis
D. Move along a helical path around x – axis

*Explanation*: $\vec{F} = q(\vec{v} \times \vec{B}), \theta = 180°$

$F = qvB \sin\theta = 0$

**Answer: A**

99. To covert a galvanometer into an ammeter, one has to connect
A. A small resistance in series
B. A small resistance in parallel
C. A high resistance in series
D. A high resistance in parallel

**Answer: B**

100. A charged particle is situated in a region of space and it experiences a force only when it is in motion. It can be deduced that the region encloses
A. Both an electric field and a gravitational field.
B. Both an electric and a magnetic field.
C. A magnetic field only
D. An electric field only

*Explanation*: F = qv B sin θ, it is non – zero only when the particle is in motion.

**Answer: C**

101. Weber amper per metre is equal to
A. Joule B. Newton C. Henry D. Watt

*Explanation*: $F = BIL = \frac{\Phi}{A} IL$

$$F = \frac{Wb}{m^2} Am \Rightarrow \frac{WbA}{m} = N$$

**Answer: B**

102. An electron continues to move in a straight line while passing through magnetic field. Choose the correct statement.
A. It is never possible.

Generated by CamScanner from intsig.com

B. It is possible if B is perpendicular to the direction of the motion of the electron.
C. It is possible if B is parallel to the direction of motion of the electron.
D. It is possible if B is at 45° to the direction of motion of the electron.

*Explanation*: When charge moves parallel to magnetic field, magnetic force is zero. **Answer: C**

103. A proton beam is going from north to south and an electron beam is going from south to north. Neglecting the earth's magnetic field, the electron beam will be deflected
A. Towards the proton beam
B. Away from the proton beam
C. Upwards
D. Downwards

*Explanation*: The two beams constitute like currents. Like currents attract. **Answer: A**

104. Two parallel long wires, each carrying a current of 1A are 0.5m apart. The force per unit length exerted by one wire on the other is

A. $\frac{\mu_o}{4\pi}$ B. $\frac{\mu_o}{3\pi}$ C. $\frac{\mu_o}{2\pi}$ D. $\frac{\mu_o}{\pi}$

*Explanation*: $F = \frac{\mu_o I_1 I_2 \ell}{2\pi r} \Rightarrow \frac{F}{\ell} = \frac{\mu_o (1)(1)}{2\pi(0.5)} = \frac{\mu_o}{\pi}$ **Answer: D**

105. Protons are shot perpendicular to a magnetic field.
A. Magnetic field will have no influence on the motion of protons.
B. Protons will continue to move in the same direction but will gain momentum.
C. Protons will continue to move in the opposite direction but will gain momentum
D. They will bend in an arc of a circle.

*Explanation*: The magnetic force shall become centripetal force. **Answer: D**

106. If a copper rod carries a direct current, the magnetic field associated with the current will be
A. Only inside the rod
B. Only outside the rod
C. Both inside and outside the rod
D. Neither inside nor outside the rod. **Answer: C**

Generated by CamScanner from intsig.com

107. A current is flowing north along a power line. The direction of the magnetic field above it, neglecting the earth's field is
A. North B. East C. South D. West
**Hint,** use the right hand rule.
**Answer: B**

108. Two long wires distance r apart exert a force F on one another when current through each wire is I. If the distance between them is doubled and the current is decreased to $I/3$, the force they exert on one another will be
A. $F/6$ B. $F/9$ C. $2F/3$ D. $F/18$

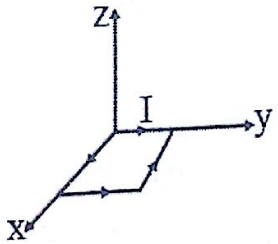
*Explanation*: $F = \frac{\mu_o I_1 I_2 \ell}{2\pi r} = \frac{\mu_o I^2 \ell}{2\pi r}$

$$F_2 = \frac{\mu_o \left(I/3\right)^2 \ell}{2\pi(2r)} = \frac{1}{18}\left(\frac{\mu_o I^2 \ell}{2\pi r}\right) = \frac{1}{18}F_1$$

**Answer: D**

109. Which one of the following is not deflected by a magnetic field?
A. positron C. $\beta$ – rays
B. $\alpha$ – particle D. $\gamma$ – rays
*Explanation*: $\gamma$ – rays are uncharged. **Answer: D**

110. A rectangular coil of cross – sectional area A is placed in a uniform magnetic field $\vec{B}$ directed along z axis. What is the torque on the coil?
A. ABI along +ve y – axis
B. ABI along –ve y – axis.
C. ABI along +ve x – axis.
D. Zero.




*Explanation*: $\tau = NI(\vec{A} \times \vec{B})$
Since $\vec{A} // \vec{B} \therefore \tau = 0$
**Answer: D**

111. An electron enters a region where magnetic (B) and electric (E) fields are mutually perpendicular to one another. Then
A. It will always move in the direction of B.
B. It will always move in the direction of E.
C. It always possess circular motion.
D. It can go undeflected.

Generated by CamScanner from intsig.com

107. A current is flowing north along a power line. The direction of the magnetic field above it, neglecting the earth's field is
A. North B. East C. South D. West
**Hint,** use the right hand rule.
**Answer: B**

108. Two long wires distance r apart exert a force F on one another when current through each wire is I. If the distance between them is doubled and the current is decreased to $I/3$, the force they exert on one another will be
A. $F/6$ B. $F/9$ C. $2F/3$ D. $F/18$

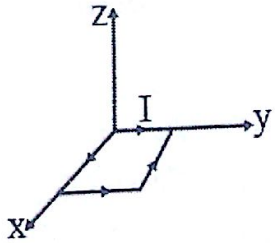
*Explanation*: $F = \frac{\mu_o I_1 I_2 \ell}{2\pi r} = \frac{\mu_o I^2 \ell}{2\pi r}$

$$F_2 = \frac{\mu_o \left(I/3\right)^2 \ell}{2\pi(2r)} = \frac{1}{18}\left(\frac{\mu_o I^2 \ell}{2\pi r}\right) = \frac{1}{18}F_1$$

**Answer: D**

109. Which one of the following is not deflected by a magnetic field?
A. positron C. $\beta$ – rays
B. $\alpha$ – particle D. $\gamma$ – rays
*Explanation*: $\gamma$ – rays are uncharged. **Answer: D**

110. A rectangular coil of cross – sectional area A is placed in a uniform magnetic field $\vec{B}$ directed along z axis. What is the torque on the coil?
A. ABI along +ve y – axis
B. ABI along –ve y – axis.
C. ABI along +ve x – axis.
D. Zero.




*Explanation*: $\tau = NI(\vec{A} \times \vec{B})$
Since $\vec{A} // \vec{B} \therefore \tau = 0$ **Answer: D**

111. An electron enters a region where magnetic (B) and electric (E) fields are mutually perpendicular to one another. Then
A. It will always move in the direction of B.
B. It will always move in the direction of E.
C. It always possess circular motion.
D. It can go undeflected.

Generated by CamScanner from intsig.com

*Explanation*: If $F_m = F_E$, Bev = eE, then the electron shall move undeflected. **Answer: D**

112. A charged particle enters in a strong magnetic field. Then its kinetic energy
A. Increases C. Remains constant
B. Decreases D. First increase and then become constant.
*Explanation*: Since the work done by magnetic field is zero, the kinetic energy remains constant.
**Answer: C**

113. A magnetic field
A. Always exerts a force on a charged particle.
B. Never exerts a force on a charged particle.
C. Exerts a force, if the charged particle is moving across the magnetic field lines.
D. Exerts a force, if the charged particle is moving along the magnetic field lines. **Answer: C**

114. A proton and an electron both moving with the same velocity v enter into a region of magnetic field directed perpendicular to the velocity of the particles. They will now move in circular orbits such that
A. Their time periods will be same.
B. The time period for proton will be higher.
C. The time period for electron will be higher.
D. Their orbital radii will be same.
*Explanation*: $T = \frac{2\pi m}{Bq} \Rightarrow T \sim m$
Since proton has larger mass, its period is large. **Answer: B**

115. A current – carrying rectangular coil is placed in a uniform magnetic field. In which orientation, the coil will not tend to rotate?
A. The magnetic field is parallel to the plane of the coil.
B. The magnetic field is perpendicular to the plane of the coil.
C. The magnetic field is at 45° with the plane of the coil.
D. Always in any orientation.
*Explanation*: $\tau = I(\vec{A} \times \vec{B}) = IAB \sin\theta$
$\tau = 0$ when A and B are parallel. **Answer: B**

Generated by CamScanner from intsig.com

116. An electron is travelling horizontally towards east. A magnetic field in vertically downward direction exerts a force on the electron along
A. East B. West C. North D. South
*Explanation*: Assume a positive charge is travelling due west and then apply Fleming's left hand rule. **Answer: D**

117. An electron enters a magnetic field whose direction is perpendicular to the velocity of the electron. Then,
A. The speed of the electron will increase.
B. The speed of the electron will decrease.
C. The speed of the electron will remain the same.
D. The velocity of the electron will remain the same.
*Explanation*: Magnetic field cannot change the speed of the charged particle. **Answer: C**

118. The magnetic induction at any point due to a long straight wire carrying a current is
A. Proportional to the distance from the wire.
B. Inversely proportional to the distance from wire.
C. Inversely proportional to the square distance from wire.
D. Does not depend on distance.

*Explanation*: $B = \frac{\mu_o I}{2\pi r} \Rightarrow B \sim \frac{1}{r}$ **Answer: B**

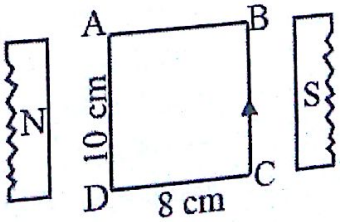
119. The magnetic field induction at a distance of 4cm from a long current carrying wire is $10^{-3}$T. The magnetic field induced at a distance of 12cm from the same wire is
A. $9\times10^{-3}$T C. $3.33\times10^{-3}$T
B. $3\times10^{-3}$T D. $1.11\times10^{-3}$T

*Solution*: $B = \frac{\mu_o I}{2\pi r} \Rightarrow B \sim \frac{1}{r}$

$\Rightarrow \frac{B_1}{B_2} = \frac{r_2}{r_1}$; Thus, $B_2 = \frac{B_1 r_1}{r_2} = \frac{10^{-3}(4)}{12} = 3.33\times10^{-3}$T

**Answer: C**

120. A 100 turns coil carries a current of 2A in a magnetic field of 0.2Wb/m$^2$. The torque acting on the coil is
A. 0.32Nm tending to rotate AD out of the page.
B. 0.32Nm tending to rotate AD into the




Generated by CamScanner from intsig.com

page.

C. 0.0032Nm tending to rotate AD out of the page.

D. 0.0032Nm tending to rotate AD into the page.

***Solution***:

$\tau = NBIA = 100 \times 0.2 \times 2 \times 0.1 \times 0.08$

$= 0.32Nm$

Applying Fleming's left hand rule, we will observe side AD rotating out of the page. **Answer: A**

121. A millimeter of range 10mA has a coil of resistance 1Ω. To use it as a voltmeter of range 10v, the resistance that must be connected in series with it is:

A. 9Ω B. 99Ω C. 999Ω D. 1000Ω

***Solution***: $R = \frac{V}{I_C} - R_C = \frac{10}{10 \times 10^{-3}} - 1 = 999\Omega$ **Answer: C**

122. A magnetic field of $0.004\,\hat{k}$ T exerts a force of $(4i + 3j)10^{-10}$N on a particle having a charge of $10^{-9}$C and moving in x – y plane. The velocity of the particle is

A. (75j + 100j)m/s C. (-75i + 100j) m/s

B. (75i – 100j) m/s D. (-75i – 100j) m/s

***Solution***: $\vec{F} = q(\vec{v} \times \vec{B}) \Rightarrow (4i + 3j)10^{-10} = 10^{-9}(v \times 0.004k)$

$\Rightarrow 0.4i + 0.3j = (v_x i + v_y i) \times 0.004k = -0.004v_y j + 0.004v_x i$

$\Rightarrow \vec{v} = -75i + 100j$ **Answer: C**

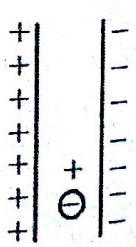
123. In an ammeter, 4% of the main current is passing through galvanometer. If the galvanometer is shunted with a 5Ω resistance, then the resistance of the galvanometer is

A. 4Ω B. 5Ω C. 20Ω D. 120Ω

***Solution***: $R_S = \frac{R_C I_C}{I - I_C} \Rightarrow R_C = \frac{R_S(I - I_C)}{I_C}$

$$R_C = \frac{5(100-4)}{4} = 120\Omega$$ **Answer: D**

124. A potential difference of 600v is applied across the plates of a parallel plate capacitor plates being separated by 3mm. An electron projected vertically parallel to the plates with a velocity of $2 \times 10^6$ m/s moves un-deflected between the plates. The magnetic field in the region between the plates is:



A. 0.1T B. 0.2T C. $2 \times 10^6$T D. 600T

Generated by CamScanner from intsig.com

**Solution:** $Bqv = Eq$

$$B = \frac{E}{v} = \frac{V/d}{v} = \frac{600}{3\times10^{-3}\times2\times10^{6}}$$

$$B = 0.1T$$

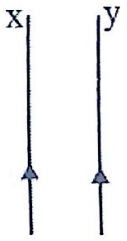
**Answer: A**

125. A particle of charge q and mass m moving with a velocity $\vec{v}$ along the x-axis enters the region $x > 0$ with uniform magnetic field B along the $\hat{k}$ direction. The particle will penetrate in this region in the x – direction up to a distance d equal to:

A. Zero B. $\frac{mv}{qB}$ C. $\frac{2mv}{qB}$ D. Infinity

**Solution:** $F_C = F_m \Rightarrow \frac{mv^2}{r} = Bqv \Rightarrow r = \frac{mv}{Bq}$ **Answer: B**

126. Two long parallel wires x and y carry currents of 3A and 5A respectively. The force experienced per unit length by x is $5 \times 10^{-5}$N/m to the right. The force per unit length experienced by wire Y is:




A. $5 \times 10^{-5}$N/m to the left
B. $3 \times 10^{-5}$N/m to the right
C. $3 \times 10^{-5}$N/m to the left
D. $5 \times 10^{-5}$N/m to the right

**Hint:** Think of Newton's 3rd law **Answer: A**

127. A proton and an α-particle enter a uniform magnetic field perpendicular with the same speed. If proton takes 25μs to make 5 revolution, then the period for the α-particle would be:

A. 50μs B. 25μs C. 10μs D. 5μs

**Solution:** From the relation, $F_m = F_C \Rightarrow Bqv = \frac{mv^2}{r}$

$$Bq = \frac{mv}{r}, \text{ but } \frac{v}{r} = \omega$$

$$Bq = m\omega, \ \omega = \frac{2\pi}{T}$$

$$Bq = \frac{m2\pi}{T} \Rightarrow T = \frac{2\pi m}{Bq}$$

$$T_p = \frac{2\pi m_p}{Bq_p} \text{ and } T_\alpha = \frac{2\pi m\alpha}{Bq\alpha}$$

Generated by CamScanner from intsig.com

$$\frac{T_\alpha}{T_p} = \frac{m_\alpha q_p}{m_p q_\alpha}, \text{ but } m_\alpha = 4m_p \text{ and } q_\alpha = 2q_p$$

$$T_\alpha = 2\left(\frac{25}{5}\right) = 10\mu s$$

**Answer: C**

128. Two straight parallel wires, both carrying 10A in the same direction, attract each other with a force of $1 \times 10^{-3}$N. If both currents are doubled, the force of attraction will be:

A. $1 \times 10^{-3}$N C. $4 \times 10^{-3}$N
B. $2 \times 10^{-3}$N D. $0.25 \times 10^{-3}$N

***Solution***: $F = \frac{\mu_o I_1 I_2 \ell}{2\pi r} = \frac{\mu_o (10)(10)\ell}{2\pi r} \Rightarrow \frac{100\mu_o \ell}{2\pi r} = 10^{-3} = 10^{-3}$

$$\frac{\mu_o \ell}{2\pi r} = 10^{-5} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (*)$$

And, $F = \frac{\mu_o (20)(20)\ell}{2\pi r} = 400\frac{\mu_o \ell}{2\pi r} = 400(10^{-5}) = 4 \times 10^{-3}$N

**Answer: C**

129. If 10% of the current passes through a moving coil galvanometer of resistance 99Ω, then the shunt resistance will be:

A. 9.9Ω B. 11Ω C. 10Ω D. 9Ω

***Solution***: $R_s = \frac{R_c I_c}{I - I_c} = \frac{99(0.1)I}{I - (0.1)I} = \frac{9.9I}{0.9I} = 11\Omega$ **Answer: B**

130. A current of 5A is flowing in a wire of length 1.5m. A force of 7.5N acts on it when it is placed in a uniform magnetic field of 2T. The angle between the magnetic field and the direction of the current is

A. 30° B. 45° C. 60° D. 90°

***Solution***: $F = BI\ell \sin\theta \Rightarrow \sin\theta = \frac{F}{BI\ell} = \frac{7.5}{2 \times 5 \times 1.5} = 0.5$

$\theta = 30°$

**Answer: A**

131. A straight wire of length 0.5m and carrying a current of 1.2A placed in a uniform magnetic field of induction 2T. The magnetic field is perpendicular to the length of the wire. The force on the wire is

A. 2.4N B. 1.2N C. 3.0N D. 2.0N

***Solution***: $F = BI\ell = 2 \times 1.2 \times 0.5 = 1.2$N **Answer: B**

Generated by CamScanner from intsig.com

132. If the strength of the magnetic field produced 10cm away from an infinitely long straight conductor is $10^{-5}$T, the value of the current flowing in the conductor will be:
A. 5A B. 10A C. 500A D. 1000A

*Solution*: $B = \frac{\mu_o I}{2\pi r} \Rightarrow I = \frac{2\pi r B}{\mu_o} = \frac{2\pi(10^{-5})}{4\pi \times 10^{-7}} \times 0.1$

$I = 500A$ **Answer: C**

133. A rectangular coil 20cm × 20cm has 100 turns and carries a current of 1A. It is placed in a uniform magnetic field of 0.5T with the direction of magnetic field parallel to the plane of the coil. The magnitude of the torque required to hold this coil in this position is
A. Zero B. 200Nm C. 2Nm D. 10Nm
*Solution*: $\tau = NBIA = 100 \times 0.5 \times 1 \times 400 \times 10^{-4} = 2Nm$

**Answer: C**

134. A solenoid is 1m long and it has 4250 turns. If a current of 5A is flowing through it, what is the magnetic field at its center?
A. $5.4 \times 10^{-2}$T C. $1.35 \times 10^{-2}$T
B. $2.7 \times 10^{-2}$ T D. $0.675 \times 10^{-2}$T

*Solution*: $B = \frac{\mu_o NI}{\ell} = 4\pi \times 10^{-7} \times 4250 \times 5 = 2.7 \times 10^{-2}$T

**Answer: B**

135. The magnetic field at a distance x from a long wire carrying current I is 0.4T. The magnetic field at a distance 2x is
A. 0.8T B. 0.2T C. 0.1T D. 1.6T

*Solution*: $B_1 = \frac{\mu_o I}{2\pi r_1} = 0.4$, and

$B_2 = \frac{\mu_o I}{2\pi r_2} = \frac{\mu_o I}{2\pi(2r_1)} = \frac{B_1}{2} = 0.2T$ **Answer: B**

136. A proton and an alpha particle enter a uniform magnetic field normally with the same velocity. The period of revolution of the $\alpha$ – particle will be:
A. Two times that of proton. C. Four times that of proton.
B. Three times that of proton. D. Same as that of proton.

*Explanation*: $T = \frac{2\pi m}{Bq}, T_\alpha = \frac{2\pi(4m_P)}{B(2q)} = 2T$ **Answer: A**

Generated by CamScanner from intsig.com

137. Magnetic field induction at the center of coil of 50 turns, radius 0.5m and carrying a current of 2A is

A. $0.5 \times 10^{-5}$T C. $3 \times 10^{-5}$T

B. $1.25 \times 10^{-4}$T D. $4 \times 10^{-5}$T

***Solution***: $B = \frac{\mu_o NI}{2r} = \frac{4\pi \times 10^{-7} \times 50 \times 2}{2 \times 0.5} = 1.25 \times 10^{-4}$T

**Answer: B**

138. A power line lies along the east – west direction and carries a current of 10A. The force per meter due to the earth's magnetic field of $10^{-4}$T is

A. $10^{-5}$N/m B. $10^{-4}$N/m C. $10^{-3}$N/m D. $10^{-2}$N/m

***Solution***: $F = BI\ell \Rightarrow \frac{F}{\ell} = BI = 10^{-4} \times 10 = 10^{-3}$ N/m **Answer: C**

139. A proton moving with a speed of $10^7$m/s enters the region of a uniform, perpendicular magnetic field of 0.4T. The radius of the circular path described by proton is (mass of proton=$1.6 \times 10^{-27}$kg)

A. 0.25mm B. 0.25m C. 4mm D. 4.25mm

***Solution***: $F_C = F_m \Rightarrow \frac{mv^2}{r} = Bqv$

$\Rightarrow \frac{mv}{r} = Bq, r = \frac{mv}{Bq} = \frac{(1.6 \times 10^{-27})(10^7)}{(0.4)(1.6 \times 10^{-19})} = 0.25$m **Answer: B**

140. A proton of energy 200 Mev enters the magnetic field of 5T. If direction of field is from south to north and motion is upward, what is the force acting on it

A. Zero C. $3.2 \times 10^{-8}$N

B. $1.6 \times 10^{-10}$N D. $1.6 \times 10^{-6}$N

***Solution***: $\frac{1}{2}mv^2 = ve \Rightarrow v = \sqrt{\frac{2Ve}{m}}$, and $F = Bqv = 1.6 \times 10^{-10}$N

**Answer: B**

141. An electric charge does not experience a force due to a magnetic field

A. If it is stationary

B. If it moves parallel to the magnetic field.

C. In both the above two cases.

D. If it moves perpendicular to the field.

**Experience**: $F = Bqv \sin\theta = 0$, if $v = 0$ or $\theta = 0°$ **Answer: C**

Generated by CamScanner from intsig.com

142. A charged particle is moved along a magnetic field line. The magnetic force on the particle is
A. Along its velocity. C. Perpendicular to its velocity.
B. Opposite to its velocity. D. Zero
*Explanation*: $\theta = 0° \Rightarrow F = Bqv \sin\theta = 0$ **Answer: D**

143. A wire is placed parallel to the lines of force in a magnetic field and a current flow in the wire. Then:
A. The wire will experience a force along the direction of magnetic field.
B. The wire will not experience any force.
C. The wire will experience a torque.
D. The wire will experience a perpendicular force.
*Explanation*: $F = BI\ell \sin\theta = 0$, since $\ell \,//\, B$. **Answer: B**

144. A proton is projected horizontally east ward in a uniform magnetic field which is horizontally and south ward in direction. The proton will be deflected
A. Upward C. South ward
B. Down ward D. North ward
**Hint:** Apply Fleming's left hand rule. **Answer: B**

145. A stragith conductor carrying current is shown. The direction of the magnetic field at P is
A. Normal to the plane of the paper, downwards.
B. Normal to the plane of the paper, upward.
C. To the left of the plane of the paper.
D. To the left of the plane of the paper.

**Hint:** Apply Right hand rule. **Answer: A**

146. A vertical wire carries a current in upward direction. An electron beam sent horizontally towards the wire will be deflected
A. To the right C. Upward
B. To the left D. Downward
**Hint:** Apply Fleming's left hand rule. **Answer: C**

147. An electron is moving with a speed of $10^8$m/s perpendicular to a uniform magnetic field B. Suddenly, the magnetic field is reduced to $B/2$. What is the radius of the path if the original radius is R?
A. No change B. $R/2$ C. 2R D. $R/9$

Generated by CamScanner from intsig.com

*Explanation*: $Bqv = \frac{mv^2}{R} \Rightarrow Bq = \frac{mv}{R}$

$$R = \frac{mv}{Bq}, \text{ when } B_2 = B/2$$

$$R_2 = \frac{mv}{B/2\, q} = \frac{2mv}{Bq} = 2R$$ **Answer: C**

148. A beam of charged particles is passing through a magnetic field. The work done on the beam by the magnetic field
A. Is zero
B. Will depend on the velocity of the beam.
C. Will depend on the deflection of the beam.
D. Will depend on the strength of the field. **Answer: A**

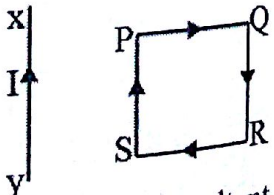
149. A 10 turn circular coil of radius 2cm carries a current of 0.5A. The magnetic induction at the center of the coil is
A. $1.57 \times 10^{-6}$T C. $1.57 \times 10^{-4}$T
B. $1.57 \times 10^{-5}$T D. 1.57T

*Solution*: $B = \frac{\mu_o NI}{2r} = \frac{4\pi \times 10^{-7} \times 10 \times 0.5}{2 \times 2 \times 10^{-2}} = 1.57 \times 10^{-4}$T

**Answer: C**

150. Which one of the following is the correct unit of magnetic flux?
A. Tesla B. $Wbm^{-1}$ C. $Wbm^{-2}$ D. $Tm^2$
*Explanation*: $\phi = BA = Tm^2$ **Answer: D**

151. A long straight wire xy lines in the same plane as a square loop of wire PQRS which is free to move. If the wire and loop carry steady currents, what will be the effect on the loop?
A. It will move towards wire xy.
B. It will move away from wire xy.
C. It will rotate about axis xy.
D. It will be unaffected.




*Explanation*: Wire xy and side PS will attract while, wire xy and side QR will repel and the resultant force is attraction. **Answer: A**

## Unit 14: Conceptual Questions

1. An electron moves with constant velocity through a region of space that is free of electric field. Can one conclude that the magnetic field is zero in this region? Explain

Generated by CamScanner from intsig.com

2. A proton and an electron moving horizontally enter a region where there is a uniform magnetic field that is directed perpendicular to their velocities. Explain the nature of their motion.
3. If a straight section of wire carries a current vertically upward (North) in a magnetic field that is directed horizontally toward the East. What is the direction of the magnetic force on the wire?
4. Why does the picture on a television screen become distorted when a magnet is brought near the screen (**Caution**: you should NOT Do this practically as it may cause permanent damage on the TV)
5. A circular loop, a square, and a rectangular loop all have the same perimeter. Which will have the maximum magnetic moment and torque when placed in a uniform magnetic field?

## Unit 14: Problems

### Magnetic force on moving charges

6. A proton moves perpendicular to a uniform magnetic field at a speed of $10^7$m/s in the z – direction and experiences an acceleration of $2 \times 10^{13}$m/s$^2$ in the x direction. Determine the magnitude and direction of the field.
7. What is the acceleration of a proton moving with a speed of 6.5 m/s at right angle to a magnetic field of 1.2T?
8. An electron moving with a velocity $\vec{v} = (2i + 3j - 7k)$m/s enters a magnetic field $\vec{B} = (4i - 11j)$T. Find the force on the electron.

### The motion of charged particle in a magnetic field

9. Find the radius of a proton's orbit when it moves perpendicular to a magnetic field of 0.4T with a speed of $6.3 \times 10^{-5}$m/s.
10. A proton with kinetic energy of $4 \times 10^{-6}$J moves perpendicular to a magnetic field of 0.3T. What is the radius of its circular path?
11. A 6μC particle with a mass of $2.8 \times 10^{-5}$kg moves perpendicular to a 1.2T magnetic field in a circular path of radius 9m.
    A. How fast is the particle moving?
    B. How long will it take the particle to complete one orbit?

### Magnetic force on current carrying wire

12. A 0.75m long wire carries a current of 2.4A in the x direction and is placed in a uniform magnetic field of 1.6T in the z direction. What is the magnetic force on the wire?

DIGITAL LIBRARY FOR ETHIOPIAN STUDENTS
Digital Library for Ethiopian students
Link: https://telegram.me//digitallibrary4ethiopianstudents or digitallibrary4ethiopianstudents
YouTube Digital[0,1] Ethiopia
Generated by CamScanner from intsig.com

13. A 3m long wire carrying a current of 15A is placed at 30° to a uniform magnetic field of 2.5T. What is the magnetic force on the wire?
14. A wire of mass 20 g/cm carries a 3A current horizontally due south. What minimum magnetic field is needed to lift the wire upward?

**Torque and Magnetic moment**

15. A coil of 18 turns has an area of $600cm^2$ and bears a current of 0.3A. If it is placed with its plane parallel to a magnetic field of 2T. Find
    a) The magnetic dipole moment.
    b) The torque on the coil.
16. A rectangular loop of 300 turns is 12cm wide and 18cm high. What is the current in this loop if the maximum torque in a magnetic field of 0.45T is 24Nm?

**Magnetic field produced by an electric current**

17. Find the magnetic field 12cm from a long straight wire that carries a current of 6.28A.
18. Two long, parallel wires separated by 0.6m carry currents of $I_1$= 8A and $I_2$ = 20A in the same direction.
    a) What is the magnitude of the magnetic field at the midway between the wires?
    b) At what point on the line joining the wires is the magnetic field zero?
19. Two long parallel wires separated by a distance of 10cm carry currents of $I_1$ = 5A and $I_2$ = 8A in the same direction. What is the force per unit length exerted on each wire by the other?

## Unit 14: Explanation to Questions

1. No. If the electron moves in the same or opposite direction to the magnetic field, the magnetic force on it is zero. Thus, the electron moves at constant velocity.
2. The proton moves in a circular path of bigger radius upwards. While the electron moves in a circular path downward at smaller radius than the proton.
3. Using the right hand rule, the force is directed away from you or into the page.
4. It is because, the magnetic field produces a magnetic force on the electron moving toward the screen that produce the image. Thus,

Generated by CamScanner from intsig.com

the magnetic force deflects the electrons, and the result is a distorted image.

5. Magnetic moment and torque depends on cross – sectional area, hence the greater the area, the greater the moment and torque for a proper orientation.

## Unit 14: Solution to Problems

6. $F = qvB = ma \quad B = \dfrac{ma}{qv} = \dfrac{(1.67\times10^{-27})(2\times10^{13})}{(1.6\times10^{-19})(10^{7})}$

$B = 2.1T$ in the $-y$ direction

7. $F = qvB = ma \Rightarrow a = \dfrac{qvB}{m} = \dfrac{(1.6\times10^{-19})(6.5)(1.2)}{1.67\times10^{-27}} = 7.5\times10^{8}\ m/s^2$

8. $\vec{F} = q(\vec{v}\times\vec{B}) = (12.3i + 4.5j - 1.6k)\times10^{-8}N$

9. $r = \dfrac{mv}{qB} = \dfrac{(1.67\times10^{-27})(6.3\times10^{5})}{(1.6\times10^{-19})(0.4)} = 1.6mm = 1.6\times10^{-3}\ m$

10. $r = \dfrac{mv}{qB}$ and $kE = \dfrac{1}{2}mv^2 \Rightarrow v = \sqrt{\dfrac{2kE}{m}}$, then $r = 2.7\times10^{-2}m$

11. a) $v = \dfrac{Bqr}{m} = \dfrac{(1.2)(6\times10^{-6})(9)}{2.8\times10^{-5}} = 2.3m/s$

b) $T = \dfrac{2\pi r}{v} = \dfrac{(2\pi)(9)}{2.3} = 2.4\,sec$

12. $F = I(\vec{\ell}\times\vec{B}) = 2.4(0.75i\times 1.6\hat{k}) = (-2.88j)N$

13. $F = I\ell B\sin\theta = (15)(3)(2.5)(\sin 30°) = 56.2N$

14. $F = W \Rightarrow I\ell B = mg \Rightarrow B = \left(\dfrac{m}{\ell}\right)\left(\dfrac{g}{I}\right) = \left(\dfrac{20\times10^{-3}}{10^{-2}}\right)\left(\dfrac{10}{3}\right) = 6.5T$

East

15. a) $\mu = NIA = (18)(0.3)(600\times10^{-4}) = 3.2\times10^{-2}Am^2$

b) $\tau = \mu\times B = \mu B\sin\theta = (3.2\times10^{-2})(2) = 6.5\times10^{-2}N.m$

16. $\tau_{max} = NIAB \Rightarrow I = \dfrac{\tau_{max}}{NAB} = \dfrac{24}{(300)(12\times10^{-2})(18\times10^{-2})(0.45)T} = 8A$

17. $B = \dfrac{\mu_o T}{2\pi r} = \dfrac{(4\pi\times10^{-7})(6.3)}{2\pi(12\times10^{-2})} = 105T$

18. a) $B_1 = \dfrac{\mu_o I_1}{2\pi r} = \dfrac{(4\pi\times10^{-7})(8)}{2\pi(0.3)} = 5.3\times10^{-6}T$ Inward

Generated by CamScanner from intsig.com

$$B_2 = \frac{\mu_o I_2}{2\pi r} = \frac{(4\pi \times 10^{-7})(20)}{2\pi(0.3)} = 1.3 \times 10^{-6} \text{ Outward}$$

$B = B_1 - B_2 = 4 \times 10^{-6}$ in ward

b) $B = 0 \Rightarrow B_1 - B_2 = 0$

$$B_1 = B_2 \Rightarrow \frac{\mu_o I_1}{2\pi r_1} = \frac{\mu_o I_2}{2\pi r_2} \Rightarrow \frac{I_1}{r_1} = \frac{I_2}{r_2}$$

$$\frac{r_1}{r_2} = \frac{I_1}{I_2} = \frac{8}{20} \Rightarrow \frac{r_1}{r_2} = \frac{2}{5}, \text{ but } r_1 + r_2 = 0.6$$

$5r_1 = 2r_2$ and $r_1 = 0.6 - r_2$, hence $r_1 = 0.17$m and $r_2 = 0.43$m

19. $$F = \frac{\mu_o I_1 I_2 \ell}{2\pi r} \Rightarrow \frac{F}{\ell} = \frac{\mu_o I_1 I_2}{2\pi r} = \frac{(4\pi \times 10^{-7})(5)(8)}{2\pi(0.1)} = 8 \times 10^{-5} \text{ N/m}$$

Generated by CamScanner from intsig.com

# Unit 15

# Electromagnetic Induction and A.C Circuit

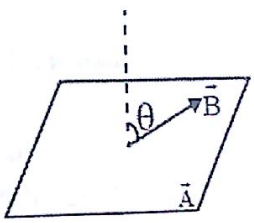
Because nature is often symmetric, the discovery that electric current produce magnetic field led scientists to discover a changing magnetic field can produce an electric current. The phenomenon of electromagnetic induction made possible the invention of a host of important applications in our everyday life, starting from motors, generators and transformers to microphones, speakers, computer disk drives, the magnetic guitar, security devices and so much more.

## 15.1 Phenomena of Electromagnetic Induction

In order to understand the phenomena of electromagnetic induction, first we need to be clear about magnetic flux.

**Magnetic Flux:** Is a measure of the number of field lines that cross a given area.

Suppose that a magnetic field $\vec{B}$ crosses a surface area A as shown. Then, the magnetic flux, $\phi$ is given by




$$\phi = \vec{B}.\vec{A} = BA\cos\theta$$

Where $\theta$ is the angel between the magnetic field and an imaginary line that is normal (perpendicular) to the given area. The SI unit of magnetic is the Weber (Wb), named after the German physicist Wilhelm Weber. Hence,

$$1Wb = 1\ Tm^2$$

- **Important points to remember**
  - When the magnetic field is parallel to the surface, $\theta = 90°$, no field line cross the surface. Thus the flux is zero. $\phi = 0$
  - When the magnetic field is perpendicular to the surface, $\theta = 0°$, most of all lines cross the surface. In a case like this, the flux is maximum. $\phi = BA$.

Generated by CamScanner from intsig.com

## Conceptual Example

1. A loop of wire is placed in a uniform magnetic field. For what orientation of the loop is the magnetic flux
   a) Maximum
   b) Minimum (zero)

   ***Explanation:***
   a) The flux is maximum when the magnetic field line is perpendicular to the area or parallel to the normal, $\theta = 0°$.
   b) The flux is minimum or zero when the field line is parallel to the area or perpendicular to the normal, $\theta = 90°$.

## Illustrative Examples

2. A magnetic field of 0.4T is directed perpendicular to a circular loop of radius 20cm. Find the flux through the loop.

   ***Solution:*** $\phi = BA\cos\theta$, here, $\theta = 0°$ and $A = \pi r^2$

   $$\phi = BA = (0.4)(\pi \times (0.2)^2) = 0.05T.m^2 = 0.05Wb$$

3. A square loop 3m on a side is placed in a magnetic field of 0.8T and makes 60° with the normal. Calculate the flux

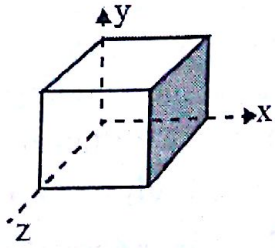
   ***Solution:*** $\phi = BA\cos\theta = (0.8)(3)^2\cos 60° = 3.6Tm^2 = 3.6Wb$

4. A solenoid of 8 cm diameter and 50cm long has 250 turns and carries a current of 10A. Calculate the flux through the solenoid.

   ***Solution:*** $B = \mu_o\left(\frac{N}{L}\right)I = (4\pi\times10^{-7})\left(\frac{250}{0.5}\right)(10) = 6.3\times10^{-3}T$

   $$\phi = BA = B\pi r^2 = (6.3 \times 10^{-3})(\pi)(0.04)^2 = 3.2 \times 10^{-5}Wb$$

5. A cube of length 2.5cm is placed in a uniform magnetic field of $\vec{B} = (5i + 4j + 3k)T$. Calculate the flux through the shaded face of the cube.

   ***Solution:***

   $\phi = B.A = (5i + 4j + 3k).(2.5 \times 10^{-2})^2 i$

   $\phi = 0.003125 = 3.1 \times 10^{-3}$ Wb

### Induced Electromotive Force

In 1819, Hans Christian Oersted discovered that an electric current produces a magnetic field. Because nature is often symmetric, the discovery that electric current produce magnetic field led Scientists to suspect that magnetic field could produce electric current. Indeed, experiments conducted by Michael Faraday in England and Joseph Henry in the United States in 1831 showed that a changing magnetic field could produce an induced current due to an induced emf.

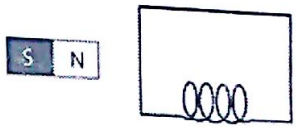
Generated by CamScanner from intsig.com

**Induced emf**: Is an emf produced when a magnetic field changes near a wire or a coil.

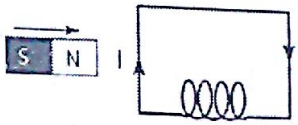
Thus, a changing magnetic field induces an emf in the coil, and the emf leads to an induced current. This means that, if a magnetic field changes near a wire, the electrons in the wire will feel a force which will tend them to move through the wire. This movement of electrons is due to an induced emf and if the wire is part of a circuit, the emf induces or brought about a current.

- **Important points to remember**

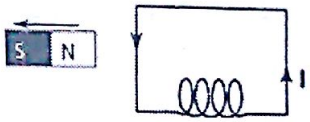
- When there is no relative motion between the magnet and the coil, there is no induced emf or current in the coil.




- When the magnet moves towards the coil there is a changing field and hence an induced emf and current.




- When the magnet moves away from the coil there is an induced emf and current but in the reversed direction.




- A current would also be created if the magnet were held stationery and the coil were moved. Because the magnetic field at the coil would be changing as the coil approached or receded from the magnet. Only relative motion between the magnet and the coil is needed to generate a current, it does not matter which one moves.

## 15.2 The Law of Electromagnetic Induction

Two scientists are given credit for the discovery of electromagnetic induction: the Englishman Michael Faraday and the American Joseph Henry. Although Henry was the first to observe electromagnetic induction, Faraday investigated it in more detail and published his findings first. Consequently, the law that describes the phenomenon bears his name.

**Faraday's Law of Electromagnetic Induction** states that "The magnitude of the induced emf in a circuit is directly proportional to the time rate of change of magnetic flux."

Mathematically,

Generated by CamScanner from intsig.com

$\boxed{\varepsilon = \frac{-\Delta\phi}{\Delta t}}$, where $\Delta\phi$ is change of flux

For a coil of wire that has N turns, the induced emf is given by

Faraday's law $\longrightarrow \boxed{\varepsilon = \frac{-N\Delta\phi}{\Delta t}} \equiv \boxed{\varepsilon = -N\frac{\phi_f - \phi_i}{t_f - t_i}}$

☛ The negative sign (−) indicates that the induced emf opposes the change in magnetic flux. The magnitude of the induced emf is obtained by

$$|\varepsilon| = N\left|\frac{\Delta\phi}{\Delta t}\right| = N\left|\frac{\phi_f - \phi_i}{t_f - t_i}\right| = N\left|\frac{B_f A\cos\theta - B_i A\cos\theta}{t_f - t_i}\right|$$

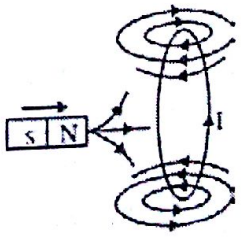
The direction of the induced emf or current is determined by the Estonian physicist Heinrich Lenz and is indicated by the minus (−) sign in Faraday's law.

**Lenz's Law** states that: "The direction of the induced emf current is such as to oppose the change creating it."

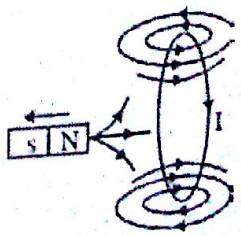
Lenz's law follows from the law of conservation of energy. If the flux through a coil is to the left and increasing, the induced emf will try to put flux through the coil to the right. If the flux is to the left and decreasing, the induced emf will try to put flux through the coil to the left. The induced emf always tries to counter balance the change in flux.

As an example, consider a bar magnet that is moved into and away from a conducting ring.

a) If the north pole of the magnet approaches the ring, a current is induced that tends to opposite the motion of the magnet. The current in the ring creates a magnetic field that has a north pole, facing the north pole of the magnet. There is now a repulsive force acting on the magnet, opposing its motion.




b) If the magnet is moved away from the ring, the induced current creates a south pole facing the north pole of the magnet. The resulting attractive force again opposes the magnet's motion.



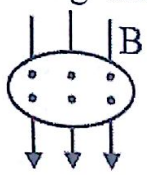

Generated by CamScanner from intsig.com

## Important Points to remember

- An emf is induced in a coil when there is a change of magnetic flux. And, the magnetic flux can change when
  - There is relative motion between a magnet and a coil.
  - There is changing current in a nearby coil.
  - There is changing orientation between the magnetic field and the coil.
- The magnitude of an induced emf depends on:
  - The strength of magnetic field; stronger magnetic field means larger emf.
  - The speed at which the coil move through the magnetic field; greater speed means greater emf.
  - The number of turns of the coil; the more the number of turns, the greater the emf. Ok?

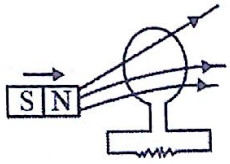
## Conceptual Examples

6. A magnetic field through a horizontal coil is increasing and downward. What will be the direction of the induced current in the coil.



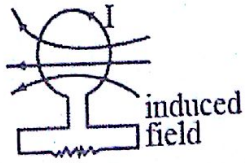

*Explanation*: This time, the flux is increasing. Thus, the direction of the current according to lenz's law is to oppose the increasing flux. Since the applied field is downward and the flux is increasing, the induced current flows in the direction that generates a magnetic field upwards. By the right hand rule, the current is counter clockwise.

7. A magnet is approaching a loop of wire as shown. What is the direction of the induced current?



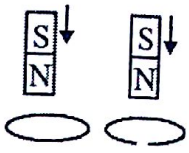

*Explanation*: Since the magnetic field at the loop is increasing as the magnet approaches, to oppose the increase in flux, the direction of the induced magnetic field must be opposite to the field of the magnet.

Since the field of the magnet passes through the loop from left to right, the induced field must pass through the loop from right to left. To create such a field, the induced current must be directed counter – clockwise as shown here.




Generated by CamScanner from intsig.com

8. Two magnets are dropped from rest through conducting rings as shown. Notice that the ring on the right has a small break in it, whereas the ring on the left is a closed loop. Which magnet will move at greater acceleration when dropped.




***Explanation***: As the magnet on the left approaches the ring it induces a circulating current. According to Lenz's law this current produces a magnetic field that exerts a repulsive force on the magnet to oppose its motion. In contrast, the ring on the right has a break, so it cannot have a circulating current. As a result, it exerts no force on its magnet. Therefore, the magnet on the right falls with the acceleration of gravity, the magnet on the left falls with a smaller acceleration. Thus, the magnet on the right has greater acceleration. Ok!

## Illustrative Examples

9. A coil of wire has 20 turns each of area $1.5 \times 10^{-3}m^2$. A magnetic field perpendicular to the coil change, from 0.05T to 0.06T in 0.1 sec.
   a) Find the average induced emf during this time.
   b) What would be the average induced emf if the field decreases from 0.06T to 0.05T in 0.1 sec?

***Solution***: a. $\varepsilon = -N\dfrac{(\phi_f - \phi_i)}{t_f - t_i} = -N\dfrac{(B_f A\cos\theta - B_i A\cos\theta)}{t_f - t_i}$

$$\Rightarrow \varepsilon = -NA\cos\theta\frac{(B_f - B_i)}{t_f - t_i} = (-20)\,(1.5\times10^{-3})\cos0°$$

$$\left(\frac{0.06-0.05}{0.1-0}\right)$$

$$\varepsilon = -3 \times 10^{-3}V = -3mV$$

b. $\varepsilon = -NA\cos\theta\dfrac{(B_f - B_i)}{t_f - t_i} = -20(1.5 \times 10^{-3})\cos 0°$

$$\left(\frac{0.05-0.06}{0.1-0}\right)$$

$$\varepsilon = 3 \times 10^{-3}V = +3mV$$

10. A coil of wire of area $0.02m^2$ has 50 turns. At t = 0 sec, the coil is oriented such that the normal to its surface is parallel to a constant magnetic field of 0.18T. The coil is then rotated through an angle of 30° in a time of 0.1 sec.

Generated by CamScanner from intsig.com

a) Determine the average induced emf.
b) What would be the induced emf if the coil were returned to its initial orientation in the same time of 0.1 sec?

*Solution*: a. $\varepsilon = -N\dfrac{(\phi_f - \phi_i)}{t_f - t_i} = -N\dfrac{(B_f A\cos\theta_i - B_i A\cos\theta\, i)}{t_f - t_i}$

$$\varepsilon = -NBA\left(\frac{\cos\theta_f - \cos\theta_i}{t_f - t_i}\right)$$

$$= -(50)(0.18)(0.02)\left(\frac{\cos 30° - \cos 0°}{0.1 - 0}\right)$$

$$\varepsilon = +0.24V$$

b. $$\varepsilon = -NBA\left(\frac{\cos\theta_f - \cos\theta_i}{t_f - t_i}\right)$$

$$= -(50)(0.18)(0.02)\left(\frac{\cos 0° - \cos 30°}{0.1 - 0}\right)$$

$$\varepsilon = -0.24V$$

11. A 0.25T magnetic field is perpendicular to a circular loop of wire with 50 turns and a radius of 15 cm. If the magnetic field is reduced to 0.01T in 0.12 sec, what is the magnitude of the induced emf?

*Solution*: $|\varepsilon| = \left|-N\dfrac{\Delta\phi}{\Delta t}\right|$, $A = \pi r^2 = \pi(0.15)^2 = 0.07\ m^2$

$\phi_i = B_i A\cos\theta = (0.25)(0.07)\cos 0° = 0.018Wb$
$\phi_f = B_f A\cos\theta = (0.01)(0.07)\cos 0° = 0.007Wb$

$$|\varepsilon| = N\left|\frac{\phi_f - \phi_i}{\Delta t}\right| = 50\frac{|0.007 - 0.018|}{0.12} = 4.6V$$

12. The induced emf in a single loop of wire has a magnitude of 1.48V when the magnetic flux is changed from $0.85T.m^2$ to $0.11\ T.m^2$. How much time is required for this change in flux?

*Solution*: $|\varepsilon| = N\left|\dfrac{\phi_f - \phi_i}{\Delta t}\right| = \Delta t = N\left|\dfrac{\phi_f - \phi_i}{\varepsilon}\right|$

$$\Delta t = \left|\frac{0.11 - 0.85}{1.48}\right| = 0.5\,sec$$

Generated by CamScanner from intsig.com

13. A coil formed by wrapping 50 turns of wire in the shape of a square is positioned in a magnetic field so that the normal to the plane of the coil makes an angle of 37° with the direction of the field. When the magnetic field is increased uniformly from 200mT to 600mT in 4 sec, an emf of magnitude 160mV is induced in the coil. What is the total length of the wire in the coil? (UEE: 2006)

A. 100m B. 200m C. 300m D. 400m

***Solution***: $|\varepsilon| = N\left|\frac{\phi_f - \phi_i}{t_f - t_i}\right| = N\left|\frac{B_f A - B_i A}{t_f - t_i}\right|$

$$|\varepsilon| = NA\left|\frac{600\times10^{-3} - 200\times10^{-3}}{4}\right|\cos 37°$$

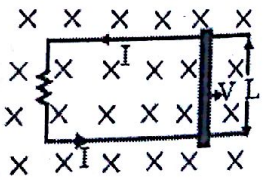
$$160 \times 10^{-3} = 50A\left(\frac{400\times10^{-3}}{4}\right) = 3993.18\times10^{-3}A$$

$$\Rightarrow A = \frac{160}{3993.18} = 0.04 = S^2 \Rightarrow S = 0.2m$$

$$\ell = N(4 \times S) = 50(4 \times 0.2) = 40m$$

## 15.3 Motional Electromotive Force

In a coil, an emf is induced when the magnetic field changes with time. An emf is also induced on a rod that moves in a magnetic field, which is usually called *motional emf.*




Consider a metal rod of length L moving to the right with a velocity $\vec{v}$ perpendicular to a uniform magnetic field $\vec{B}$. Each charge q in the rod experiences a magnetic force, F = qVB, and an electric force, F = Eq, and when the two forces balance, we have

$$Eq = qvB$$

But, we know that, $E = \frac{v}{d} = \frac{\varepsilon}{L}$

$$\Rightarrow \frac{\varepsilon}{L}q = qvB$$

$$\boxed{\varepsilon = VBL} \leftarrow \text{motional emf}$$

or, $|\varepsilon| = N\left|\frac{\Delta\phi}{\Delta t}\right| = \frac{B\Delta A}{\Delta t} = \frac{BL\Delta x}{\Delta t} = BLV, N = 1$ for a rod

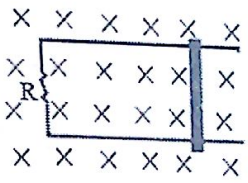
Generated by CamScanner from intsig.com

**Keep in mind that;**

- The separated charges on the ends of the moving conductor give rise to an induced emf, called a motional emf because it originates from the motion of charges through a magnetic field. The emf exists as long as the rod moves. If the rod is brought to a halt, there is no induced emf.
- Using the right hand rule, for a rod that moves to the right in a magnetic field directed into the page, the induced current is counter clock wise as shown above.

## Conceptual Example

14. What is the direction of the induced current through the resistor as the conducting rod moves, in the magnetic field?

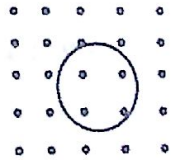



a) To the right
b) To the left

*Explanation*:

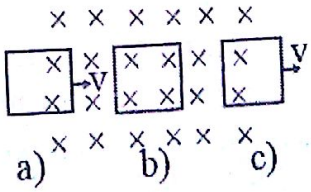
a) According to Lenz's law, the induced current must be counterclockwise so as to produce a counter acting magnetic field directed out of the page.
b) With the same reasoning, clockwise.

15. Consider a system in which a metal ring is falling out of a magnetic field pointing out of the page. What can you say about the direction of the induced current in the ring?



*Explanation*: As the ring falls out of the field, the flux decrease, thus, according to Lenz's law, the induced current opposes the decrease in flux by generating more field lines out of the page. To do this the current must be counterclockwise.

16. A rectangular loop moves into a uniform magnetic field $\vec{B}$ with a velocity $\vec{v}$. What is the direction of the induced current on the loop?




A. As it enters the magnetic field.
B. As it moves within the field.
C. As it leaves the field.

*Explanation*:

A. As it moves into the field, the flux is increasing, and the induced current opposes this hence it generates a magnetic field directed out of the page. To do this, the current must be counter clockwise.

Generated by CamScanner from intsig.com

B. In this case, the flux is not changing and there is no induced current.

C. Now, the flux is decreasing. To oppose this, the induced current generates a magnetic field directed into the page. To accomplish this, the current must be clock wise.

## Illustrative Examples

17. A metal rod of 0.1m moves at a speed of 2.5 m/s through a magnetic field of 0.6T. If the total resistance of the circuit is 0.03Ω. Find

a) The induced emf.
b) The induced current.
c) The force acting on the rod.

***Solution***: a. $\varepsilon = vBL = (2.5)(0.6)(0.1) = 0.15V$

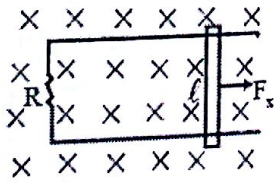
b. $I = \frac{V}{R} = \frac{0.15}{0.03} = 5A$

c. $F = ILB = (5)(0.1)(0.6) = 0.3N$

18. A 1.2m conducting bar moves to the right at a speed of 2m/s. If the resistance of the loop is 6Ω. Calculate

a) The applied force required to move the bar.
b) At what speed should the bar be moved to produced current of 0.5A in the circuit.

***Solution***: a. $\varepsilon = B\ell v = (2.5)(1.2)(2) = 6V$



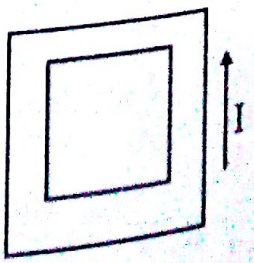

and, $I = \frac{\varepsilon}{R} = \frac{6}{6} = 1A$

$\Rightarrow F = BI\ell = (2.5)\ (1)\ (1.2) = 3N$

b. $\epsilon = B\ell V$, but $\varepsilon = IR$

$\Rightarrow IR = B\ell V \Rightarrow V = \frac{IR}{B\ell} = \frac{(0.5)(6)}{(2.5)(1.2)} = 1m/s$

9. Two rectangular loops of wire lie in the plane of this paper. If the current I in the outer loop is counterclockwise and increases with time, which one of the following statements is correct about the current induced in the inner loop? (UEE 2006)



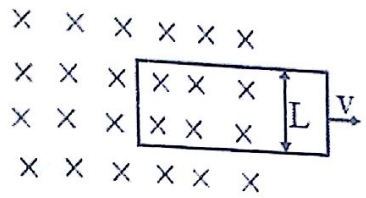

A. Induced in the clock wise direction.
B. Induced in the counterclockwise direction.
C. The induced current in the inner loop creates the same magnetic field as the outer loop.
D. It is zero.

Generated by CamScanner from intsig.com

*Explanation*: The current in the outer loop creates an increasing flux. The magnetic field due to the outer loop is out of the page. According to Lenz's law the induced current in the inner loop must produce a field into the page to oppose the increasing flux. To do so, the current in the inner loop must be clockwise.

**Answer: A**

20. A rectangular loop of wire of width L and resistance R. One end of the loop is in a uniform magnetic field of strength B directed perpendicular into the plane of the page. The loop is pulled to the right at a constant speed v. What are the magnitude and direction of the induced current in the loop? (UEE 2007)




A. I = BLVR and clockwise.

B. $I = BL\,V/R$ and clockwise.

C. I = BLVR and counterclockwise.

D. $I = BL\,V/R$ and counterclockwise.

*Explanation*: We know, $\varepsilon = BLV$ and, $I = \frac{V}{R} = \frac{\varepsilon}{R} = \frac{BLV}{R}$

Since the flux is decreasing as the loop moves out of the field, to oppose this, the induced current generates a magnetic field directed into the page. To do this, the current must be clockwise.

**Answer: B**

## 15.4 Mutual Inductance and Self Inductance

We know that an emf can be induced in a coil by keeping the coil stationary and moving a magnet nearby, or by moving the coil near a stationary magnet.

Another important method of inducing an emf is using two coils or a single coil.

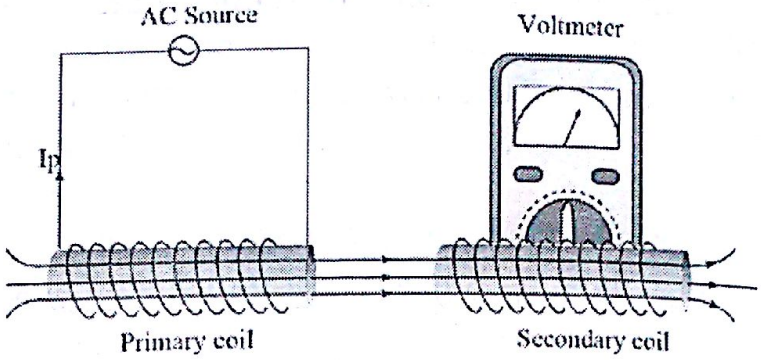
### Mutual Inductance

Here, two coils of wire are placed close to each other, the primary coil and the secondary coil.

**Mutual Induction:** Is the effect in which a changing current in one circuit induces an emf in another circuit

Generated by CamScanner from intsig.com

The primary coil is the one connected to an AC generator, which send an alternating current $I_P$ through it. The secondary coil is not attached to a generator, although a voltmeter is connected across it to register any induced emf.




The current – carrying primary coil is an electromagnet and creates a magnetic field in the surrounding region. If the two coils are close to each other, a significant fraction of this magnetic field penetrates the secondary coil and produce a magnetic flux. The flux is changing since the current in the primary coil and its associated magnetic field are changing. Because of the change in flux, an emf is induced in the secondary coil.

According to Faraday's law of electromagnetic induction, the average emf induced in the secondary coil is proportional to the change in flux through it, which in turn is proportional to the magnetic field produced by the changing current in the primary coil.
Thus, we can write

$$N_s\phi_s = MI_P \Rightarrow M = \frac{N_s\phi_s}{I_P}$$

Substituting this question in to Faraday's law;

$$\varepsilon_s = -N_s\frac{\Delta\phi_s}{\Delta t} = -\Delta\frac{(N_s\,\phi_s)}{\Delta t}, \text{ but } N_s\phi_s = MI_P$$

$$\varepsilon_s = \frac{-\Delta(MI_P)}{\Delta t} = -M\frac{\Delta I_P}{\Delta t}$$

$$\boxed{\varepsilon_s = -M\frac{\Delta I_P}{\Delta t}} \longleftarrow \text{emf induced in the secondary coil due to a changing current in the primary coil.}$$

Where; M is proportionality constant and is called *mutual inductance.*

- The unit of mutual inductance is $\frac{V.s}{A}$ and is called *henry* (H) in honor of the American physicist Joseph Henry.

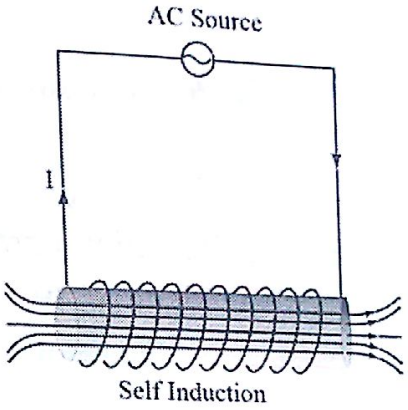
Generated by CamScanner from intsig.com

## Self-Inductance

In this case, a coil induces an emf on itself.

**Self-Induction**: Is the effect in which a changing current in a circuit induces an emf in the very same circuit, that is on itself

For instance, consider a coil connected to an AC generator. The alternating current create an alternating magnetic field that, in turn, creates a changing flux through the coil, and the change in flux induces an emf in the coil. Since flux is proportional to the magnetic field, and the magnetic field is proportional to the current, it follows that;



Self Induction

$$N\phi = LI \Rightarrow L = \frac{N\phi}{I}$$

From Faraday's law, the induced emf is

$$\varepsilon = -N\frac{\Delta\phi}{\Delta t} = \frac{-\Delta(N\phi)}{\Delta t}, \text{ but } N\phi = LI$$

$$\varepsilon = -\Delta\frac{(L\Delta I)}{\Delta t} = -L\frac{\Delta I}{\Delta t}$$

$$\boxed{\varepsilon = -L\frac{\Delta I}{\Delta t}} \leftarrow \text{emf induced in a coil due to a change in its current}$$

Where; L is a proportionality constant and is called *self-inductance* or simply inductance of the coil.

- The unit of self-inductance is henry (H)

**Important points of remember**

Factors which determine inductance are primarly;

- Magnetic flux
- Number of turns on a coil.
- Current through the coil, that is decreasing current results in an increasing inductance.

### Conceptual Example

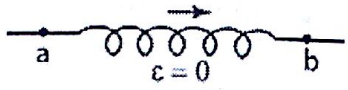
21. Explain the nature of the potential difference in the following cases.
    a) Inductor with constant current flowing from a to b.

Generated by CamScanner from intsig.com

b) Inductor with increasing current flowing from a to b.
c) Inductor with decreasing current flowing from a to b
d) Resistor with constant current flowing from a to b.

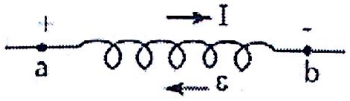
*Explanation*: a. There is no potential difference

$$V_{ab} = L\frac{\Delta I}{\Delta t} = 0$$



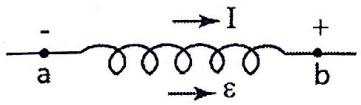

b. Potential drops from a to b

$$V_{ab} = L\frac{\Delta I}{\Delta t} < 0$$



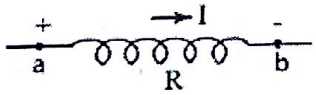

c. Potential increases form a to b

$$V_{ab} = L\frac{\Delta I}{\Delta t} > 0$$




d. Potential drops from a to b

$$V_{ab} = IR > 0$$




**Remember:** The potential difference across a resistor depends on the current and the potential difference across an inductor depends on the rate of change of the current.

## Illustrative Examples

22. Which of the following statement is NOT true? (UEE 2003)
A. An electric current can be induced in a circuit with a change in the cross – sectional area of the wire.
B. The induced emf in a circuit is directly proportional to the time rate of change of magnetic flux.
C. Emf can be induced in a circuit by change in the area enclosed by the loop.
D. An electric current can be induced in a circuit with a changing of magnetic field.

**Answer: A**

23. Which device functions with the principle of mutual induction? (UEE 2002)
A. Electromagnetic
B. Transformer
C. Electric motor
D. Transistor

*Explanation*: In a transformer, the change in flux on one coil induced emf on the other.

**Answer: B**

Generated by CamScanner from intsig.com

24. Two coils, held in fixed positions have mutual inductance of 100μH. What is the peak voltage in one when the sinusoidal current given by I(t) = 10A sin (1000t) flows in the other? (UEE 2004)

A. 1μV B. 100V C. 10V D. 1V

*Solution*: $\varepsilon = L\frac{\Delta I}{\Delta t} = L\frac{dI}{dt} = 100(10^{-6})\frac{d}{dt}(10 \sin 1000t)$

$\varepsilon = \cos(1000t)$, $\varepsilon$ = max when $t = 0$, $\cos 0° = 1$

$\varepsilon = 1V$

**Answer: D**

25. An emf of 8V is induced in a coil when the current in it changes at the rate of 32 A/s. Compute the inductance of the coil.

*Solution*: $|\varepsilon| = L\left|\frac{\Delta I}{\Delta t}\right| \Rightarrow 8 = L(32) \Rightarrow L = 0.25H$

26. The current in an inductor of 40μH increases from zero to 6A in 0.3 second. What is the magnitude and direction of the induced emf?

*Solution*: $|\varepsilon| = L\left|\frac{\Delta I}{\Delta t}\right| = 40\times10^{-6}\left|\frac{6}{0.3}\right| = 8\times10^{-4}\,V$

Since the current is increasing according to Lenz's law, the direction of the emf is opposite to the current.

27. Find the self-inductance in a coil where the magnetic field is 30mT, the area of the coil is 5 $cm^2$, there are 50 turns on the coil and the current through the coil is 1.5A?

*Solution*: $|L| = \left|-\frac{\phi N}{I}\right| = \left|-\frac{BAN}{I}\right| = \left|-\frac{30\times10^{-3}\times5\times10^{-4}\times50}{1.5}\right|$

$L = 5\times10^{-4}H$

28. Two coils have mutual inductance of $3.25 \times 10^{-4}$H. The current in the first coil increases at a rate of 830 A/s. What is the magnitude of the induced emf in the second coil?

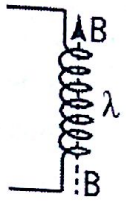
*Solution*: $\varepsilon_2 = \frac{M\Delta I_1}{\Delta t} = (3.25 \times 10^{-4})\,(830) \simeq 0.3V$

**Inductance of a Solenoid**

Consider a coil of N turns and length $\ell$ in a circuit. For a fixed area A and changing current I. Faraday's law gives,

Generated by CamScanner from intsig.com

$$\varepsilon = -N\frac{\Delta\phi}{\Delta t} = -NA\frac{\Delta B}{\Delta t} \text{ and } B = \mu_o \frac{N}{\ell} I$$




Also, $\varepsilon = -L\frac{\Delta I}{\Delta t}$ equating the two expressions for ε

We get, $L = \frac{\mu_o N^2 A}{\ell}$

For a coil with n turns per unit length this expression simplifies to

$$\boxed{L = \mu_o n^2 A}$$

**Illustrative Example**

29. Find the inductance of an air cored solenoid of 250 turns per unit length and area $2.5cm^2$.

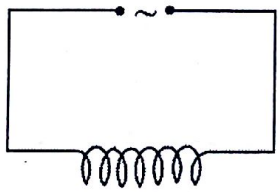
***Solution***: $L = \mu_o n^2 A = (4\pi \times 10^{-7})(250)^2(2.5 \times 10^{-4})$

$L \simeq 2.0 \times 10^{-5} H$

**Energy Stored in an Inductor**

Consider an indicator in a circuit as shown below. When a current flows through the inductor, the induced voltage as we all know is

$$\varepsilon = -L\frac{\Delta I}{\Delta t}$$



When the current flows through the indicator, there is potential energy stored by the magnetic field, $PE_B$.

The power that must be supplied to the inductor to drive current at any time is given by

$$P = IV = I\varepsilon = -LI\frac{\Delta I}{\Delta t}$$

The energy stored in the inductor is then obtained by integrating the power.

$$\text{Since, Power} = \frac{\text{work}}{\text{time}} = \frac{\text{energy transformed}}{\text{time taken}}$$

$\Rightarrow$ energy = (power)(time)

In our case,

$$P = -LI\frac{\Delta I}{\Delta t} = \frac{\Delta P_{E_B}}{\Delta t} = \frac{dP_{P_{E_B}}}{dt}$$

Generated by CamScanner from intsig.com

$$\Rightarrow P_{E_B} = P\Delta t = Pdt = \int LIdI$$

$$\boxed{P_{E_B} = \frac{1}{2}LI^2} \leftarrow \text{energy stored in inductor}$$

## Illustrative Examples

30. Find the energy stored in an inductor of 250 turns and area of $2.5cm^2$ when a current of 2.5A flows thought it.

*Solution*: $L = \mu_o n^2 A = (4\pi \times 10^{-7})(250)^2(2.5 \times 10^{-4})$

$$L = 2 \times 10^{-5}H$$

$$P_{E_B} = \frac{1}{2}LI^2 = \frac{1}{2} \times 2 \times 10^{-5} \times (2.5)^2 \quad P_{E_B} = 6.25 \times 10^{-5}J$$

**Magnetic Energy Density**

It is the energy per unit volume of a magnetic field

$$U_B = \frac{\text{energy}}{\text{volume}} = \frac{P_{E_B}}{V}$$

Now, consider an inductor of length $\ell$ and cross section area A.

We know, energy stored in an inductor is

$$P_{E_B} = \frac{1}{2}LI^2$$

We also know that $L = \frac{\mu_o N^2 A}{\ell}$ and $B = \frac{\mu_o NI}{\ell}$ up on substituting these values into $P_{E_B} = \frac{1}{2}LI^2$ and simplification, we get

$$P_{E_B} = \frac{AB^2\ell}{2\mu_o}$$

The volume is $A\ell$, hence the energy density

$$U_B = \frac{\text{energy}}{\text{volume}} = \frac{P_{E_B}}{V} = \frac{\frac{AB^2\ell}{2\mu_o}}{A\ell}$$

This simplifies to

$$\boxed{U_B = \frac{B^2}{2\mu_o}} \leftarrow \text{magnetic energy density}$$

Generated by CamScanner from intsig.com

**Illustrative Example**

31. Find the energy density for an inductor with a magnetic field of 0.5T. The permeability of free space is, $\mu_o = 4\pi \times 10^{-7} Tm^2/A$.

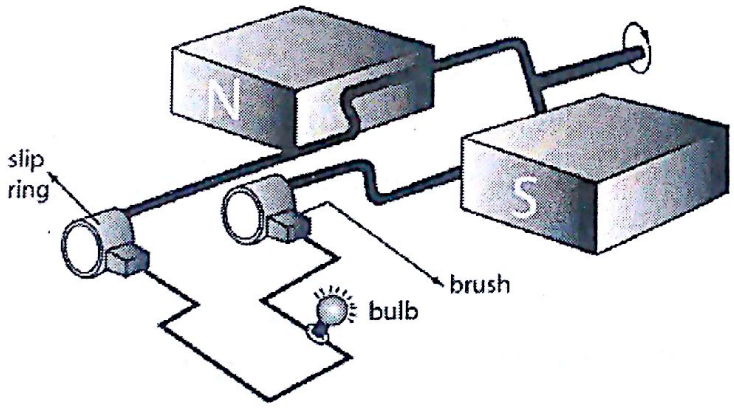
***Solution***: $U_B = \frac{B^2}{2\mu_o} = \frac{(0.5)^2}{2 \times 4\pi \times 10^{-7}} = 2.5 \times 10^8 J/m^3$

## 15.5 Electric Generator

An electric generator is a device that converts mechanical energy to electrical energy.

The mechanical energy can be supplied by any of a number of sources, such as falling water in a hydroelectric dam, steam in a coal fired power plant, or gasoline.

Imagine, that a coil of area A is located in a magnetic field between the poles of a magnet. The coil then experiences a torque which make it rotate with angular speed ω. As it rotates the flux is changing over time and hence an emf is induced. According to Faraday the emf induced is given by




$$\varepsilon = -N\frac{\Delta\phi}{\Delta t} = -N\frac{\Delta BA\cos\theta}{\Delta t}$$

For a small change of time, the above relation is differentiated.

$$\varepsilon = -NBA\frac{d}{dt}\cos\theta = -NBA(-\sin\theta)\frac{d\theta}{dt}$$

$$\varepsilon = NBA\sin\theta\frac{d\theta}{dt}, \text{ but } \theta = \omega t = 2\pi ft$$

$$\varepsilon = NBA\sin 2\pi f\, t(2\pi f), \text{ but } 2\pi f = \omega$$

$$\boxed{\varepsilon = NBA\,\omega\sin\omega t} \leftarrow \text{emf in an AC generator}$$

**Illustrative Example**

32. The coil in an electric generator has 100 turns and an area of $2.5 \times 10^{-3} m^2$. It is described that the maximum emf of the coil be

Generated by CamScanner from intsig.com

120V when it rotates at the rate of 60 cycles per second. Find the strength of the magnetic field required for this generator.

*Solution*: $\varepsilon = NBA\ 2\pi f \sin 2\pi ft$

$\varepsilon_{max} = NBA^2 2\pi f$, f = 60 cycles/sec = 60Hz

$$B = \frac{\varepsilon}{NA2\pi f} = \frac{120}{(100)(2.5\times10^{-3})(2\pi)(60)}$$

$B = 1.3T$

33. A rectangular coil 25cm by 35cm has 120 turns and is rotating with an angular sped of 190 rad/s in a magnetic field of 33mT. Find the maximum emf produced in this coil which is part of a generator.

*Solution*: $\varepsilon = NBA\ \omega \sin \omega t$

$\varepsilon_{max} = NBA\ \omega = (120)(33 \times 10^{-3})(0.25 \times 0.35)(190)$

$\varepsilon_{max} = 65V$

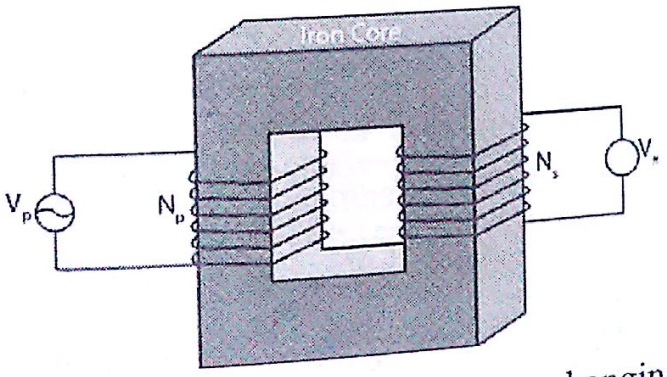
## 15.6 Transformer

One of the most important applications of mutual induction and self-induction takes places in a transformer.

**Transformer:** Is a device used to change voltage. That is, it is designed to increase or decrease an AC voltage.

For instance, to energize cell phone batteries, a transformer plays a role in reducing the 220V to a much smaller value, typically between 3 and 9V. In another example, a picture tube in a television needs about 15000V to accelerate the electron beam, and a transformer is used to obtain this high voltage from the 220V at a wall socket.

A transformer consists of an iron core on which two coils are wound: a primary coil with $N_P$ turns and a secondary coil with $N_S$ turns. The primary coil is the one connected to the AC generator.




The alternating current in the primary coil establishes a changing magnetic field in the iron core. Because iron is easily magnetized, it greatly enhances the magnetic field and guides the field lines to the secondary coil. Since the magnetic field is changing, the flux through

Generated by CamScanner from intsig.com

the primary and secondary coils is also change, and the consequently, an emf is induced in both coils.

In the primary coil, the induced emf $\varepsilon_P$ is due to self-induction and is specified by Faraday's law as:

$$\varepsilon_P = -N_P \frac{\Delta\phi}{\Delta t} \quad \ldots\ldots\ldots\

**Important points to remembers**

- If $N_S$ is greater than $N_P$, the secondary (output) voltage, $V_s$ is greater than the primary (input) voltage, $V_P$ and the transformer is called *step up transformer*.

  $\boxed{N_S > N_P \Rightarrow V_S > V_P}$ ← Step up transformer
- If $N_s$ is less than $N_P$, the secondary voltage is less than the primary voltage and the transformer is called *step down transformer*.

  $\boxed{N_S < N_P \Rightarrow V_S < V_P}$ ← Step down transformer
- A transformer that steps up the voltage simultaneously steps down the current, and a transformer that steps down the voltage steps up the current.
- A transformer neither step up nor step down power since energy is conserved.

## Conceptual Examples

34. Explain the difference between direct current, DC and alternating current, AC.

    ***Explanation***: Direct current, DC has a constant value. It is the type of current that is obtained from cells or batteries. And if you look at it on an oscilloscope, it would take the form of a straight horizontal line. In contrast, alternating current, AC is constantly varying. It is the type of current that is transmitted from power stations and if you look at it oscilloscope, it would take a sinusoidal wave form.

35. A transformer operates with an AC source not with DC. Why?

    ***Explanation***: It is because, if a DC source is used in the primary coil, it produces a flux that does not change in time, and thus no emf is induced in the secondary coil.

36. Whenever electricity, which is in the form of alternating current is transmitted from power stations through cables, to domestic use such as factories and homes, there is power lose in the transmission lines due to heating. How could such power loses be minimized?

    ***Explanation***: One possible solution is to use transformers that step up the voltage to high level while reducing the current. A smaller current means less power loss, since $P = I^2R$.

Generated by CamScanner from intsig.com

## Illustrative Examples

37. A transformer converts the mains supply from 220V down to 2V. There are 160 turns on the secondary coil. How many turns does the primary coil have?

***Solution:*** $\frac{V_S}{V_P} = \frac{N_S}{N_P} \Rightarrow N_P = \frac{V_P}{V_S} N_S = \left(\frac{220}{2}\right)(160) = 17600$ turns

38. An ideal transformer steps up 8V to 2000V. If the secondary coil carries 2A. Find the current in the primary coil.

***Solution:*** $\frac{I_P}{I_S} = \frac{V_S}{V_P} \Rightarrow I_P = \left(\frac{V_S}{V_P}\right)(I_S) = \left(\frac{2000}{8}\right)(2) = 500A$

39. A transformer increase a 300V to 600V. The current in the primary coil is 6A, while the current in the secondary coil is 2A. Calculate the efficiency of the transformer.

***Solution:*** $\text{efficiency} = \frac{\text{output power}}{\text{input power}}$

$$\eta = \frac{I_S V_S}{I_P V_P} = \frac{(600)(2)}{(300)(6)} = 0.67 \Rightarrow \eta = 67\%$$

40. A step up transformer is used to (UEE 2005)

A. Increase the power
B. Decrease the power
C. Increase the voltage
D. Decrease the voltage

**Answer: C**

41. A 6V battery is connected across the primary coil of a transformer having 50 turns. If the secondary coil of the transformer ha s100 turns, what voltage appears across the secondary? (UEE 2006)

A. 24V B. 12V C. 6V D. 3V

***Solution:*** $\frac{V_S}{V_P} = \frac{N_S}{N_P} \Rightarrow V_S = \left(\frac{N_S}{N_P}\right)(V_P) = \left(\frac{100}{50}\right)(60) = 12V$

**Answer: B**

## 15.7 Alternating Current (a.c)

Thomas Edison favored direct current, dc that does not vary with time. George Westinghouse favored alternating current, ac that varies sinusoidally. Present day household and industrial power systems operate with alternating current. Any appliance that you plug into a wall socket uses ac, and many battery powered devices such as radios make use of the dc supplied by the battery to amplify alternating current.

Generated by CamScanner from intsig.com

Circuits in modern communication equipment including television, also make extensive use of ac.

**Root mean square (rms) values in ac circuits**

In an AC circuit, both the voltage and the current are varying continuously, thus their average values are zero. A more useful type of average or mean is the root mean square, or rms for short.

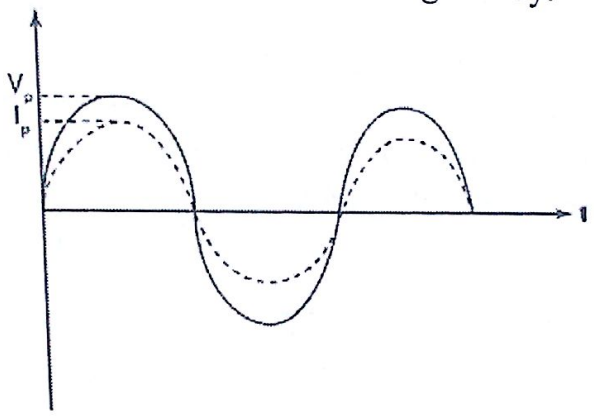
**Root mean square (rms)** value: Is a value for the current or voltage that would be equivalent to the effective steady value.

In an AC source the current and the voltage at any time are given by:

$I = I_P \sin \omega t$

$V = V_P \sin \omega t$

Where: $I_P$ and $V_P$ are the peak (maximum) values.




For example, we can see that, $I^2 = I_p^2 \sin^2 \omega t$ varies between 0 and $I_p^2$, hence, the mean value of $I^2$ is half of $I_p^2$.

$$I_{av}^2 = \frac{1}{2} I_P^2$$

$$I_{rms} = \frac{1}{\sqrt{2}} I_P$$

$$\Rightarrow \boxed{I_{rms} = 0.707\ I_P}$$

In the same manner, the root mean square value of the voltage is,

$$V_{rms} = \frac{1}{\sqrt{2}} V_P$$

$$\boxed{V_{rms} = 0.707\ V_P}$$

The half cycle average current is given by the relation:

$$\boxed{I_{avg} = \frac{2}{\pi} I_P = 0.637 I_P}$$

Where, the constant $\frac{2}{\pi}$ is the average value of cos ωt or sin ωt.

Generated by CamScanner from intsig.com

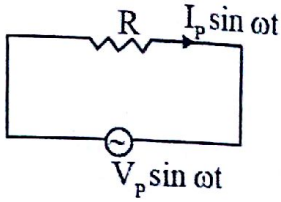
## Illustrative Example

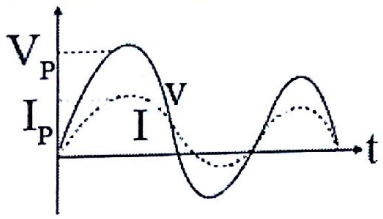
42. A given circuit operates with an rms voltage of 120V. What is the maximum or peak voltage in this circuit?

*Solution*: $V_{max} = \sqrt{2}V_{rms} = \left(\sqrt{2}\right)(120) = 170\,V$

### Resistive Circuits and Alternating Current

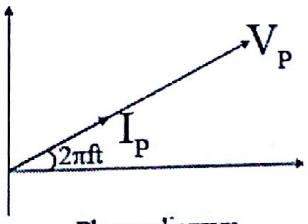
Consider a resistor in an AC circuit. The voltage and current versus time diagram, that is the Phasor diagram shows that, the voltage and the current are in phase. That is, when the voltage reaches its maximum value, at that very same moment, the current also reaches its maximum value. Also, when the voltage decreases, the current also decreases. This can be demonstrated as follows:




$V = V_p \sin \omega t$

$I = I_p \sin \omega t$




Phasor diagram

## Illustrative Example

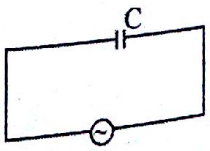
43. An output voltage of v = 120 sin ωt is applied across a 40Ω resistor. Calculate the peak current in the circuit.

*Solution*: $I_P = \frac{V_P}{R} = \frac{120}{40} = 3A$

### Capacitive Circuits and Alternating Current

Consider a capacitor in an AC Circuit. The generator supplies an rms voltage, $V_{rms}$ given by

$$\boxed{V_{rms} = I_{rms} X_C}$$




Where, $X_C$ is called *capacitive reactance,* which is the opposition of the capacitor to the flow of current through it. $X_C$ is found experimentally to be inversely proportional to both the frequency and the capacitance. That is;

Generated by CamScanner from intsig.com

$$X_C = \frac{1}{2\pi fC} = \frac{1}{\omega C}$$

## Illustrative Example

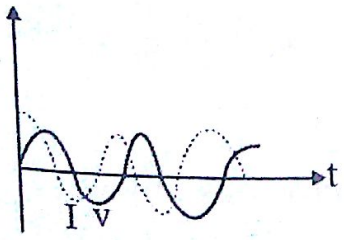
4. Find the capacitive reactance of a 22μF capacitor in a 60Hz circuit.

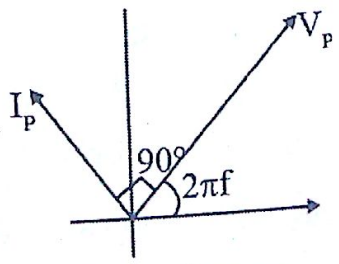
*Solution*: $X_C = \frac{1}{\omega C} = \frac{1}{2\pi fC} = \frac{1}{2\pi(60)(22\times10^{-6})} = 120\Omega$

**Remember**: In a capacitive circuit, the current has a maximum value at the start. It is a maximum because, there is no charge on the capacitor at the start and hence no capacitor voltage to oppose the generator voltage. When the capacitor is fully charged, the capacitor voltage completely opposes the generator voltage. The result is that, the current decreases to zero. And, when the capacitor voltage decreases, the charges flow out of the capacitor in a direction opposite to that of the charging current and thus results in a negative current. Thus, voltage and current are 90° or $\frac{\pi}{2}$ radian out of phase. Hence, there is a phase difference of $\frac{\pi}{2}$ between the current and the voltage.

More specifically, if the voltage varies according to $V = V_P \sin 2\pi ft$, the current varies as $I = I_P \sin\left(2\pi ft + \frac{\pi}{2}\right) = I_P \cos 2\pi$ ft. Since the current reaches its maximum value before the voltage does, it is said that;

- In a capacitive circuit, the current leads the voltage by a phase angle of $\frac{\pi}{2}$ or 90°. This can easily be visualized graphically as follows:




Phasor diagram

The fact that the current and voltage for a capacitor are 90° out of phase, power being the product of current and voltage, the instantaneous power is positive meaning that the generator is delivering energy to the

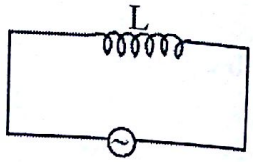
Generated by CamScanner from intsig.com

capacitor at one time and it is negative means returning energy to the generator at another time. Thus, the power alternates between positive and negative values for equal periods of time. In other words, the capacitor alternately absorbs and releases energy. As a result;

- The average power, and hence, the average energy used by a capacitor in an AC circuit is zero.

**Inductive Circuits and Alternating Current**

Consider an inductor in an AC circuit. The generator supplies and rms voltage, $V_{rms}$ given by




$$\boxed{V_{rms} = I_{rms}X_L}$$

Where; $X_L$ is called *inductive reactance*, which is the resistance of the inductor to the flow of current through it. $X_L$ is found experimentally to be directly proportional to the frequency and inductance and hence,

$$\boxed{X_L = 2\pi fL = \omega L}$$

**Illustrative Example**

45. A 21mH inductor is connected to an AC generator with an rms voltage of 24V and a frequency of 60Hz. What is the rms current in the inductor?

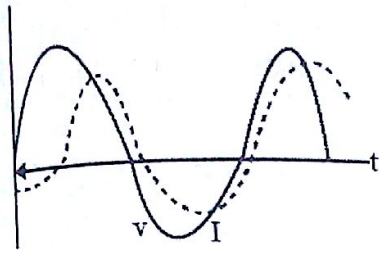
***Solution***: $I_{rms} = \dfrac{V_{rms}}{X_L} = \dfrac{V_{rms}}{\omega L} = \dfrac{V_{rms}}{2\pi fL} = \dfrac{24}{2\pi(60)(21\times10^{-3})} = 3A$

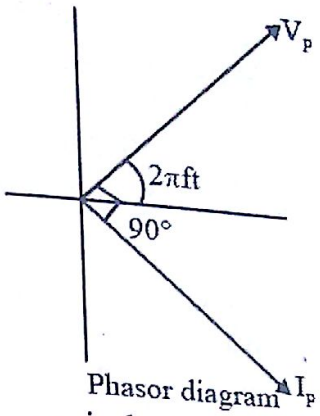
Here also, the voltage and the current are 90° or $\dfrac{\pi}{2}$ radian out of phase. More specially, if the voltage varies as, $V = V_P \sin 2\pi ft$, the current fluctuates as $I = I_P \sin\left(2\pi ft - \dfrac{\pi}{2}\right) = -I_p \cos 2\pi ft$. In this case, the current reaches its maximum after the voltage does, and it is said that;

- In an inductive circuit, the current lags behind the voltage by a phase angle of $\dfrac{\pi}{2}$ or 90°.

This can easily be visualized graphically as follows:

Generated by CamScanner from intsig.com

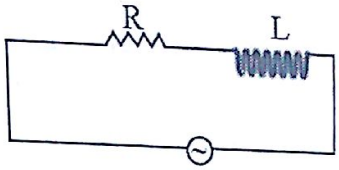



Phasor diagram

As in the case of the capacitive circuit, an inductor also alternately absorbs and releases energy for equal period of time. Thus, the average power and hence the average energy used by an inductor in an AC circuit is zero.

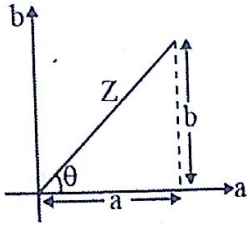
**Resistor and Inductor in an AC circuit**




Consider a circuit containing an inductor and a resistor in an AC circuit. In this case, the total opposition to the flow of current is the combination of the resistance from the resistor and the reactance from the inductor. This opposition is called *impedance*.

**Impedance (Z):** Is the total opposition to the flow of current in an AC circuit when there is a resistor, capacitor or inductor combinations.

To analyze an AC circuit containing different components, we use the concept of complex numbers. The basic idea of complex number is given below




A complex number is of the form

$$Z = a + ib$$

Where a: is real part

b: is imaginary part

The magnitude of the complex number is found by Pythagoras' theorem as:

$$Z = \sqrt{a^2 + b^2}$$

and, the angle θ can be obtained using

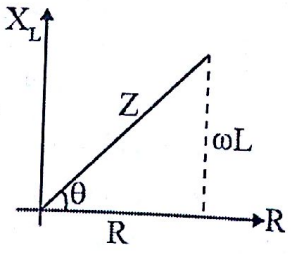
$$\tan\theta = \frac{b}{a}$$

DIGITAL LIBRARY FOR ETHIOPIAN STUDENTS
Digital Library for Ethiopian students
Link: https://telegram.me//digitallibrary4ethiopianstudents or digitallibrary4ethiopianstudents
YouTube Digital[0,1] Ethiopia
Generated by CamScanner from intsig.com

So that, $\theta = \tan^{-1}\left(\frac{b}{a}\right)$

Now, let's turn our attention to our central point of discussion. The total impedance in the resistor and inductor circuit in the form of complex number is given by

$$Z = R + iX_L = R + i\omega L$$




And, its magnitude is

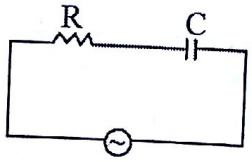
$$\boxed{Z = \sqrt{R^2 + \omega^2 L^2}}$$

And, the phase difference is

$$\boxed{\theta = \tan^{-1}\left(\frac{X_L}{R}\right)}$$

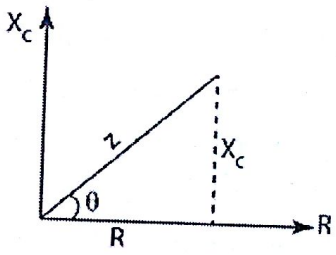
**Resistor and Capacitor in an AC circuit**

Consider the circuit shown:




The impudence, Z is given by

$$\boxed{Z = \sqrt{R^2 + X_C^2} = \sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}}$$

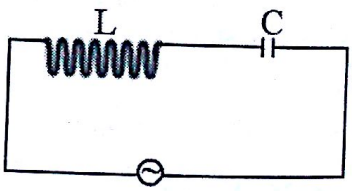



And, the phase angle is obtained by

$$\theta = \tan^{-1}\left(\frac{X_C}{R}\right)$$

**Inductor and Capacitor in an AC circuit**

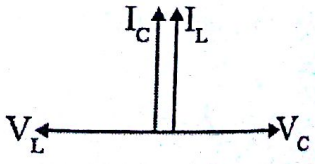
Consider the circuit shown. Since $V_L$ and $V_C$ are in opposite directions, the impudence of this circuit is given by:




$$\boxed{Z = \sqrt{(X_L)^2 - (X_C)^2}}$$

When $X_L = X_C$ that is,

$$\omega_L = \frac{1}{\omega C}$$




$$2\pi fL = \frac{1}{2\pi fC}$$

Solving for the frequency,

$$\boxed{f = \frac{1}{2\pi\sqrt{LC}}}$$ ⟵ This frequency is called *resonant frequency*.

Generated by CamScanner from intsig.com

## Conceptual Examples

46. What would be the effect on current if the frequency of the emf is increased in a purely?
a) Resistive b) Inductive c) Capacitive AC circuit

*Explanation*: a. No effect, R is not affected by frequency.

b. Decreases $X_L = \omega L = 2\pi fL$ and $I = \frac{V}{X_L} = \frac{V}{2\pi fL}$

c. Increase $X_C = \omega C = 2\pi fL$ and $I = \frac{V}{X_C} = 2\pi fCV$

## Illustrative Examples

47. In a purely inductive AC circuit L = 35mH and the rms voltage is 165V. What is the inductive reactance and the rms current in ohm and Ampere if the frequency is 40Hz respectively? (UEE 2001)
A. 8.79, 19.77 C. 7.79, 20.77
B. 7.69, 20.77 D. 9.89, 18.77

*Solution*: $X_L = 2\pi fL = 2(3.14)(40)(35 \times 10^{-3}) = 8.79\Omega$

$$I_{rms} = \frac{V_{rms}}{X_L} = \frac{165}{8.79} = 18.77A$$ **Answer: A**

48. The peak voltage of an alternating emf is 141 volts. The effective value of the voltage in volts is (UEE 2002)
A. 70.5 B. 100 C. 141 D. 200

*Solution*: $V_{rms} = 0.707\ V_P = (0.707)(141) \simeq 100V$ **Answer: B**

49. The voltage output of a generator is given by v = 300 sin ωt. What is the rms current in the circuit when the generator is connected to 100Ω resistor? (UEE 2002)
A. 1.5A B. 2.12A C. 1.12A D. 1.41A

*Solution*: $I = \frac{V}{R} = \frac{300\sin\omega t}{100} = 3\sin\omega t$

$I_{rms} = 0.707 I_P = (0.707)(3) = 2.12A$

**Answer: B**

50. What is the inductance of an LC circuit that oscillates at 120Hz when the capacitance is 8μF? (UEE 2003)
A. 0.22μH B. 0.22H C. 2.19H D. 2.19μH

*Solution*: At resonance, $X_L = X_C$

$$\Rightarrow 2\pi fL = \frac{1}{2\pi fC} \Rightarrow L = \frac{1}{4\pi^2 f^2 C} = 0.22H$$ **Answer: B**

Generated by CamScanner from intsig.com

51. The phase angle in an alternating current circuit is zero degree when the circuit contains (UEE: 2003)
A. Resistance only C. Inductance only
B. Capacitance only D. Resistance and inductance
***Explanation***: In a purely resistive circuit, the current and voltage are in phase hence, the phase angle is zero.
**Answer: A**

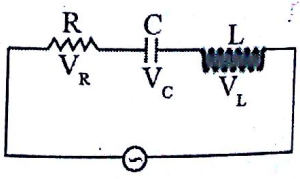
52. The impendence of an 8 microfarad capacitor at 1000Hz is? (UEE: 2003)
A. 600 ohm B. 22.7 ohm C. 19.9 ohm D. 0.44 ohm
***Solution***: $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(1000)(8\times10^{-6})} = 19.9\Omega$ **Answer: C**

53. Which statement is correct about a purely inductive circuit? (UEE: 2004)
A. The current lead the voltage by $\pi/2$
B. The current is in phase with the voltage.
C. The current lags behind the voltage by $\pi/2$.
D. The current lags by $\pi/2$ at low frequencies and leads by $\pi/2$ at high frequencies.
**Answer: C**

54. What is the main reason why alternating current is used to transmit electricity? (UEE: 2004)
A. To minimize power gain.
B. To minimize energy loss.
C. To maximize voltage gain.
D. To maximize current gain.
**Answer: B**

**Combining Resistor, Capacitor and Inductor**

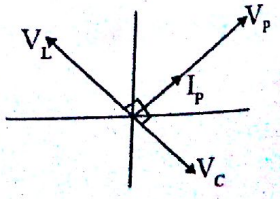
Consider the RLC circuit shown since $V_L$ and $V_C$ point in opposite direction, they give a resultant of $V_L - V_C$ which is perpendicular to $V_R$ and may be combine with it to give a total voltage $V_P$. Using Pythagoras theorem, we get




$$V_P^2 = V_R^2 + (V_L - V_C)^2$$

The corresponding rms voltage is,

$$V_{rms}^2 = V_R^2 + (V_L - V_C)^2$$




Generated by CamScanner from intsig.com

Where; $V_R = I_{rms}R$, $V_C = I_{rms}X_C$ and $V_L = I_{rms}X_C$

So we have; $V_{rms} = I_{rms}\sqrt{R^2 + (X_L - X_C)^2}$

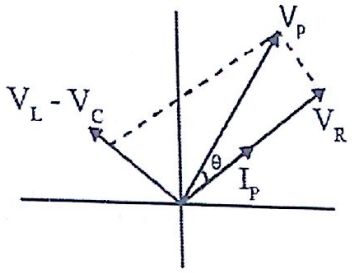
Therefore, for the RLC circuit, it follows that;

$$V_{rms} = I_{rms}Z$$

Where the impendence Z of the RLC circuit is

$$\boxed{Z = \sqrt{R^2 + (X_L - X_C)^2}}$$

The phase angel between the current and the voltage across an RLC circuit is the angel θ which is given by,




$$\boxed{\tan\theta = \frac{X_L - X_C}{R}} \text{ or } \boxed{\tan\theta = \frac{V_L - V_C}{V_R}}$$

**Average power in an AC circuit**

On the average, only the resistance consumes power. That is, $\boxed{P = I_{rms}^2 R}$

but, from the above figure, $\cos\theta = \frac{V_R}{V_P} = \frac{I_{rms}R}{I_{rms}Z} = \frac{R}{Z}$ so that, $R = Z\cos\theta$.

Therefore, $P = I_{rms}^2 Z\cos\theta = I_{rms}(I_{rms}Z)\cos\theta \Rightarrow \boxed{\boxed{P = I_{rms}V_{rms}\cos\theta}}$

The term cos θ is called the *power factor* of the circuit.

- **Important points to remember**
  - If no resistance is present, R = 0 and $\cos\theta = \frac{R}{Z} = 0$

    consequently, $P = I_{rms}V_{rms}\cos\theta = 0$.
  - For the phase angle θ, $\tan\theta = \frac{X_L - X_C}{R}$
    i. If $X_L > X_C$, then θ is positive and the voltage leads the current
    ii. If $X_L < X_c$, then θ is negative, so the voltage lags the current.

Generated by CamScanner from intsig.com

## Summary of A.C Circuit Elements

| Circuit element | Impedance | Average power | Phase Angle |
|---|---|---|---|
| R | $Z = R$ | $P_{av} = I_{rms}^2 R$ | $\theta = 0°$ |
| C | $Z = X_C = \frac{1}{2\pi fC}$ | $P_{av} = 0$ | $\theta = -90°$ |
| L | $Z = X_L = 2\pi fC$ | $P_{av} = 0$ | $\theta = +90°$ |
| RC | $Z = \sqrt{R^2 + X_C^2}$ | $P_{av} = I_{rms} V_{rms} \cos\theta$ | $-90° < \theta < 90°$ |
| RL | $Z = \sqrt{R^2 + X_L^2}$ | $P_{av} = I_{rms} V_{rms} \cos\theta$ | $0 < \theta < 90°$ |
| LC | $Z = \sqrt{X_L^2 - X_C^2}$ | $P_{av} = 0$ | |
| RLC | $Z = \sqrt{R^2 + (X_L - X_C)^2}$ | $P_{av} = I_{rms} V_{rms} \cos\theta$ | +ve $(X_L > X_C)$<br>−ve $(X_C > X_L)$ |

### Illustrative Examples

55. A capacitance C = 2μF, and inductance L = 10H and a resistance R = 5Ω are in series in a circuit. When an rms current of 2A flows, the power in watts in the circuit is (UEE: 2002)

A. 100 B. 50 C. 20 D. 2

***Solution***: $P = I_{rms}^2 R = (2)^2(5) = 20\,W$ **Answer: C**

56. In RLC circuit, givne R = 7.6Ω, L = 2.2mH and C = 1.8μF. What is the frequency of the oscillation of the circuit? (UEE 2004)

A. $2.5 \times 10^3$ Hz
B. $15.72 \times 10^3$ Hz
C. $12.5 \times 10^3$ Hz
D. $25.0 \times 10^3$ Hz

***Solution***: $f = \frac{1}{2\pi\sqrt{L_C}} = 2.5 \times 10^3\,Hz$ **Answer: A**

57. A series RLC circuit has a resistance of 45Ω and an impedance of 75Ω. What average power is delivered to this circuit when $V_{rms} = 210V$? (UEE: 2005)

A. 573W B. 353W C. 255W D. 247W

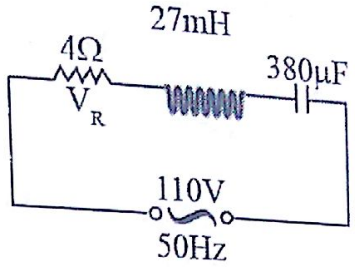
***Solution***: $I_{rms} = \frac{V_{rms}}{Z} = \frac{210}{75} = 2.8A$

$P = I_{rms}^2 R = (2.8)^2(45) = 353\,W$ **Answer: B**

Generated by CamScanner from intsig.com

58. Determine the total impedance for the circuit shown below (UEE: 2006)

A. 0Ω
B. 4Ω
C. 27Ω
D. 31Ω




**Solution:** $Z = \sqrt{R^2 + (X_L - X_C)^2}$

$$Z = \sqrt{(4)^2 + \left(2\pi \times 50 \times 27 \times 10^{-3} - \frac{1}{2\pi \times 50 \times 380 \times 10^{-6}}\right)^2}$$

$Z \simeq 4\Omega$

**Answer: B**

59. A series RLC circuit has a total impedance of 75Ω. If the resistance is R = 45Ω and the average power delivered to the circuit is P = 352.8W. What is the root mean square value of the AC voltage source? (UEE: 2007)

A. 83V B. 116V C. 210V D. 297V

**Solution:** $P = I_{rms}V_{rms}\cos\theta$, where $\cos\theta = \frac{R}{Z}$

$$P = (I_{rms})(V_{rms})\left(\frac{R}{Z}\right) \Rightarrow V_{rms} = \frac{PZ}{I_{rms}R}, \text{ again } P = I_{rms}^2 R$$

$$I_{rms} = \sqrt{\frac{P}{R}} = \sqrt{\frac{352.8}{45}} = 2.8$$

Hence, $V_{rms} = \frac{(352.8)(75)}{(2.8)(45)} = 210V$

**Answer: C**

## Additional Examples

### Electromagnetic Induction

60. If a surface is parallel to magnetic lines of force, then the magnetic flux linked with it is:

A. Small but not zero
B. Zero
C. Infinite
D. Very large

**Answer: B**

**Explanation:** BA cos 90° = 0

61. Whenever magnetic lines of force move across a conductor, there is produced an electric field. This is referred to as:

A. Lenz's law
B. Faraday's law
C. Ohm's law
D. Gauss's law

**Answer: B**

Generated by CamScanner from intsig.com

62. The energy stored in inductance is 1J when a current of 0.1A is established in it. What is the self–inductance of the coil?
A. 200H B. 50H C. 25H D. 3H

*Solution*: $E = \frac{1}{2}LI^2 \Rightarrow L = \frac{2E}{I^2} = \frac{2\times 1}{(0.1)^2} = 200H$ **Answer: A**

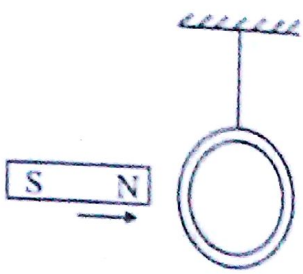
63. The mutual inductance of two coils of is 6mH. If the current flowing in one coil is 2A, what is the induced emf in the second coil?
A. 12mv B. 3mv C. 3V D. zero

*Explanation*: Since the current is constant, and

$$\varepsilon = M\frac{\Delta I}{\Delta t} \Rightarrow \varepsilon = 0$$

**Answer: D**

64. A copper ring is suspended by a thread in a vertical plane. The north pole of a magnet is brought near the ring in horizontal direction. What will be the effect on the ring?




A. Ring will be attracted towards the magnet.
B. Ring will be repelled away.
C. Ring will make simple harmonic motion.
D. No change in the position of ring.

*Hint*: Use Lenz's law. **Answer: B**

65. The average emf induced in a coil in which the current changes from 2A to 4A in 0.05 second is 8V. What is the self-inductance of the coil?
A. 0.1H B. 0.2H C. 0.4H D. 0.8H

*Solution*: $\varepsilon = L\frac{\Delta I}{\Delta t} \Rightarrow L = \frac{(\varepsilon)(\Delta t)}{\Delta I} = \frac{(8)(0.05)}{(4-2)} = 0.2H$ **Answer: B**

66. When a metallic sphere is moved in a magnetic field, it gets heated due to
A. Direct current C. Alternating current
B. Eddy current D. Additional current

**Answer: B**

67. When a magnet is being moved towards a coil, the induced emf does not depend on.
A. The number of turns of the coil.
B. The motion of the magnet.

Generated by CamScanner from intsig.com

C. The magnetic moment of the magnet.
D. The resistance of the coil.

*Hint:* $\varepsilon = N\frac{\Delta\phi}{\Delta t}$

**Answer: D**

68. Keeping all other factors same, the number of turns of a coil is doubled. What will the magnetic flux linked to the coil be?
A. doubled B. halved C. tribled D. quadrupled

**Answer: A**

69. The current passing through a coil of 5H is decreasing at the rate of 2A/s. What is the emf developed in the coil?
A. 10V B. −10V C. 2.5V D. −2.5V

*Solution:* $\varepsilon = -L\frac{\Delta I}{\Delta t} = (-5)(-2) = 10\,v$

**Answer: A**

70. Lenz's law is consequence of the law of conservation of:
A. Charge B. Momentum C. Mass D. Energy

**Answer: D**

71. The north pole of a long horizontal bar magnet is being brought closer to a vertical conducting coil along the perpendicular direction. The direction of the induced current in the conducting coil will be:
A. Horizontal B. Vertical C. Clockwise D. Anticlockwise

*Explanation:* The face of the coil facing the magnet has to acquire north polarity to repel the magnet. This is possible if the induced current is anti – clockwise.

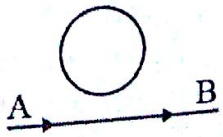
**Answer: D**

72. A coil of area $80cm^2$ and 50 turns is rotating with 2000 rev/min about an axis perpendicular to a magnetic field of 0.05T. The maximum value of the emf developed in it is:
A. $200\pi V$ B. $\frac{10\pi}{3}V$ C. $\frac{4\pi}{3}V$ D. $\frac{2}{3}V$

*Solution:* $\varepsilon = NBA\omega = (50)(0.05)(80\times10^{-4})\left(2\pi\times\frac{2000}{60}\right) = \frac{4\pi}{3}v$

**Answer: C**

73. An electron moves along the line AB, which lies in the same plane as a circular loop of conducting wire as shown. What will be the direction of current induced if any, in the loop?
A. No current will be induced.
B. The current will be clockwise.
C. The current will be anticlockwise.




Generated by CamScanner from intsig.com

D. The current will change direction as the electron pass by.

***Hint:*** Apply Lenz's law. **Answer: C**

74. A coil of area A is kept perpendicular to a magnetic field B, and is rotated by 180°, the change in flux will be

A. BA B. Zero C. 2BA D. 4BA

***Explanation:*** $\Delta\phi = BA\cos 0 - BA\cos 180° = 2BA$ **Answer: C**

75. "The direction of the induced current in a circuit is always such that it opposes the cause due to which it is produced." This law is named as

A. Ohm's law
B. Lenz's law
C. Kirchhoff's law
D. Faraday's law

**Answer: B**

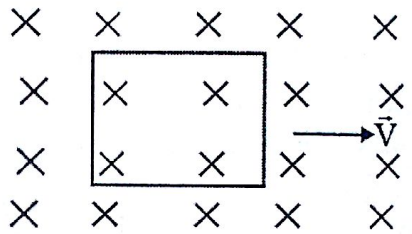
76. A conducting square loop of side L and resistance R moves in its plane with a uniform velocity v perpendicular to a constant magnetic field. The current induced in the loop is

A. $\frac{B\ell v}{R}$ clockwise

B. $\frac{B\ell v}{R}$ anticlockwise

C. $\frac{2B\ell v}{R}$ anticlockwise

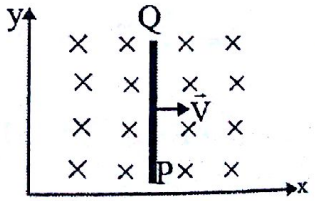
D. Zero




***Explanation:*** Since the flux is not changing, there is no induced emf and hence no induced current. **Answer: D**

77. A conducting rod PQ moves parallel to x – axis in a uniform magnetic field pointing into the page, along negative z axis. The point P of the rod gets:

A. Positively charged
B. Negatively charged
C. Neutral
D. First positively charged and then negatively charged.




***Hint:*** Use Fleming's right hand rule and find the direction of the induced emf or use $\vec{F} = q(\vec{v} \times \vec{B})$. **Answer: B**

78. A magnet is moved with a fast speed towards a coil at rest. If the induced emf, current and charge in the coil are E, I and Q respectively, and if the speed of the magnet is doubled, the incorrect statement is:

Generated by CamScanner from intsig.com

A. E increases
B. I increases
C. Q remains the same
D. Q increases

**Answer: D**

79. The back emf in a DC motor is maximum when
A. The motor has picked up maximum speed.
B. The motor has just started moving.
C. The speed of motor is still on the increase.
D. The motor has just been switched off.

*Hint:* $\varepsilon = NBA\omega$

**Answer: A**

80. The armature current in a DC motor is maximum when the motor has
A. Picked up maximum speed.
B. Just started.
C. Intermediate speed.
D. Just been switched off.

*Explanation:* $I = \frac{V - \varepsilon}{R}$, at the start $\varepsilon = 0$

$\Rightarrow I_{start} = \frac{V}{R}$, which is maximum

**Answer: B**

81. A transformer reduces the main from 200V to 11V. If the primary coil draws 5A of current and the secondary coil draws 90A, the efficiency of the transformer is:
A. 95% B. 90% C. 88% D. 40%

*Solution:* Input power $= V_P I_P = (220)(5) = 1100W$

Output power $= V_S I_S = (11)(90) = 990W$

$$\text{efficiency} = \frac{\text{output power}}{\text{Input power}} = \frac{990}{1100} = 90\%$$

**Answer: B**

82. The core of a transformer is lamented to reduce energy losses due to:
A. Eddy currents
B. Resistance in winding
C. Applied voltage
D. Unknown

**Answer: A**

83. The armature of a DC motor has 20Ω resistance. It draws a current of 1.5A when run by 200V DC supply. The value of back emf induced will be:
A. 150V B. 170V C. 180V D. 190V

*Solution:* $I = \frac{v - \varepsilon}{R} \Rightarrow \frac{200 - \varepsilon}{20} = 1.5, \varepsilon = 190V$

**Answer: D**

Generated by CamScanner from intsig.com

84. If the angular speed or rotation of an armature of alternating current generator is doubled, then the induced electromotive force will be:
A. Twice B. Four times C. Halved D. Eight times
*Explanation*: $\varepsilon = NBA\omega$ **Answer: A**

85. The function of a main transformer is to convert:
A. One direct voltage to another direct voltage of different magnitude.
B. One alternating voltage to another alternating voltage of different magnitude.
C. A high value alternating voltage to a high value direct voltage.
D. A low value alternating voltage to a high value direct voltage.

**Answer: B**

86. In an ideal transformer, the most important function of the soft – iron core is:
A. To reduced eddy currents.
B. To improve the flux – linkage between the primary and secondary coils.
C. To dissipate the heat generated by the two coils.
D. To eliminate the back emf produced in the secondary coil.
*Explanation*: Soft – iron has a high permeability and hence it provides relatively large magnetic field with in the core for a given current and thus improves the flux – linkage between the primary and secondary coils. **Answer: B**

87. In a transformer, the number of turns of primary coil and secondary coil are 5 and 4 respectively. If 220V is applied on the primary coil, then the ratio of primary to the secondary current is
A. 9: 5 B. 5: 9 C. 5:4 D. 4: 5
*Solution*: $\frac{I_P}{I_S} = \frac{V_S}{V_P} = \frac{N_S}{N_P}$

Since $\frac{N_S}{N_P} = \frac{4}{5}$ and $\frac{I_P}{I_S} = \frac{N_S}{N_P} = \frac{4}{5}$

Hence $\frac{I_P}{I_S} = \frac{4}{5} = 4:5$

**Answer: D**

DIGITAL LIBRARY FOR ETHIOPIAN STUDENTS
Digital Library for Ethiopian students
Link: https://telegram.me//digitallibrary4ethiopianstudents or digitallibrary4ethiopianstudents
YouTube Digital[0,1] Ethiopia
Generated by CamScanner from intsig.com

88. A coil of wire has 600 turns and a self–inductance of 108mH. The self–inductance of 2nd similar coil of 500 turns will be:
A. 74mH B. 75mH C. 76mH D. 77mH

**Solution:** $L = \frac{N\phi}{I} \Rightarrow L \sim N$

$$\frac{L_1}{L_2} = \frac{N_1}{N_2} \Rightarrow L_2 = \frac{(L_1)(N_2)}{N_1} = \frac{(108)(500)}{600} = 75mH$$

**Answer: B**

89. The mutual inductance of two circuits A and B is 3mH and their respective resistances are 10Ω and 4Ω. How much current should change in 0.02 second in circuit A so that the induced current is B should be 0.006A.
A. 0.24A B. 1.6A C. 0.18A D. 0.16A

**Solution:** Induced emf in B, $\varepsilon mf_B = (I)(R) = 0.006 \times 4 = 0.024$

and, $\varepsilon = M\frac{\Delta I}{\Delta t} \Rightarrow \Delta I = \frac{(0.024)(0.02)}{(0.003)} = 0.16A$

**Answer: D**

90. The mutual inductance between two coils is 1.25H. If the current in the primary changes at the rate of 80A/s, then the induced emf in the secondary is:
A. 12.5V B. 64V C. 0.016V D. 100V

**Solution:** $\varepsilon = M\frac{\Delta I}{\Delta t} = (1.25)(80) = 100\,V$

**Answer: D**

91. The magnetic flux linked with a coil is changed from 1Wb to 0.1Wb in 0.1 second. The induced emf is
A. 9V B. 10V C. 0.009V D. 0.9V

**Solution:** $\varepsilon = -\frac{\Delta\phi}{\Delta t} = \frac{-(0.1-1)}{0.1} = 9V$

**Answer: A**

92. A straight conductor of length 4m moves at a speed of 10 m/s when the conductor makes an angle of 30° with the direction of magnetic field of induction 0.1T. Then the induced emf is
A. 1V B. 2V C. 4V D. 8V

**Solution:** $\varepsilon = B\ell V \sin\theta = (0.1)(4)(10)\sin 30° = 2V$

**Answer: B**

93. A 50 turn circular coil has a radius of 3cm. It is kept in a magnetic field that increases from 0.1T to 0.35T in 2ms. The average induced emf in the coil is
A. 1.77V B. 17.7V

Generated by CamScanner from intsig.com

*Solution*: $|\varepsilon| = \left|-\frac{NA\Delta B}{\Delta t}\right| = \frac{50(\pi \times (3\times10^{-2})^2)(0.35-0.1)}{2\times10^{-3}}$

$\varepsilon = 17.7V$ **Answer: B**

A coil having an area of $2m^2$ is placed in a magnetic field which changes from $1Wb/m^2$ to $4Wb/m^2$ in 2 second. What is the induced emf in the coil?

A. 4V B. 3V C. 1.5V D. 2V

*Solution*: $|\varepsilon| = \left|-\frac{\Delta\phi}{\Delta t}\right| = \frac{A\Delta B}{\Delta t} = \frac{2(4-1)}{2} = 3V$ **Answer: B**

A magnet is moved towards a coil (a) quickly (b) slowly, then the induced emf is

A. Larger in case (a)
B. Smaller in case (a)
C. Equal in both cases
D. Larger or small depending upon the radius of the coil.

*Explanation*: The more the rate of change of magnetic flux, the larger is the induced emf. **Answer: A**

If a current of 2A gives rise a magnetic flux of $5 \times 10^{-5}Wb$ through a coil having 100 turns, then the magnetic energy stored in the medium surrounding the coil is

A. 5J B. $5 \times 10^{-7}J$ C. $5 \times 10^{-3}J$ D. 0.5J

*Solution*: $E = \frac{1}{2}LI^2$, but $L = \frac{N\phi}{I}$

$$\Rightarrow E = \frac{1}{2}\left(\frac{N\phi}{I}\right)(I^2) = \frac{1}{2}N\phi I = \frac{1}{2}(100)(5\times10^{-5})(2) = 5\times10^{-3}J$$

**Answer: C**

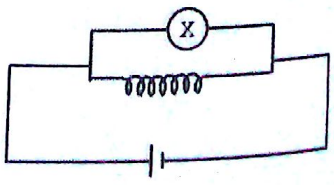
When the number of turns in a coil is doubled without any change in the length of the coil, its self – inductance becomes:

A. Four times B. Doubled C. Halved D. Tripled

*Hint*: $L = \frac{N\phi}{I}$ **Answer: B**

A bulb of 100W is connected in parallel with an ideal inductance of 1H. This arrangement is connected to a 90V battery through a switch. On pressing the switch, the

A. Bulb does not glow.
B. Bulb glows.
C. Bulb glows after a short time and




Generated by CamScanner from intsig.com

*Solution*: $|\varepsilon| = \left|-\frac{NA\Delta B}{\Delta t}\right| = \frac{50(\pi \times (3\times10^{-2})^2)(0.35-0.1)}{2\times10^{-3}}$

$\varepsilon = 17.7V$ **Answer: B**

94. A coil having an area of $2m^2$ is placed in a magnetic field which changes from $1Wb/m^2$ to $4Wb/m^2$ in 2 second. What is the induced emf in the coil?

A. 4V B. 3V C. 1.5V D. 2V

*Solution*: $|\varepsilon| = \left|-\frac{\Delta\phi}{\Delta t}\right| = \frac{A\Delta B}{\Delta t} = \frac{2(4-1)}{2} = 3V$ **Answer: B**

95. A magnet is moved towards a coil (a) quickly (b) slowly, then the induced emf is

A. Larger in case (a)
B. Smaller in case (a)
C. Equal in both cases
D. Larger or small depending upon the radius of the coil.

*Explanation*: The more the rate of change of magnetic flux, the larger is the induced emf. **Answer: A**

96. If a current of 2A gives rise a magnetic flux of $5 \times 10^{-5}$Wb through a coil having 100 turns, then the magnetic energy stored in the medium surrounding the coil is

A. 5J B. $5 \times 10^{-7}$J C. $5 \times 10^{-3}$J D. 0.5J

*Solution*: $E = \frac{1}{2}LI^2$, but $L = \frac{N\phi}{I}$

$\Rightarrow E = \frac{1}{2}\left(\frac{N\phi}{I}\right)(I^2) = \frac{1}{2}N\phi I = \frac{1}{2}(100)(5\times10^{-5})(2) = 5\times10^{-3}J$

**Answer: C**

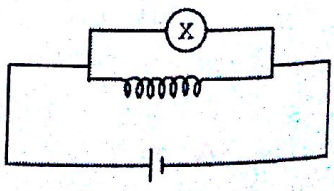
97. When the number of turns in a coil is doubled without any change in the length of the coil, its self – inductance becomes:

A. Four times B. Doubled C. Halved D. Tripled

*Hint*: $L = \frac{N\phi}{I}$ **Answer: B**

98. A bulb of 100W is connected in parallel with an ideal inductance of 1H. This arrangement is connected to a 90V battery through a switch. On pressing the switch, the

A. Bulb does not glow.
B. Bulb glows.
C. Bulb glows after a short time and




Generated by CamScanner from intsig.com

then continues to glow.

D. Bulb glows for a short time and then stops glowing.

*Explanation*: The inductance acts as a short – circuit. The whole of the current flows through the inductor. Since no current flows through the bulb, it does not glow.

**Answer: A**

99. The magnetic flux through each turn of a 100 turn coil is $(t^3 - 2t)10^{-3}$ Wb. The induced emf at t = 2 sec is?

A. −4V B. −1V C. 1V D. 4V

*Solution*: $\varepsilon = -N\frac{\Delta\phi}{\Delta t} = -\frac{d\phi}{dt} = \frac{dN}{dt}(t^3 - 2t) \times 10^{-3}$

$= -(3t^2 - 2)(10^{-3})(100)$

$\varepsilon = [3(2)^2 - 2](10^{-3})(100) = -1V$

**Answer: B**

100. A coil having 500 square loops, each of side 10cm, is placed normal to a magnetic field which increases at a rate of 1 T/s. The induced emf in volt is?

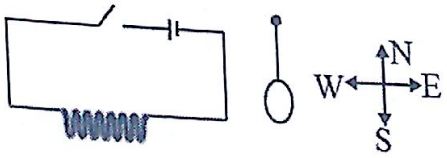
A. 0.1 B. 0.5 C. 1.0 D. 5.0

*Solution*: $\varepsilon = -N\frac{\Delta\phi}{\Delta t} = \frac{NA\Delta B}{\Delta t} = (500)(10 \times 10^{-2})^2(1) = 5V$

**Answer: D**

101. An aluminum ring hangs vertically from a thread with its axis pointing east – west. What is the initial motion of the rign when the current in the coil is switched on?

A. Towards West
B. Towards East
C. Towards South
D. Towards North




*Explanation*: As the switch is closed, magnetic flux linkage is developed from the west hence, according to Lenz's law, the ring moves East.

**Answer: B**

102. In a step up transformer, the turns ratio is 1:10. A resistance of 200Ω connected across the secondary is drawing a current of 0.5A. What is the primary voltage and current?

A. 50V, 1A B. 10V, 5A
C. 25V, 4A D. 20V, 2A

*Solution*: $V_S = I_S R_S = (0.5)(200) = 100V$

$\frac{V_P}{V_S} = \frac{N_P}{N_S} = \frac{1}{10} \Rightarrow V_P = \frac{V_S}{10} = \frac{100}{10} = 10V$

Generated by CamScanner from intsig.com

Again, $\frac{I_P}{I_S} = \frac{N_S}{N_P} = 10 \Rightarrow I_P = 10I_S = (10)(0.5) = 5\,A$

**Answer: B**

103. The number of turns in the primary coil of a transformer is 200 and the number of turns in the secondary coil is 10. If 240V AC is applied to the primary, the output from the secondary will be:
A. 48V B. 24V C. 12V D. 6V

***Solution***: $\frac{V_S}{V_P} = \frac{N_S}{N_P} \Rightarrow V_S = \left(\frac{N_S}{N_P}\right)(V_P) = \left(\frac{10}{200}\right)(240) = 12V$

**Answer: C**

104. An electric motor runs on a DC source of emf 200V and draws a current of 10A. If the efficiency is 40%, then the resistance of the armature is
A. 5Ω B. 12Ω C. 120Ω D. 160Ω

***Solution***: $\text{Efficiency} = \frac{\text{Output power}}{\text{Input power}}$

$$\Rightarrow \frac{\text{Out put power}}{(200)(10)} = \frac{40}{100} \Rightarrow P_{\text{out put}} = 800W$$

Loss = Input power – out put power = (200) (10) – 800 = 1200W
Now, $1200 = 10 \times 10R \Rightarrow R = 12\Omega$

**Answer: B**

105. A DC motor has an internal resistance of 4Ω and is operating at 220V and draws 5A current. How much back emf is produced?
A. 80V B. 160V C. 200V D. 120V

***Solution***: $\varepsilon = v - Ir = 220 - (5)(4) = 200V$

**Answer: C**

106. When an electric motor is run at 120V, 10A current flows through it and the induced back emf is 115V. What will be the current flowing in the coil at the time of switch off?
A. 230A B. 240A C. 10A D. Zero

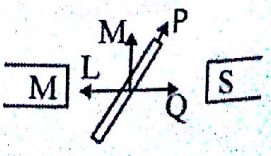
***Solution***: $I = \frac{V - \varepsilon}{R} \Rightarrow R = \frac{V - \varepsilon}{I} = \frac{120 - 115}{10} = 0.5\Omega$

At the time of switch off

$$I = \frac{\varepsilon}{R} = \frac{115}{0.5} = 230A$$

**Answer: A**

107. An emf will be induces between the ends of the conductor shown when it moves in the direction of




Generated by CamScanner from intsig.com

A. P B. Q
C. L D. M

*Explanation:* Along P, V is parallel to $\ell$, along L and Q, V is parallel to B. But, along M, B, $\ell$ and V are mutually perpendicular.

**Answer: D**

108. The equivalent inductance of two inductors is 2.4H when connected in parallel and 10H when connected in series. The difference between the two inductances is:
A. 2H B. 3H C. 4H D. 5H

*Solution:* In parallel, $\frac{1}{L_P}=\frac{1}{L_1}+\frac{1}{L_2}\Rightarrow L_P=\frac{L_1L_2}{L_1+L_2}=2.4$

In series, $L_S = L_1 + L_2 = 10$

Finally, $L_1 - L_2 = 2H$

**Answer: A**

109. The magnetic flux linked with the coil varies with time as $\phi = 3t^2 + 4t + 9$. What is the magnitude of the induced emf at t = 2 seconds?
A. 9V B. 16V C. 3V D. 4V

*Solution:* $|\varepsilon|=\left|\frac{\Delta\phi}{\Delta t}\right|=\frac{d\phi}{dt}=\frac{d}{dt}(3t^2+4t-9)=6t+4$

at $t = 2$ sec

$\varepsilon = 6(2) + 4 = 16V$

**Answer: B**

110. Two coil A and B having 300 and 600 turns are placed near each other. On passing a current of 3A in A, the flux linked with A is $1.2 \times 10^{-4}$Wb and with B it is $9 \times 10^{-5}$Wb. The mutual inductance of the system is
A. $2 \times 10^{-5}$H C. $4 \times 10^{-5}$H
B. $3 \times 10^{-5}$H D. $6 \times 10^{-5}$H

*Solution:* $M=\frac{\phi}{I}=\frac{9\times10^{-4}}{3}=3\times10^{-5}\,H$

**Answer: B**

111. A train is moving towards north with a speed of 180 km/hr. If the vertical component of the earth's magnetic field is $0.2 \times 10^{-4}$T, the emf induced in the axle 1.5m long is
A. 15mv B. 1.5mv C. 5.4mv D. 54mv

*Solution:* $\xi = Blv = (0.2 \times 10^{-4})\,(1.5)\left(180\times\frac{1000}{3600}\right)$

$= 1.5 \times 10^{-3}V = 1.5mV$

**Answer: B**

Generated by CamScanner from intsig.com

112. A cylindrical magnet is kept along the axis of a circular coil. If the magnet is rotated about its axis, then:
A. A current will be induced in the coil.
B. No current will be induced in the coil.
C. Both an emf and a current will be induced in the coil.
D. Only an emf will be induced in the coil.
*Explanation*: There is no change in magnetic flux linked with the coil.
**Answer: B**

113. A horizontal straight conductor, placed along south – north direction falls under gravity, then there is
A. An induced current from south to north direction.
B. An induced current from north to south direction.
C. No induced emf along the length of the conductor.
D. An induced emf along the length of the conductor.
**Answer: C**

114. To induce emf in a coil, the magnetic flux linkage
A. Must remain constant C. must decrease
B. Must increase D. Must increase or decrease
**Answer: D**

115. The average emf induced in a coil in which a current changes from 2A to 4A in 0.05 second is 8V. The self-inductance of the coil is
A. 0.1H B. 0.2H C. 0.4H D. 0.8H
*Solution*: $\varepsilon = L\frac{\Delta I}{\Delta t} \Rightarrow L = \frac{(\Delta t)(\varepsilon)}{\Delta I} = \frac{(0.05)(8)}{4-2} = 0.2H$
**Answer: B**

116. The flux in a closed circuit of a resistance 10Ω varies with time according to the equation.
$\phi = 6t^2 - 5t + 1$
What is the magnitude of the induced current at t = 0.25 second?
A. 1.2A B. 0.8A C. 0.6A D. 0.2A
*Solution*: $I = \frac{V}{R}, V = -\frac{d\phi}{dt} = (6t^2 - 5t + 1) \Rightarrow v = -12t + 5$
Thus, $v = -12(0.25) + 5 = 2V$
$\Rightarrow I = \frac{2}{10} = 0.2A$
**Answer: D**

Generated by CamScanner from intsig.com

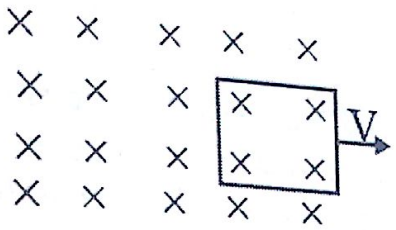
117. A conducting square loop of side $\ell$ and resistance R is being pulled out of the magnetic field with speed V. The current induced in the loop is?

A. Zero

B. $\frac{BV\ell}{R}$ clockwise

C. $\frac{BV\ell}{R}$ anticlockwise

D. $\frac{2BV\ell}{R}$ clockwise




***Explanation:*** $I = \frac{V}{R} = \frac{BV\ell}{R}$, and its direction is clockwise based on Lenz's law.

**Answer: B**

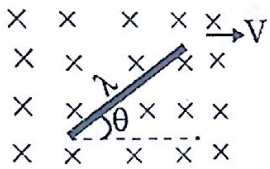
118. A rod of length $\ell$ is taken out of magnetic field with a velocity V in the direction shown. The emf induced in the rod is

A. $BV\ell$

B. $BV\ell \cos\theta$

C. $BV\ell \sin\theta$

D. Zero




***Hint:*** $\varepsilon = B\ell_1 V = BV\ell \sin\theta$

**Answer: C**

119. An infinitely long cylindrical conducting rod is kept along + z–axis. A constant magnetic field is also present along + z–axis. Then current induced will be

A. Zero
B. Along +z–axis
C. Along clockwise as seen from +z
D. Along anticlockwise seen from +z

***Hint:*** $\varepsilon = \vec{v}.\vec{\ell} \times \vec{B} = \text{zero}$

**Answer: A**

120. A conducting wire is moved towards the right in a magnetic field. The direction of induced current in the wire is as shown. The direction of the magnetic field must be:

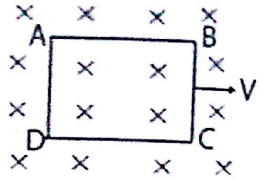
A. In the plane of paper pointing towards the right.
B. In the plane of paper pointing towards the left.
C. Perpendicular to the plane of paper and downwards.
D. Perpendicular to the plane of paper and upwards.

***Hint:*** From Lenz's law

**Answer: C**

Generated by CamScanner from intsig.com

121. A metallic square loop ABCD is moving on its own plane with uniform velocity in a uniform magnetic field. An electric field is induced




A. In AD but not in BC
B. In BC but not in AD
C. Neither in AD nor in BC
D. In both AD and BC

***Explanation***: Same emf is induced in sides AD and BC and points A and B are at the same potential, also points C and D are at the same potential. Thus, no current is induced in the square.

**Answer: C**

## Alternates Current

122. Given $I = I_o \sin\left(\omega t - \frac{\pi}{2}\right)$ and $V = V_o \sin \omega t$. Power is

A. VI B. $\frac{1}{2}$VI C. $V_oI_o$ D. Zero

***Explanation***: Phase difference is $\frac{\pi}{2}$. So power is zero.

**Answer: D**

123. A resistance of 40Ω and an inductance of 95.5 mH are connected in series in 50 cycle/second AC circuit. The impudence of this combination is

A. 30Ω B. 40Ω C. 50Ω D. 60Ω

***Solution***: $X_L = 2\pi fL = 2 \times 3.14 \times 50 \times 95.5 \times 10^{-3} = 30\Omega$

$$Z = \sqrt{R^2 + X_L^2} = \sqrt{40^2 + 30^2} = 50\Omega$$ **Answer: C**

124. The power factor in an AC series L – R circuit is

A. $\sqrt{R^2 + L^2\omega^2}$ C. $R\sqrt{R^2 + L^2\omega^2}$

B. $L/R$ D. $\frac{R}{\sqrt{R^2 + L^2\omega^2}}$

***Hint***: Power factor, $\cos\theta = \frac{R}{\sqrt{R^2 + L^2\omega^2}}$ **Answer: D**

125. The equation of an alternating current is $I = 50\sqrt{2}$ sin 400πt. The frequency and the root mean square of the current are respectively.

A. 200H, 50A C. 200H, $50\sqrt{2}$ A

Generated by CamScanner from intsig.com

B. 400H, $50\sqrt{2}$ A D. 50H, 200A

**Hint:** $\omega t = 400\pi t$ or $2\pi ft = 400\pi ft \Rightarrow f = 200Hz$ **Answer: A**

126. A coil of inductance L has an inductive reactance of $X_L$. The coil is made from super conducting material and has no resistance. The rate at which power is dissipated in the coil is:

A. Zero B. $IX_L$ C. $I^2X_L$ D. $IX_L^2$

**Explanation:** No power is consumed in a super conductor and in a purely inductive circuit. **Answer: A**

127. If L and R denote inductance and resistance respectively, which of the following has the dimensions of frequency?

A. $R/L$ B. $L/R$ C. $\sqrt{R/L}$ D. $\sqrt{L/R}$

**Hint:** time = $L/R$, hence frequency = $R/L$ **Answer: A**

128. Given: V = 100 sin (100t) and I = 100 sin $\left(100t + \frac{\pi}{3}\right)$. Power is

A. 10000W B. 10W C. 2.5W D. 5W

**Solution:** $P = VI\cos\theta = \left(\frac{V_o}{\sqrt{2}}\right)\left(\frac{I_o}{\sqrt{2}}\right)\cos\frac{\pi}{3}$

$$= \frac{100 \times 100 \times 10^{-3}}{2} \times \frac{1}{2} = 2.5W$$

**Answer: C**

129. The average value of alternating current over one cycle is

A. $\frac{I_{max}}{\sqrt{2}}$ B. $\sqrt{2}I_{max}$ C. $0.63I_{max}$ D. Zero

**Answer: D**

130. The peak voltage of 220V AC main is:

A. 155.6V B. 220V C. 311V D. 440V

**Solution:** $V_o = \sqrt{2}V = \left(\sqrt{2}\right)(220) = 311V$ **Answer: C**

131. An AC source is connected to resistive circuit. Which of the following is true?

A. Current leads ahead of voltage in phase.
B. Current lags behind voltage in phase.
C. Current and voltage are in the same phase.
D. Any of the above may be true depending on the resistance value.

**Answer: C**

Generated by CamScanner from intsig.com

132. Current in the LCR circuit becomes extremely large when:
A. Frequency of AC supply in increased.
B. Frequency of AC supply is decreased.
C. Inductive reactance becomes equal to capacitive reactance.
D. Inductance becomes equal to capacitance.

**Answer: C**

133. The inductive reactance of an inductor of $\frac{1}{\pi}$H at 50 Hz frequency is

A. $\frac{50}{\pi}\Omega$ B. $\frac{\pi}{50}\Omega$ C. $100\Omega$ D. 5 sec

***Solution***: $X_L = L\omega = 2\pi fL = 2(50)\,(\pi)\left(\frac{1}{\pi}\right) = 100\Omega$

**Answer: C**

134. A coil of inductance 40H is connected in series with a resistance of 8Ω and the combination is joined to the terminals of 2V battery. The time constant of the circuit is:
A. 40 sec B. 20sec C. 8 sec D. 5 sec

***Solution***: $t = \frac{L}{R} = \frac{40}{8} = 5\,\text{sec}$ **Answer: D**

135. A group of electric lamps having a total power rating of 900W is supplied by an AC voltage v = 300 sin (314t + 60). The rms value of the current is
A. 12A B. $12\sqrt{2}$A C. $6\sqrt{2}$A D. 6A

***Solution***: $900 = \frac{300}{\sqrt{2}} I\cos 60° \Rightarrow I = 6\sqrt{2}\text{A}$ **Answer: C**

136. An AC source is of $\frac{200}{\sqrt{2}}$V, 50 Hz. The value of voltage after $\frac{1}{600}$ second from the start is:

A. 200V B. $\frac{200}{\sqrt{2}}$V C. 100V D. 50V

***Solution***: $V = V_o \sin \omega t = \sqrt{2}\left(\frac{200}{\sqrt{2}}\right)\sin 2\pi \times 50 \times \frac{1}{600} = 200\sin\frac{\pi}{6}$

$V = 100\text{V}$ **Answer: C**

Generated by CamScanner from intsig.com

137. A coil having inductance of $\frac{1}{\pi}$H is connected in series with 300Ω. If 20V from a 200 cycle source is impressed across the combination, the phase angle between V and I is

A. $\tan^{-1}\left(\frac{5}{4}\right)$ C. $\tan^{-1}\left(\frac{3}{4}\right)$

B. $\tan^{-1}\left(\frac{4}{5}\right)$ D. $\tan^{-1}\left(\frac{4}{3}\right)$

*Solution*: $\tan\theta = \frac{L\omega}{R} = \frac{\frac{1}{\pi}\times 2\pi\times 200}{300} = \frac{4}{3} \Rightarrow \theta = \tan^{-1}\left(\frac{4}{3}\right)$

**Answer: D**

138. The magnetic field energy in an inductor changes from maximum value to minimum value in 5ms when connected to an AC source. The frequency of the source is

A. 20Hz B. 50Hz C. 200Hz D. 500Hz

*Solution*: $\frac{T}{4} = 5\times10^{-3} \Rightarrow T = 2\times10^{-2}$ sec

$$\Rightarrow f = \frac{1}{T} = \frac{1}{2\times10^{-2}} = 50\text{Hz}$$

**Answer: B**

139. 5 cm long solenoid having 10Ω resistance and 5mH inductance is joined to a 10V battery. At steady state, the current through the solenoid is

A. 5A B. 1A C. 2A D. Zero

*Solution*: $I = \frac{V}{R} = 1\text{A}$ **Answer: B**

140. An inductive coil has a resistance of 100Ω. When an AC signal of frequency 1000Hz is fed to the coil, the applied voltage leads the current by 45°. What is the inductance of the coil?

A. 10mh B. 12mH C. 16mH D. 20mH

*Solution*:

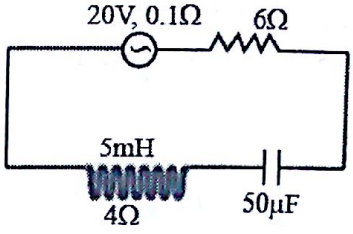
$$X_L = R \Rightarrow 2\pi fL = R \Rightarrow L = \frac{R}{2\pi f} = \frac{100}{2\times3.14\times1000} = 16\text{mH}$$

**Answer: C**

Generated by CamScanner from intsig.com

141. In the circuit shown, the AC soruce has voltage 20 cos (ωt) with ω = 2000 rad/s. The amplitude of the current will be nearest to

A. 2A
B. 3.3A
C. $2/\sqrt{5}$ A
D. $\sqrt{5}$A




***Solution***: $L\omega = (5 \times 10^{-3})(2000) = 10\Omega$

$$\frac{1}{C\omega} = \frac{1}{(50\times10^{-6})(2000)} = 10\Omega$$

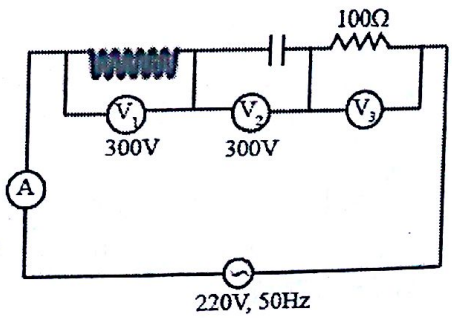
Since, $L\omega = \frac{1}{L\omega} \Rightarrow Z = R = 6 + 0.1 + 4 = 10.1\Omega$

$$I_o = \frac{v}{t} = \frac{20}{10.1} \simeq 2A$$

**Answer: A**

142. In the circuit shown below, what will be the reading of the voltmeter $V_3$ and ammeter A?

A. 800V, 2A
B. 300V, 2A
C. 220V, 2.2A
D. 100V, 2A




***Solution***: Voltage across L and C cancel out. So, the voltage across R is 220V. $I_o = \frac{v}{R} = \frac{220}{100} = 2.2A$ **Answer: C**

143. The reactance of a coil when used in a power supply of 220V 50Hz is 50Ω. The inductance of the coil is

A. 2.2H B. 0.22H C. 1.6H D. 0.16H

***Solution***: $X_L = 2\pi fL \Rightarrow 50 = 2\times\frac{22}{7}\times50\times L \Rightarrow L = 0.16H$

**Answer: D**

144. An rms voltage of 110V is applied across a series circuit having a resistance 11Ω and impedance 22Ω. Power consumed is

A. 275W B. 366W C. 550W D. 1100W

Generated by CamScanner from intsig.com

**Solution:** $P = VI\cos\theta = V\left(\frac{V}{Z}\right)\left(\frac{R}{Z}\right) = \frac{110\times110\times11}{22\times22} = 275w$

145. An inductor, capacitor and resistor are connected in series to an AC supply, the p.d across the inductor, capacitor and resistor are 90V, 60V and 40V respectively. The supply voltage is

**Answer: A**

A. 190V B. 100V C. 130V D. 50V

**Solution:** $V = \sqrt{V_R^2 + (V_L - V_C)^2} = \sqrt{40^2 + (90-60)^2} = 50V$

146. In an LR circuit, an inductance of 0.1H and a resistance of 1Ω are connected in series with an ac source of v = 5 sin 10t. The phase difference between the current and applied voltage will be:

**Answer: D**

A. $\pi/2$ B. $\pi/4$ C. $\pi/6$ D. 0

**Solution:** $\tan\theta = \frac{\omega L}{R} = \frac{(10)(0.1)}{1} = 1 \Rightarrow \theta = \pi/4$

147. The current and voltage in an AC circuit are given by, $I = I_o \sin \omega t$, $V = V_o \sin \omega t$. The power consumed in the circuit is

**Answer: B**

A. $V_oI_o$ B. $\frac{V_oI_o}{2}$ C. $\frac{V_oI_o}{\sqrt{2}}$ D. Zero

**Explanation:** $P = VI\cos\theta$, here $\cos\theta = 1$

$$P = \left(\frac{V_o}{\sqrt{2}}\right)\left(\frac{I_o}{\sqrt{2}}\right) = \frac{V_oI_o}{2}$$

**Answer: B**

148. A generator produces a voltage that is given by V = 240 sin 120t. The frequency and rms voltage are

A. 19Hz and 120V C. 60Hz and 240V
B. 19Hz and 170V D. 754Hz and 170V

**Solution:** $\omega = 2\pi f = 120 \Rightarrow f \simeq 19Hz$ and $V_{rm} = \frac{240}{\sqrt{2}} \simeq 170V$

**Answer: B**

149. An AC circuit consists of an inductor of inductance 0.5H and a capacitor of capacitance 8μF in series. The current in the circuit is maximum when the angular frequency is

A. 5000 rad/s C. 2000rad/s
B. 4000 rad/s D. 500 rad/s

Generated by CamScanner from intsig.com

***Solution***: $\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(0.5)(8\times10^{-6})}} = 500\text{rad/s}$ **Answer: D**

150. In a series LCR circuit, R = 10Ω and impedance Z = 20Ω. The phase angle between current and voltage is

A. 45° B. 90° C. 60° D. 30°

***Solution***: $\cos\theta = R/Z = 10/20 = \frac{1}{2} \Rightarrow \theta = 60°$ **Answer: C**

151. An rms current of 0.6A flows when an inductance of 40mH is connected to an AC source of 0.24V. The angular frequency of the source is

A. 1 rad/s B. 2 rad/s C. 5 rad/s D. 1000 rad/s

***Solution***: $V = IX_L = L\omega I$

$$\Rightarrow \omega = \frac{V}{LI} = \frac{24}{(0.6)(40\times10^{-3})} = 1000\text{rad/s}$$ **Answer: D**

152. In an AC source, the current varies according to I = 5 sin (100πt). What is the mean power developed by the current in a load resistance of 10Ω?

A. 125w B. 160 w C. 250 w D. 500 w

***Solution***: $I_{rms} = \frac{I_o}{\sqrt{2}} = \frac{5}{\sqrt{2}}$

$$P = (I_{rms})^2 R = \left(\frac{5}{\sqrt{2}}\right)^2 (10) = 125\text{w}$$ **Answer: A**

153. The voltage of an AC supply is V = 120 sin 100πt cos 100πt. The maximum voltage and frequency are

A. 120V, 100Hz C. 60V, 200Hz

B. 100V, 120Hz D. 60V, 100Hz

***Solution***: $V = \frac{120}{2}[2\sin 100\pi t\cos 100\pi t]$ or V = 60 sin 200πt

comparing with $V = V_o \sin\omega t$

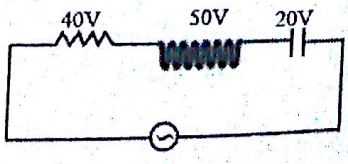
$\Rightarrow V_o = 60\text{V}$ and $\omega = 2\pi f = 200\pi \Rightarrow 100$ Hz

**Answer: D**

154. For the LCR circuit shown, what is the applied source voltage?

A. 10V B. 50V

C. 70V D. 110V




Generated by CamScanner from intsig.com

**Solution:** $V^2 = V_R^2 + (V_L - V_C)^2 = (40)^2 + (50-20)^2 \Rightarrow V = 50V$

**Answer: B**

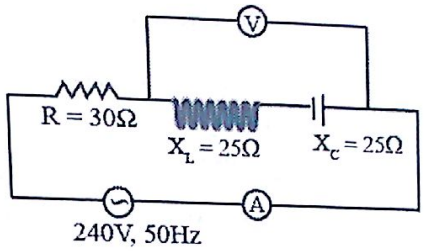
155. In a series LCR circuit the voltage across, R, C and L is 10V each. The value of applied emf is

A. 30V B. 20V C. $10\sqrt{3}$V D. 10V

**Solution:** $V^2 = V_R^2 + (V_L - V_C)^2 = 10^2 + 0 = 10V$ **Answer: D**

156. In the circuit shown, the reading of the voltmeter and ammeter will be

A. 150V, 3A
B. 0V, 3A
C. 0V, 8A
D. 150V, 8A




**Solution:** Since $X_L = X_C$, the voltmeter reading is zero. Also, the circuit is a resonant circuit. It behaves at a purely resistive circuit.

$$I = \frac{V}{R} = \frac{240}{30} = 8A$$

**Answer: C**

157. The self–inductance of a motor is 10H. In order to impart maximum power at 50Hz, it should be connected to a capacitance of

A. 8μF B. 4μF C. 2μF D. 1μF

**Solution:** For $P_{max}$, power, input impudence = output impudence

$$X_L = X_C \Rightarrow \omega L = \frac{1}{\omega C} \Rightarrow C = \frac{1}{\omega^2 L} = \frac{1}{(2\pi \times 50)^2 \times 10} = 1\mu F$$

**Answer: D**

158. The phase difference between the alternating current and the emf is $\pi/2$. Which of the following cannot be the constituent of the circuit?

A. R, L B. C C. L D. L, C

**Answer: A**

159. An alternating current circuit consists of an inductance and a resistance in series. In this circuit

A. The p.d and the current in the resistance leads the p.d of the inductance by angle of $\pi/2$.

Generated by CamScanner from intsig.com

B. The p.d and the current in the resistance lags behind the p.d across the inductance by $\pi/2$.

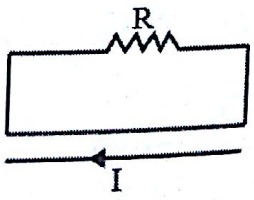
C. The p.d and the current in the resistance lags behind the p.d across the inductance by $\pi$.

D. The p.d across the resistance leads the p.d across the inductance by $\pi/2$.

**Answer: A**

## Unit 15: Conceptual Questions

1. Why is the induced emf that appears in an inductor called a back or counter emf?
2. When a DC voltage is applied to a transformer, the primary coil sometimes overheat and burn. Why?
3. What is the advantage of transmitting power at high voltages?
4. Will a transformer operate if a battery is used for the input voltage across the primary? Explain
5. An inductor is attached to an a.c source. What change will make the current to be halfed.
6. A series RLC circuit has $V_C = 5V$, $V_R = 7V$ and $V_L = 9V$. What can you say about the frequency of oscillation?
7. What will happen to the voltage of a RLC circuit at resonance frequency?
8. In a series RLC circuit, what determines whether the inductive or capacitive behavior dominates?
9. What is the phase angle when the inductive reactance equals the capacitive reactance?
10. What is the impedance of an RLC circuit at the resonance frequency?
11. What is the direction of the current induced in the resistor, when the current in the wire is zero?




## Unit 15: Problems

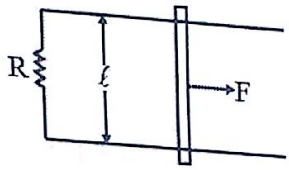
### Magnetic Flux

12. A 10 cm diameter circular loop rotating in a uniform 0.4T magnetic field. What is the magnetic flux through the loop when
A. $\theta = 0°$ B. $\theta = 30°$ C. $\theta = 60°$ D. $\theta = 90°$
13. a solenoid 4 cm in diameter and 20 cm long has 250 turns and carries a current of 15A. Calculate the magnetic flux through the circular cross – sectional area of the solenoid.

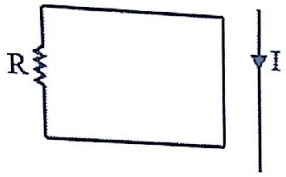
Generated by CamScanner from intsig.com

**Faraday's Law of Induction**

14. The magnetic flux through a coil of 200 turns change from 0.25Wb to zero in 0.04 sec. Calculate the emf induced in the coil.
15. Calculate the induced emf in 20 $cm^2$ circular coil having 80 turns if the magnetic field strength passing through the coil changes from 0.003T to 0.004T in 0.2 sec.
16. A square, single turn coil 0.2m on a side is placed with its plane perpendicular to a constant magnetic field. An emf of 18mV is induced in the coil winding when the area of the coil decreases at the rate of 0.1 $m^2/s$. What is the magnitude of the magnetic field?
17. Consider the arrangement shown. Assume that $R = 6\Omega$ and $\ell = 1.2m$ and that a uniform 2.5T magnetic field is directed into the page. At what speed should the bar be moved to produce a current of 0.5A in the resistor?




18. The following system shows a current carrying wire and a circuit containing a resistor. What is the direction of the induced current in the circuit?




**Self – Inductance**

19. An emf of 12V is induced in a coil when the current in it changes at a rate of 8 A/S. Calculate the inductance of the coil.
20. A steady current of 3A produces a magnetic flux of 0.006Wb in a coil of 400 turn. Calculate the inductance.
21. Find the induced emf when the current in a 50mH inductor increases from 0 to 0.02A in 0.04 sec.

**Stored Magnetic Energy**

22. How much energy is stored in 80mH inductor at an instant when the current is 3A?
23. The magnetic field inside a solenoid is 9T. What is the magnetic energy density in the field?

**Transformers**

24. The primary coil on a transformer has 1200 turns. If it is desired to step down the voltage from 220 volts to 110 volts. How many turns should the secondary coil have?
25. A transformer on a pole near a factory steps the voltage down from 3600V to 120V. The transformer is to deliver 1000kw to the factory at 90% efficiency. Find
    a) Power delivered to the primary.

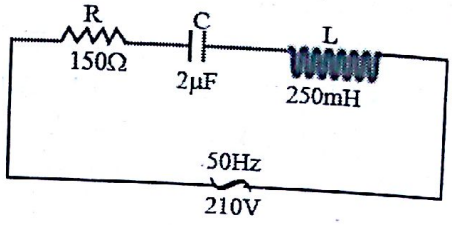
Generated by CamScanner from intsig.com

b) Current in primary

**The RLC Circuits**

26. At what frequency will the inductive reactance of 40mH equal the capacitance reactance of 80μF.

27. A series RLC circuit of R = 150Ω, L = 250mH, C = 2μF is connected across 210V operating at 50Hz. Calculate the
    a) Inductive reactance
    b) Capacitive reactance
    c) Impedance
    d) Current
    e) Phase angle
    f) Power loss




## Unit 15: Explanation to Questions

1. Because, if a battery is acting to increase the current, in the inductor, an emf is induced to oppose the increase in current. If the current is reduced, the emf tends to increase the current.
2. There is no back emf and hence the current rises to higher value.
3. It is economical since the loss $I^2R$ is low at high voltage. At high voltage, it is possible to use step –down transformers to make low voltage and high current to the end users.
4. No. A voltage is induced in the secondary only if the flux through the core changes with time.
5. $I = \frac{V}{X_L} = \frac{V}{2\pi fL}$. Thus, doubling the frequency will half the current.
6. Since $V_L > V_C \Rightarrow X_L > X_C$, this is the condition above resonance for $X_L$ increases with frequency while $X_C$ decreases.
7. At resonance, $X_L = X_C \Rightarrow V_L = V_C$ and they cancel out. The net voltage is the voltage across the resistor. Hence, the voltage is in phase with the current.
8. Frequency. At high frequency, for instance, the inductive reactance is greater than the capacitive reactance.
9. $\tan\theta = \frac{X_L - X_C}{R} = \frac{0}{R} = 0 \Rightarrow \theta = 0°$
10. $Z = \sqrt{R^2 + (X_L - X_C)^2}$. At resonance, $X_L = X_C \Rightarrow Z = R$
11. From left to right.

DIGITAL LIBRARY FOR ETHIOPIAN STUDENTS
Digital Library for Ethiopian students
Link: https://telegram.me//digitallibrary4ethiopianstudents or digitallibrary4ethiopianstudents
YouTube Digital[0,1] Ethiopia
Generated by CamScanner from intsig.com

## Unit 15: Solution to problems

12. $\phi = AB\cos\theta,\ A = \pi\frac{d^2}{4} = \pi\frac{(10\times10^{-2})}{4} = 7.85\times10^{-3}\,m^2$

a) $\phi = (7.85\times10^{-3})(0.4)\cos 0° = 3.14\times10^{-3}Wb$
b) $\phi = (7.85\times10^{-3})(0.4)\cos 30° = 2.72\times10^{-3}Wb$
c) $\phi = (7.85\times10^{-3})(0.4)\cos 60° = 1.57\times10^{-3}Wb$
d) $\phi = 0$

13. $\phi = BA = \left(\frac{\mu_o IN}{\ell}\right)(\pi r^2) = 4\pi\times10^{-7}(15)\left(\frac{250}{0.2}\right)\pi\times4\times10^{-2} = 0.003Wb$

14. $\varepsilon = N\left|\frac{\Delta\phi}{\Delta t}\right| = 200\left(\frac{0.25}{0.04}\right) = 1250V$

15. $\varepsilon = -N\frac{\Delta\phi}{\Delta t} = -N\frac{\Delta BA}{\Delta t} = -80\left(\frac{0.004-0.003}{0.2}\right)20\times10^{-4} = -0.008V$

16. $\varepsilon = N\frac{\Delta\phi}{\Delta t} = \frac{N\Delta BA}{\Delta t} \Rightarrow B = \frac{\varepsilon}{\Delta A/\Delta t} = \frac{18\times10^{-3}}{0.1} = 0.18T$

17. $\varepsilon = BV\ell \Rightarrow IR = B\ell V \Rightarrow V = \frac{IR}{B\ell} = \frac{(0.5)(6)}{(2.5)(1.2)} = 1m/s$

18. Counter clockwise

19. $\varepsilon = L\left|\frac{\Delta I}{\Delta t}\right| \Rightarrow 12 = L(8) \Rightarrow L = 1.5H$

20. $L = \frac{N\phi}{I} \Rightarrow L = \frac{(400)(0.06)}{3} = 0.8H$

21. $\varepsilon = L\left|\frac{\Delta I}{\Delta t}\right| = 50\times10^{-3}\left|\frac{0.02}{0.04}\right| = 25V$

22. $PE = \frac{1}{2}LI^2 = \frac{1}{2}(80\times10^{-3})(9) = 0.36J$

23. $U = \frac{B^2}{2\mu_o} = \frac{9^2}{2\times4\pi\times10^{-7}} = 0.32\mu J/m^3$

24. $\frac{V_P}{V_S} = \frac{N_P}{N_S} \Rightarrow N_S = \frac{V_S}{V_P}(N_P) = 1200\left(\frac{110}{220}\right) = 600$ turns

25. a) $\eta = \frac{P_{out}}{P_{in}} = \frac{V_S I_S}{V_P I_P} \Rightarrow P_{in} = \frac{P_{out}}{\eta} = \frac{1000kw}{0.9} = 1111kw$

Generated by CamScanner from intsig.com

b) $P_{in} = V_P I_P \Rightarrow I_P = \frac{P_{in}}{V_P} = \frac{1111000}{3600} \simeq 309A$

26. $f = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(40\times10^{-3})(80\times10^{-6})}} = 89Hz$

27. a) $X_L = 2\pi fL = 2\pi\,(50)(250\times10^{-3}) = 78.5\Omega$

b) $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(50)(2\times10^{-6})} = 1592\Omega$

c) $Z = \sqrt{R^2 + (X_L + X_C)^2} = \sqrt{(150)^2 + (1592 - 78.5)^2} = 1521\Omega$

d) $I = \frac{V}{Z} = \frac{210}{1521} = 0.14A$

e) $\tan\theta = \frac{X_L - X_C}{R} = \frac{78.5 - 1592}{150} = -10.09 \Rightarrow \theta = -84°$

f) $P = V_{rms} I_{rms} \cos\theta = \left(\frac{210}{\sqrt{2}}\right)\left(\frac{0.14}{\sqrt{2}}\right)\left(\frac{150}{1521}\right) = 1.4W$

Generated by CamScanner from intsig.com

# Unit 16

## Atomic Physics

We along with everything else on Earth, are made of atoms. Moreover, the study of atomic and nuclear physics plays a significant role in our everyday life. For example, laser is used in eye surgery, production of three dimensional images of an object, precise cutting of a metal, telephone communication via optical fiber. Gamma knife radiosurgery is also becoming a promising way of treating the brain and killing cancer without physical surgery. Nuclear fusion and fission are becoming enormous sources of energy. Radioactive dating is used in the determination of the age of archeological or geological samples.

### 16.1 Blackbody Radiation

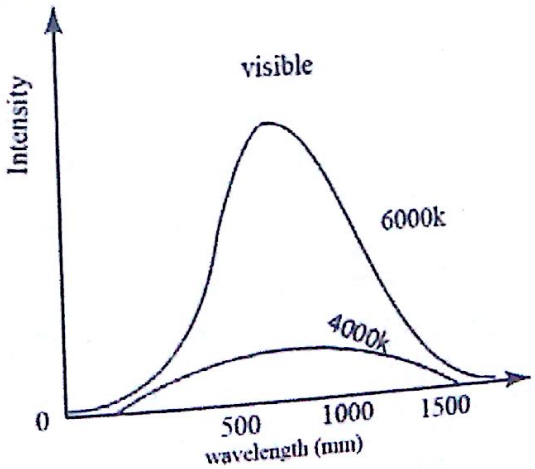
All bodies, no matter how hot or cold, continuously radiate electromagnetic waves. For instance, we see the glow of our sun which has a surface temperature of about 6000k. Certainly the human body which is about 310k does not emit enough visible light to be seen in the dark, but the body does emit electromagnetic wave in the infrared region of the spectrum. Our eyes can only see electromagnetic radiation in the visible part of the spectrum.

An ideal surface for emitting light with a continuous spectrum is one that also absorbs all wavelengths of electromagnetic radiation such an ideal surface is called a *blackbody*.

**Blackbody:** Is a body that is a perfect radiator and absorber of electromagnetic energy

A good approximation to a blackbody is a hollow box with a small apeture in one wall.

The distribution of energy, that is intensity in a blackbody depends on temperature. At higher temperature, the intensity increases and shifts to shorter wavelength, that is towards the visible region of the spectrum.




Generated by CamScanner from intsig.com

**Important Points to remember**

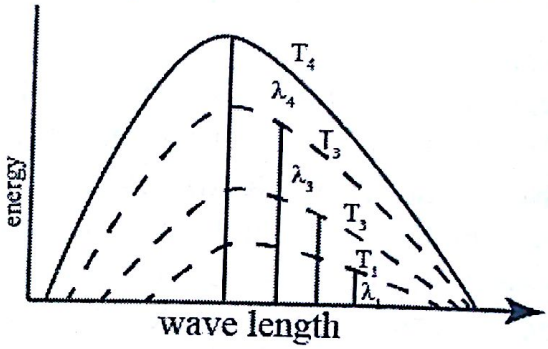
- An ideal blackbody absorbs all the light that is incident on it.
- The distribution of energy in a blackbody is independent of the material from which the blackbody is made, it depends only on the temperature.

**Conceptual Example**

1. Is a blackbody black? Explain

   ***Explanation:*** No. not necessarily. a blackbody is called "black" because, it appears black when illuminated. It do not reflect light at all. Hence, the material from which it is made need not be black at all to carry the name blackbody as long as it functions as above.

2. The energy spectrum emitted by a black body at different temperature is as shown in the figure below. Which of the following statements is NOT correct about the black body energy spectrum? (UEE: 2005)




   A. $T_1 < T_2 < T_3 < T_4$
   B. $\lambda_1 > \lambda_2 > \lambda_3 > \lambda_4$
   C. Blackbody radiate energy over the entire electromagnetic spectrum
   D. The wavelength corresponding to the maximum intensity increases with temperature

**Answer: D**

## 16.2 The Photoelectric Effect

In 1883 Thomas Edison has discovered thermionic emission in which electrons are emitted when a metal is heated. And in 1887 the German physicist Heinrich Hertz first discovered the emission of electrons from a metal when light is shone on it. The emitted electrons are called *photoelectrons* and the process of emission of photoelectrons is called *photoelectric effect.*

**Photoelectric Effect:** Is the emission of electrons from a metal surface when light is shone on the surface.

In 1905 Albert Einstein proposed that light is a collection of discrete packets or bundles of energy called *photons* or *quanta*.

**Photon:** Is a packet or bundle of energy that constitute light

Generated by CamScanner from intsig.com

The energy of a photon of light of frequency f is given by

$\boxed{E = hf}$ ← Energy of a photon

Where: h = planck's constant with a value $h = 63\times10^{-34}$ J

f = frequency of light

For instance, the light energy given off by a light bulb is carried by photons. The brighter the bulb, the greater is the number of photons emitted per second.

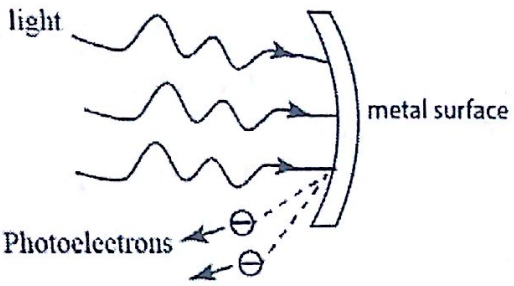
From the relation of speed of radiation, frequency and wavelength, $c = f\lambda$ we have.

$f = \frac{c}{\lambda}$ substitute this in E = hf

$$E = h\left(\frac{c}{\lambda}\right) \Rightarrow \boxed{E = \frac{hc}{\lambda}}, \quad c = 3\times10^{8}\, m/s$$

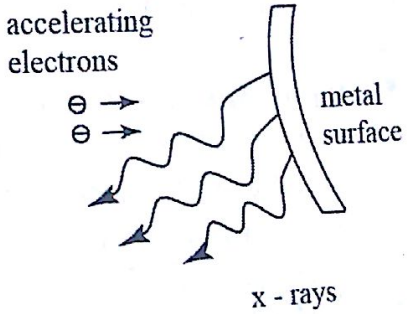
## Conceptual Example

3. Can we say that $x$ – rays are photoelectrons? Explain




***Explanation***: No, $x$ – rays and photoelectrons are produced in opposite manners.

Photoelectrons are produced when light, specially ultraviolet light strikes a metal surface. However.




$x$ – rays are produced when high – speed electrons are suddenly slowed down such as when a metal target is struck by electrons that have been accelerated

According to Einstein, when light shines on a metal a photon give its energy to an electron in the metal. Thus, a minimum energy is needed by the electron to escape. That means work must be done to free and pull out the electron from the metal surface. And the work needed is called *work function*.

**Work function**: Is the minimum amount of energy necessary to eject an electron from a metal surface.

Generated by CamScanner from intsig.com

If a photon has energy in excess of the work needed to remove an electron, the excess appears as kinetic energy of the ejected electron. Thus, the least strongly held electrons are ejected with the maximum kinetic energy $k_{E\ max}$.

Einstein applied the conservation of energy principle and proposed the following relation to describe the photoelectric effect.

Photon energy = Maximum kinetic energy of electron + Work function

$$\Rightarrow \boxed{hf = KE_{max} + \emptyset}$$

Where: $\emptyset$ (phi) is called the work function. To escape from the metal, the electron must overcome the barrier or stoping potential $v_o$. Thus, the most energetic electron would have a kinetic energy of.

$\boxed{KE_{max} = ev_o}$, Therefore

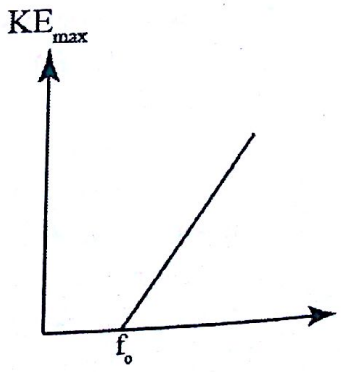
$$\boxed{hf = ev_o + \emptyset}$$

**Stopping potential, $v_o$:** Is the voltage required to stop ejected photoelectrons from reaching the anode.

Photons can eject electrons from a metal when the light frequency is above a minimum, value, $f_o$ called *cutoff* or *threshold frequency*.

**Cutoff or threshold frequency:** Is the minimum light frequency that can eject an electron, from a metal.

For frequencies above the cutoff frequency, ejected electrons have a maximum kinetic energy, $KE_{max}$ that is linearly related to the frequency




**Important points to remember**

- To eject electrons, the incident light must have a frequency greater than a certain minimum value called the cutoff frequency. If the frequency of the light is less than the cutoff frequency, no electron is emitted, no matter how intense the beam may be.
- The maximum kinetic energy of the photoelectrons is independent of the light intensity. As the light intensity increases more photons per second strike the metal, and consequently more electrons per second are ejected.
- The maximum kinetic energy of the photoelectrons increase with increasing light frequency.

Generated by CamScanner from intsig.com

- At the cutoff frequency, the electron departs from the metal with no kinetic energy
- The energy of a photon is usually expressed in electron volt. $1 ev = 1.6 \times 10^{-19} J$

## Conceptual Example

4. Why do the kinetic energy of photoelectrons is independent of the light intensity.

***Explanation:*** It is because, the light energy is concentrated in packets rather than sepread out in waves. If the frequency is high enough, no time is needed for the electron to acquire sufficient energy to escape the metal.

## Illustrative Example

5. Find the energy of one photon of visible light of frequency $6.3 \times 10^{14}$ Hz

***Solution***: $E = hf = (6.63 \times {}^{-34})(6.3 \times 10^{14}) = 4.2 \times 10^{-19} J$

6. Find the energy of one photon of visible light of wavelength 670 nm.

***Solution***: $E = hf = h\left(\frac{c}{\lambda}\right) = \left(6.63\times10^{-34}\right)\left(\frac{3\times10^{8}}{670\times10^{-9}}\right) = 3\times10^{-19} J$

7. The work function for potassium is 2.24ev. If potassium is illuminated with light of wavelength 480nm, find the maximum kinetic energy of the photoelectrons.

***Solution***: $E = hf = h\left(\frac{c}{\lambda}\right) = 6.63\times10^{-34}\left(\frac{3\times10^{8}}{480\times10^{-9}}\right)$

$E = 4.14 \times 10^{-19} J$. In electron volt. We have

$\frac{4.14\times10^{-19}}{1.6\times10^{-19}} = 2.59 \text{ ev}$

$E = K_{Emax} + \emptyset \Longrightarrow K_E = E - \emptyset = 2.59 \text{ ev} - 2.24 \text{ ev} = 0.35 \text{ev}$

8. A photon of energy 5ev impart all its energy to an electron that leaves a metal surface with 2.1ev of kinetic energy. What is the work function of the metal?

***Solution:*** $E = K_{Emax} + \emptyset \Longrightarrow \emptyset = E - K_E = 5ev - 2.1 \text{ ev}$

$\emptyset = 2.9 \text{ ev}$

9. A key phenomenon that demonstrates the particle nature of light is: (UEE: 2003)

A. Photoelectric effect
B. Doppler effect
C. Dispersion of light
D. Interference

**Answer: A**

Generated by CamScanner from intsig.com

10. Which of the following is correct about the meaning of photon? (UEE: 2005)
    A. quantum of matter
    B. positively changed particle
    C. negatively changed particle
    D. quantum of light **Answer: D**

11. The intensity of a beam of light is increased but the light's frequency is unchanged. Then which of the following is true?
    A. The photons travel faster
    B. Each photon has more energy
    C. The photons are larger
    D. There are more photons per second **Answer: D**

12. Sodium and copper have work function of 2.3ev and 4.5ev respectively. The ratio of their wave length is
    A. 4: 1 B. 1: 4 C. 1: 2 D. 2: 1

    ***Explanation:*** $\varnothing = hf_o = h\left(\frac{c}{\lambda o}\right) \Rightarrow \lambda_o = \frac{hc}{\varnothing}$

    $$\frac{\text{wave length of sodium}}{\text{wave length of copper}} = \frac{\frac{hc}{\varnothing \text{sodium}}}{\frac{hc}{\varnothing \text{copper}}} = \frac{\varnothing \text{copper}}{\varnothing \text{sodium}} = \frac{4.5ev}{2.3ev} = 2.1$$

    **Answer: D**

13. Photon of energy 8ev is incident on a photo sensitive surface of threshold frequency $1.6\times1015_{HZ}$. Then the kinetic energy of the photoelectron is
    A. 6ev B. 1.4ev C. 1.2 ev D. 2ev

    ***Solution:*** $E = K_E + \varnothing \Longrightarrow KE = E - \varnothing$ but $\varnothing = hf_o$

    $\varnothing = hf_o = (6.63 \times 10^{-34})\,(1.6\times10^{15}) = 10.6 \times 10^{-19}J = 6.6ev = 1.4$ ev

    **Answer: B**

## 16.3 Wave – Particle Duality of Matter

In 1924, Louis de Broglie suggested that an atom's electrons, protons and neutrons possess both particle and wave properties. This is known as the *wave – particle duality of matter*.

☛ **Important Points to remember**

- The photoelectric effect provides evidence that electromagnetic waves and thus light considered as a stream of particles called *photons*.

Generated by CamScanner from intsig.com

- Interference or diffraction demonstrated by electrons prove that particles can act as waves.

## The De Broglie Wavelength

As a graduate student, in 1923, Louis de Broglie of France made the astonishing suggestion that since light waves could exhibit particle – like behavior. De Broglie proposed that all moving matter has a wavelength associated with it, just as a wave does. Which is given by

$$\lambda = \frac{h}{p}$$ ← De Broglie wavelength

Where: $\lambda$ = wave length of a particle
h = planck's constant
p = momentum of the particle

### Illustrative Example

14. Determine the de Broglie wavelength for an electron moving at a speed of 6×106m/s

*Solution*: $\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{6.63\times10^{-34}}{(9.1\times10^{-31})(6\times10^{6})} = 1.2\times10^{-10}\,m$

15. Which of the following phenomena most clearly demonstrates the wave nature of electrons (UEE: 2007)
    A. The photoelectric effect
    B. Blackbody radiation
    C. The Compton Effect
    D. Diffraction of electrons

**Answer: D**

## 16.4 Heisenberg's Uncertainty Principle

This principle is developed by the German Physicist Werner Heisenberg in 1972, and is stated as:

> It is impossible to know both the exact position and the exact velocity and therefore exact momentum of a particle at the same time.

In order to observe any particle. Light must be reflected off it. For example, to find the position and momentum of an electron, at least one photon of light must be used. The photon must hit the electron and then be reflected back to the eye. But, since the photon is massive than the electron it kicks off the electron and changes its position and speed and hence its momentum.

The change in momentum is uncertain and is in the same order as the magnitude of the photon momentum.

Generated by CamScanner from intsig.com

$$\boxed{\Delta p = \frac{h}{\lambda}}$$

And the position of the particle is uncertain at least one photon wavelength.

$$\boxed{\Delta x \geq \lambda}$$

The uncertainty in position or momentum is therefore

$$\boxed{\Delta p \, \Delta x \geq h}$$

**Illustrative Example**

16. Light of wavelength $3\times10^{-6}$m is used to measure the position of an electron of wavelength $1.73\times10^{-11}$m. Find the uncertainty in position of the electron.

***Solution***: $\Delta p = \frac{h}{\lambda} = \frac{6.63\times10^{-34}}{1.73\times10^{-11}} = 3.83\times10^{-23}$kg m/s

$$\Delta p \Delta \geq h$$

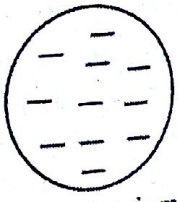
$$\Rightarrow \Delta x \geq \frac{h}{\Delta p} = \frac{6.63\times10^{-34}}{3.83\times10^{-23}}$$

$\Delta x \geq 1.7\times10^{-11}$m ← at least the wavelength

☛**Note that**: To increase the accuracy of position a photon with smaller wavelength must be used. But, decreasing the wavelength increases the frequency and thus the energy of the photon, therefore, when the photon collides with the electron, it will change the momentum of the electron by a larger amount, so if the position is more accurately determined, the momentum is less accurately determined. The opposite is also true – if the momentum is more accurately determined, then the position must be less accurately determined

## 16.5 Early Models of the Atom

In 1897, the English physicist J.J. Thomson discovered a "particle" called the *electron*, smaller in size and negatively charged. Thomson proposed that:



Thomson Atomic model

The main part of the atom was positively changed and had negatively changed electrons embedded or scattered through it.

In 1911, the New Zealand physicist Ernest Rutherford proposed the nuclear model of the atom.

Generated by CamScanner from intsig.com

$$\boxed{\Delta p = \frac{h}{\lambda}}$$

And the position of the particle is uncertain at least one photon wavelength.

$$\boxed{\Delta x \geq \lambda}$$

The uncertainty in position or momentum is therefore

$$\boxed{\Delta p\, \Delta x \geq h}$$

**Illustrative Example**

16. Light of wavelength $3\times10^{-6}$m is used to measure the position of an electron of wavelength $1.73\times10^{-11}$m. Find the uncertainty in position of the electron.

***Solution***: $\Delta p = \frac{h}{\lambda} = \frac{6.63\times10^{-34}}{1.73\times10^{-11}} = 3.83\times10^{-23}$kg m/s

$$\Delta p \Delta \geq h$$

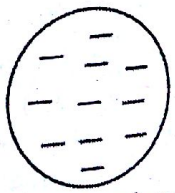
$$\Rightarrow \Delta x \geq \frac{h}{\Delta p} = \frac{6.63\times10^{-34}}{3.83\times10^{-23}}$$

$\Delta x \geq 1.7\times10^{-11}$m ← at least the wavelength

☛**Note that**: To increase the accuracy of position a photon with smaller wavelength must be used. But, decreasing the wavelength increases the frequency and thus the energy of the photon, therefore, when the photon collides with the electron, it will change the momentum of the electron by a larger amount, so if the position is more accurately determined, the momentum is less accurately determined. The opposite is also true – if the momentum is more accurately determined, then the position must be less accurately determined

## 16.5 Early Models of the Atom

In 1897, the English physicist J.J. Thomson discovered a "particle" called the *electron*, smaller in size and negatively charged. Thomson proposed that:



Thomson Atomic model

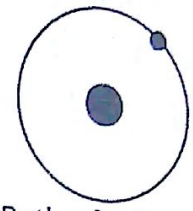
The main part of the atom was positively changed and had negatively changed electrons embedded or scattered through it.

In 1911, the New Zealand physicist Ernest Rutherford proposed the nuclear model of the atom.

Generated by CamScanner from intsig.com

According to Rutherford

All the positive changed and most of the mass of the atom is concentrated in a tiny central nucleus. Most of the atom is empty space, and electrons orbit at the edge.



Rutherford model

In 1913, a Danish physicist Niels Bohr proposed that:

Specific energy levels existed with in an atom's structure and that electrons move in circular orbits around the nucleus.

In his model, electrons that are closer to the nucleus have a lower energy state than those that are further away. It is possible for electrons to exist in different states. But, as they move from a higher energy level $E_1$ to a lower energy level $E_2$, they emit radiation. The energy of a quantum of this radiation is given by the equation.

$$E = hf = E_1 - E_2$$

Electrons can only exist in specific energy states and will not be found with energies between these states.

Bohr's model of the atom allows line spectra to be explained. Only certain frequencies are present in line spectra and each element has a unique pattern. Such spectra are produced by hot gases where atoms are far apart.

**Illustrative Example**

17. Two lines in the spectrum of a sodium lamp has wavelengths of 589nm and 589.6m. Calculate
    a. the frequency
    b. the energy change in electro volt

***Solution :*** a. $F_1 = \frac{C}{\lambda} = \frac{3\times10^8}{589\times10^{-9}} = 5.09\times10^{14}HZ$ $f_2 = 5.088\times10^{14}Hz$

b. For $\lambda_1 = 589 \times 10^{-9}m$, $E_1 - E_2 = hf = 6.63 \times10^{-34} \times 5.09\times10^{14}$

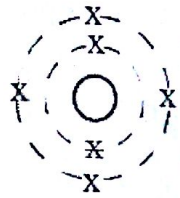
$\Delta E_1 = 3.375 \times 10^{-19} = 2.109$ ev

## 16.6 The Model of the Atom used Now

The model of the atom that we use now was proposed by Heisenberg in 1920. In Heisenberg Model:

DIGITAL LIBRARY FOR ETHIOPIAN STUDENTS
Digital Library for Ethiopian students
Link: https://telegram.me//digitallibrary4ethiopianstudents or digitallibrary4ethiopianstudents
YouTube Digital[0,1] Ethiopia
Generated by CamScanner from intsig.com

There is a central nucleus around which there are regions in which there is a high probability of finding an electron and the shape of these "probability clouds" represent the electron orbitals.



This model is represented in a simplified diagram called a *dot* and *cross* diagram. The crosses represent electrons and circles show the shells. The nucleus is shown as a circle in the middle. Now, we know that there are two types of particles in the *nuleons*. Protons, which are positively charged and neutrons which do not have any charge. The collective name of these particles is called *nucleus*. The number of protons in the nucleus determines what element the atom is. For example, carbon atoms have a different number of protons from nitrogen atoms. The number of protons in an uncharged atom is the same as the number of electrons.

☛ The elements in the periodic table are listed in order of proton number, which is also called the *atomic number* and is often given the symbol Z.

The number of neutrons must be equal or greater than the number of protons, but there are some elements that have the same number of protons but different number of neutron. For example, carbon can have atoms with 6 neutrons, atoms with 7 neutrons and atoms with 8 neutrons but all have 6 protons. Such atoms are called *isotopes*.

**Isotopes**: Are atoms of the same element that have different number of neutrons but same protons.

To describe atoms of elements, we combine the chemical symbol of the element with the atomic number Z as a subscript and the mass number, A which is the total number of protons and neutrons as a superscript.

$$^{A}_{Z}X$$

Where X = chemical symbol of the element
Z = Atomic number = number of protons
A = mass number

Mass number = number of protons + number of neutrons
A = Z + N

For instance, the chemical symbol of Aluminum is Al its atomic number is Z = 13 and its mass number is A = 27. Hence we can write it in the form of the above notation as: $^{27}_{13}Al$.

Generated by CamScanner from intsig.com

## 16.7 The Strong Nuclear Force

Gravity, electromagnetic, weak nuclear force and strong nuclear force are the four basic forces in nature, basic in a sense that all forces in nature can be explained in terms of these four.

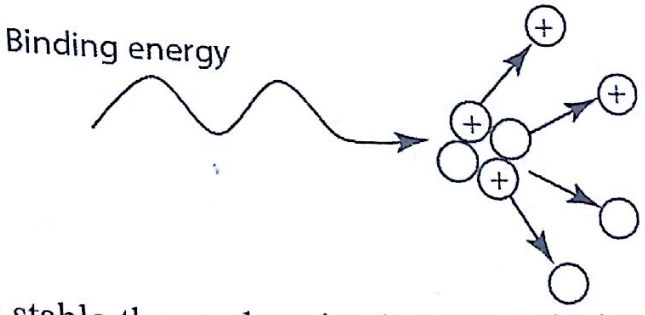
The protons and neutrons in the nucleus of an atom are held together by the strong nuclear force. It is the strongest of the four forces and strong enough to overcome the repulsion between the positive charges on protons. It is created between nucleons by the exchange of particles called *mesons*.

- The range of action of the strong nuclear force is extremely short. The nucleons must be close together about the diameter of a proton or a neutron which is about $10^{-15}$m for meson exchange to happen.

### Nuclear Binding Energy and the Mass Defect

Because of the strong nuclear force, the nucleon in a stable nucleus are held tightly together. Therefore, energy is required to separate a stable nucleus into its constituent protons and neutrons. The required energy is called the *binding energy* of the nucleus.

**Binding Energy:** Is the energy required to separate a nucleus into its constituent nucleons, that is neutrons and protons.




- The more stable the nucleus is, the greater is the amount of energy needed to break it apart.

In Einstein's theory of special relativity, energy and mass are equivalent. The rest energy $E_o$ and the mass m are related via

$$E_o = mc^2$$

Where c is the speed of light in Vacuum. Therefore, a change in the rest energy, $\Delta E_o$ is equivalent to a change in mass $\Delta m$. The binding energy is thus appears as extra mass. In other words, the sum of the individual masses of the separated protons and neutrons is greater by an amount $\Delta m$

Generated by CamScanner from intsig.com

than the mass of the stable nucleus. The difference in mass Δm is known as the *mass defect* of the nucleus.

**Mass defect**: Is the difference between the rest mass of a nucleus and the sum of the rest masses of its constituent nucleons.

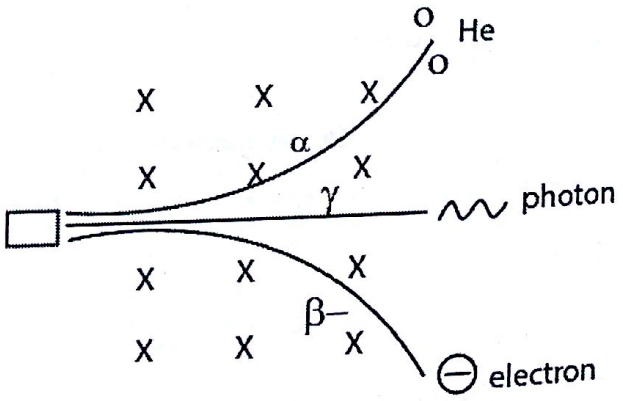
The binding energy of a nucleus can be determined from the mass defect according to

Binding energy = (Mass defect) $c^2$ = (Δm) $c^2$

## 16.8 Radioactivity

Radioactivity was discovered by the French physicist Antoine Henri Becquerel in 1896. The strong nuclear force holds the nucleons tight in the nucleus, overcoming the coulomb repulsion of protons. The balance of force is not always maintained, however, and sometimes particles or photons are emitted from the nuclei of atoms. Such unstable nuclei like uranium and radium are said to be *radioactive*.

When an unstable or radioactive nucleus disintegrates spontaneously, certain kind of particles or high energy photons are released. These particles and photons are collectively called "*rays*" Three kinds of rays are produced by naturally occurring radioactivity:




α rays, β rays and γ rays. They are named according to the first three letters of the Greek alphabet, alpha (α), beta (β), and gamma (γ), to indicate the extent of their ability to penetrate matte. α ray are the least penetrating, being blocked by about 0.01mm sheet of lead, whereas β rays penetrate lead to 0.1mm. γ rays are most penetrating and can pass through about 100mm thick of lead.

α and β rays are deflected by a magnetic field and hence consist of moving changed particles r rays are not deflected by a magnetic field and, thus are not charged.

**Alpha Decay** → emission of Helium $\left({}^{4}_{2}\mathrm{He}\right)$

When a nucleus disintegrates and produced α rays, it said to undergo α decay.

Generated by CamScanner from intsig.com

Experimental evidence shows that α rays are positively charged particles of helium nucleus. Thus, an α particle has a charge of + 2e.

- In the decaying process, the original nucleus is referred to as the *parent nucleus* (p), and the nucleus remaining after disintegration is called the *daughter nucleus* (D).

When a nucleus decays by giving off an α particle $^{4}_{2}He$, it loses two protons and two neutrons. As a result, its atomic number Z, decreases by 2, and its mass number decreases by 4. Symbolically, we can write this as.

**α Decay**

parent nucleus → Daughter nucleus + α particle (helium nucleus)

$$^{A}_{Z}P \rightarrow ^{A-4}_{Z-2}D + ^{4}_{2}He + \text{energy}$$

For example, the uranium parent is converted into the thorium daughter. The parent and daughter nuclei are different, so α decay converts one element into another, a process known as *transmutation*

$$^{238}_{92}U \rightarrow ^{234}_{90}Th + ^{4}_{2}He + \text{energy}$$

- When a nucleus releases an α particle, the nucleus also releases energy.

**Beta Decay**

The basic process that occurs in beta decay is the conversion of a neutron to a proton and an electron. Therefore we have $\beta^-$ and $\beta^+$ decays.

**$\beta^-$ decay** → emission of electron ($\bar{e}$)

when a nucleus decays by giving off an electron, its mass number is unchanged, but its atomic number z increases by 1. Symbolically, which can be represented as:

**$\beta^-$ decay**

parent nucleus → Daughter nucleus + $\beta^-$ particle (electron)

$$^{A}_{Z}P \rightarrow ^{A}_{Z+1}D + ^{o}_{-1}e + \text{energy}$$

For example, the thorium nucleus, is converted into protactinium as:

$$^{234}_{90}Th \rightarrow ^{234}_{91}Pa + ^{o}_{-1}e + \text{energy}$$

- In $\beta^-$ decay, the electron emitted does not actually exist within the parent nucleus and is not one of the orbital electrons. Instead, the electron is created when a neutron decays into a proton and an electron. Thus, the proton number of the parent nucleus increase by one. The electron escape from the atom leaving it positively charged.

Generated by CamScanner from intsig.com

**$\beta^+$ decay** → emission of positron ($e^+$)

In this process, the particle emitted by the nucleus is a positron rather than an electron. A positron has the same mass as an electron but carries a charge of +e instead of – e. In this case, the atomic mass remains unchanged but the atomic number decreases by 1, as shown below.

**$\beta^+$ decay**

$$^{A}_{Z}P \rightarrow {}^{A}_{Z-1}D + {}^{0}_{1}e + \text{energy}$$

An example of β+ emission is the decay of an isotope of nitrogen into an isotope of carbon

$$^{13}_{7}N \rightarrow {}^{3}_{6}C + {}^{0}_{+1}e + \text{energy}$$

The emitted positron does not exist within the nucleus but, rather, is created when a nuclear proton is transformed into a neutron. In the process, the proton number of the parent nucleus decreases by one.

**Gamma Decay** → emission of photon

An atom in an excited state can emit a photon when one of its electrons drops to a lower energy level. Photon as it decays to a state of lower energy. Since nuclear energies are so much greater than atomic energies, the photons given off by a nucleus are highly energetic. We refer to such high – energy photons as gamma (γ) rays.

**γ decay**

Excited energy state → Lower energy state + Photon(γ ray)

$$^{A}_{Z}P \rightarrow {}^{AP}_{Z} + \gamma \text{ ray}$$

For example, the nitrogen nucleus may decay to its ground state with the emission of a γ ray.

$$^{17}_{7}N \rightarrow {}^{14}_{7}N + \gamma$$

☛ **Notice that**: Neither the atomic number nor the mass number is changed by the emission of γ ray.

**Illustrative Example**

18. When the radioactive element $_{144}Nd_{60}$ decays to $_{140}Ce_{58}$, which one of the following particles is emitted?

A. A beta particle
B. An alpha particle
C. A proton
D. A gamma ray

*Explanation*: $_{144}Nd_{60} \rightarrow {}_{140}Ce_{58}$ is equivalent to;

$$^{A}_{Z}P \rightarrow {}^{A-4}_{z-2}P + {}^{4}_{2}He$$

**Answer: B**

Generated by CamScanner from intsig.com

## 16.9 Nuclear Decay

The nucleus doesn't "age" in any sense. Instead, a nucleus has a certain probability that, within the next second, it will spontaneously turn into a different nucleus and, in the process, eject an alpha or beta particle or gamma ray. This spontaneous transformation of the nucleus is called *nuclear decay*.

The rate at which a decay process occurs in a radioactive sample is proportional to the number of radioactive nuclei present in the sample, i.e those nuclei that have not yet decayed.

If N is the number of radioactive nuclei present at some instant, the rate of change of N is

$$\frac{dN}{dt} = -\lambda N$$

Where: The negative sign shows that the number of nuclei in the sample N, decreases in time due to the decays. $\lambda$ is called the decay constant and has different value for different nuclei

$\frac{dN}{dt} = -\lambda N$ can be written as

$$\frac{dN}{N} = -\lambda dt$$

Intigrating from initial time t = 0 to time t.

$$\int_{No}^{N} \frac{dN}{N} = -\lambda \int_{o}^{t} dt$$

$$\ln\left(\frac{N}{N_o}\right) = -\lambda t \Rightarrow \boxed{N = N_o e^{-\lambda t}}$$

Where: N represents the number of radioactive nuclei present at time t

$N_o$ represents the number of undecayed or original radioactive nuclei at t = 0

**Half – Life:** Is the time taken for half the atoms of a given nuclide within a sample to decay

At half-life, $N = \frac{N_o}{2}$, put this into the decay equation

$$N = N_o e^{-\lambda t}$$

$$\frac{N_o}{2} = N_o e^{-\lambda t} \qquad ,t = t_{1/2}$$

Generated by CamScanner from intsig.com

$$\frac{1}{2} = e^{-\lambda t_{1/2}}$$

$$\ln\frac{1}{2} = -\lambda t_{1/2}$$

$$-\ln 2 = -\lambda t_{1/2}$$

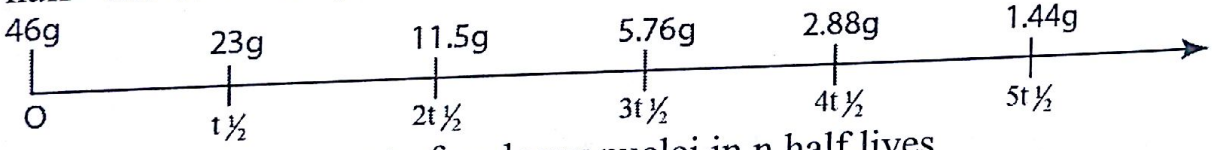
$$t_{\frac{1}{2}} = \frac{\ln 2}{\lambda}$$

$$\boxed{t_{\frac{1}{2}} = \frac{0.693}{\lambda}} \leftarrow \text{half-life}$$

☛ The number of undecayed nuclei $N = N_o\, e^{-\lambda t}$ can also be written as:

$$\boxed{N = \frac{N_o}{2^n} = N_o\left(\frac{1}{2}\right)^n}$$ , n is number of half lives $n = \frac{t}{t_{1/2}}$

For instance, the radioactive decay of tritium with a sample of 46g and half–life of $7\times10^8$ years can be shown as follows:




We can see the amount of undecay nuclei in n half lives

i. $N = N_o\left(\frac{1}{2}\right)^n = 46\left(\frac{1}{2}\right)^1 = 23g \leftarrow 1$ half life

ii. $N = N_o\left(\frac{1}{2}\right)^2 = 46\left(\frac{1}{4}\right) = 11.5g \leftarrow 2$ half life

iii. $N = N_o\left(\frac{1}{2}\right)^3 = 46\left(\frac{1}{8}\right) = 5.76g \leftarrow 3$ half life

iv. $N = N_o\left(\frac{1}{2}\right)^4 = 46\left(\frac{1}{16}\right) = 2.88g \leftarrow 4$ half life

v. $N = N_o\left(\frac{1}{2}\right)^5 = 46\left(\frac{1}{32}\right) = 1.44g \leftarrow 5$ half life

DIGITAL LIBRARY FOR ETHIOPIAN STUDENTS
Digital Library for Ethiopian students
Link: https://telegram.me//digitallibrary4ethiopianstudents or digitallibrary4ethiopianstudents
YouTube Digital[0,1] Ethiopia
Generated by CamScanner from intsig.com

## Illustrative Example

19. Radioactivity decay constant for radium is $1.07\times10^{-4}$ per years. Then its half-life period will be:

A. 2520 years B. 6476 years
C. 7000 years D. 8099 years

*Solution:* $t_{\frac{1}{2}} = \frac{1n2}{\lambda} = \frac{0.693}{1.07\times10^{-4}} = 6476$ years

20. The half-life of radium is 3.5 days. What fraction of the sample remains undecayed after 14 days?

A. $\frac{1}{2}$ B. $\frac{1}{4}$ C. $\frac{1}{8}$ D. $\frac{1}{16}$

*Solution:* $\lambda = \frac{1n2}{t_{\frac{1}{2}}} = \frac{0.693}{3.5}$

$$N = N_o e^{-\lambda t} \Rightarrow N/N_o = e^{-\lambda t} = e^{\left(-\frac{0.693}{3.5}\right)14} = \frac{1}{16}$$

**Answer: D**

21. The half – life of a radioactive isotope is 1 hour. What fraction of a sample of this isotope left after

A. 3 hour B. 1 day

*Solution:* $t_{\frac{1}{2}} = \frac{0.693}{\lambda} \Rightarrow \lambda = \frac{0.693}{t_{\frac{1}{2}}} = \frac{0.693}{1} = 0.693$

$$N = N_o e^{-\lambda t} \Rightarrow \frac{N}{N_o} = e^{-\lambda t}$$

$$\frac{N}{N_o} = e^{-0.693\,t}, \text{ but } e^{-0.693} = \frac{1}{2}$$

a. $\frac{N}{N_o} = e^{-0.693t} = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$

b. $\frac{N}{N_o} = e^{-0.693t} = \left(\frac{1}{2}\right)^{24} = \frac{1}{2^{24}}$

22. What fraction of radioactive sample has decayed after two half-lives have elapsed

A. $\frac{1}{4}$ B. $\frac{1}{2}$ C. $\frac{3}{4}$ D. $\frac{1}{8}$

Generated by CamScanner from intsig.com

***Solution***: $N = N_o\left(\frac{1}{2}\right)^n$ , $n = 2$

$$\frac{N}{N_o} = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

**Answer: A**

23. In three half-lives, the activity of radioactive sample will have decreased by

A. 25% B. 12.5% C. 87.5% D. 75%

***Solution***: $\frac{N}{N_o} = \left(\frac{1}{2}\right)^n$, $n = 3$, $\frac{N}{N_o} = \left(\frac{1}{2}\right)^3 = \left(\frac{1}{8}\right) \times 100\% = 12.5\%$

**Answer: B**

24. If 16g of radon gas with a half-life of 4 days present in balloon at $t = 0$, what is the mass of radon gas present after 24 days.

***Solution***: $n = \frac{t}{t_{1/2}} = \frac{24}{4} = 6$ , $N_o = 16g$

$$N = N_o\left(\frac{1}{2}\right)^n = 16\left(\frac{1}{2}\right)^6 = 0.25g$$

25. By emitting an α particle, radon decays into polonium with half-life of 4 days. A sample contains $6.4 \times 10^{10}$ atoms of radon. After 12 days the number of atoms left in the sample are

A. $2.1 \times 10^{10}$ atom B. $8.0 \times 10^{9}$ atom

C. $3.2 \times 10^{10}$ atom D. $0.5 \times 10^{10}$ atom

***Solution***: The number of half-lives in 12 days is

$$n = \frac{\text{total time}}{\text{time of half life}} = \frac{12 \text{ days}}{4 \text{ days}} = 3$$

$$N = N_o\left(\frac{1}{2}\right)^n = \left(6.4 \times 10^{10}\right)\left(\frac{1}{2}\right)^3 = 8 \times 10^9 \text{ atoms}$$

**Answer: B**

26. Half-life of a radioactive substance is 12 years. Calculate the time interval between 25% undecay and 6.75% undecay

***Solution***: $\frac{N}{N_o} = \left(\frac{1}{2}\right)^n \Rightarrow \frac{25}{100} = \left(\frac{1}{2}\right)^n \Rightarrow n = 2$

Since $n_1 = \frac{t_1}{t_{1/2}} \Rightarrow 2 = \frac{t_1}{12} \Rightarrow t_1 = 24$ years

Generated by CamScanner from intsig.com

Similarly, $\frac{N}{N_o}=\left(\frac{1}{2}\right)^n \Rightarrow \frac{6.75}{100}=\left(\frac{1}{2}\right)^n \Rightarrow n=4$

$$n_2=\frac{t_2}{t\,½} \Rightarrow 4=\frac{t_2}{12} \Rightarrow t_2=48 \text{ years}$$

Now, $t_2 - t_2 = 48 - 24 = 24$ years.

## 16.10 Nuclear Reaction

A nuclear reaction is said to occur whenever the incident nucleus, particle or photon causes a change to occur in a target nucleus.

In 1919, Ernest Rutherford observed that when an α particle strikes a nitrogen nucleus, an oxygen nucleus and a proton are produced. This nucleus reaction is written as:

α particle (incident) + Nitrogen (target) ⟶ Oxygen + proton

$$\Rightarrow {}^4_2He + {}^{14}_7N \rightarrow {}^{17}_8O + {}^1_1H$$

### Nuclear Fission

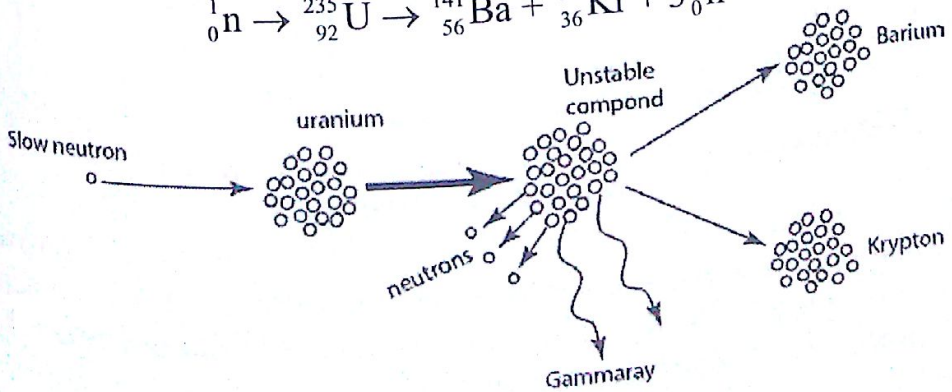
In 1939, four German Scientist found that a uranium nucleus, after absorbing a neutron, splits into two fragments, each with a smaller mass than the original nucleus. The splitting of a massive nucleus into two less massive fragments is called *nuclear fission*.

**Nuclear Fission:** Is the process by which heavy nuclei are split into two or more nuclei

For example, when uranium absorbs slow moving neutron, unstable compound nucleus is created. The compound nucleus then disintegrates into Barium, krypton and three neutrons

Slow neutron + uranium → compound nucleus (unstable) → Barium + krypton + $3\,{}^1_0n$

$${}^1_0n \rightarrow {}^{235}_{92}U \rightarrow {}^{141}_{56}Ba + {}^{92}_{36}Kr + 3\,{}^1_0n$$




Generated by CamScanner from intsig.com

- Nuclear fission is the basis for nuclear weapons and nuclear power plants.

**Nuclear Fusion**

When two light nuclei combine to form a more massive nucleus, the reaction is called *nuclear fusion*.

**Nuclear Fusion:** Is a process by which two nuclides join together to form a heavier nucleus.

- When light nuclei undergo fusion, the resulting nucleus will have a greater binding energy than the original nuclei. This means, the larger nucleus formed by fusion has less mass than the sum of the masses of the original nuclei. The mass difference appears as an energy given off by the reaction.

For example, two hydrogen isotopes fuse to form Helium and a neutron as follows

$$^{2}_{1}H + ^{3}_{1}H \rightarrow ^{4}_{2}He + ^{1}_{0}n$$

- Hydrogen bomb and the sun use the principle of fusion for their working.

**Additional Examples**

27. Of the following properties, the photon does not possess
A. rest mass B. momentum
C. energy D. frequency

**Answer: A**

28. Dual nature of radiation is shown by
A. Diffraction and reflection
B. Refraction and diffraction
C. photoelectric effect
D. photo electric effect and diffraction

**Answer: D**

29. Neglecting variation of mass with velocity, the wavelength associated with an electron having a kinetic energy E is proportional to
A. $E^{1/2}$ B. E C. $E^{-1/2}$ D. $E^{-2}$

***Explanation:*** $\lambda = \frac{h}{mv} = \frac{h}{p} = \frac{h}{\sqrt{2mE}} \Rightarrow \lambda : E^{-1/2}$

**Answer: C**

30. The photoelectric threshold frequency of a metal is $f_o$. When light of frequency $-4\ f_o$ is incident on the metal, the maximum kinetic energy of the emitted photoelectrons is

Generated by CamScanner from intsig.com

A. $4h f_o$ B. $3h f_o$

C. $\frac{h f_o}{4}$ D. $\frac{h f_o}{2}$

*Explanation*: $\frac{1}{2} m v_{max}^2 = 4hf_o - hf_o = 3hf_o$ **Answer: B**

31. Consider electrons and protons accelerated in a vacuum tube across the same potential difference. Which of the following is true?

A. protons have smaller momentum
B. electrons have smaller velocity
C. protons have larger kinetic energy
D. protons and electrons have same kinetic energy

*Explanation*: In vacuum, no collisions are supposed to occur. So, the kinetic energy of protons and electrons are the same.

**Answer: D**

32. A point source of light is used in a photo electric effect. If the source is removed farther from the emitting metal, the stopping potential

A. will increase B. will decrease
C. will remain constant D. will increase or decrease

*Explanation*: When position change, intensity changes, but has no effect on stopping potential **Answer: C**

33. An electron and a proton enter a magnetic field perpendicularly. Both have same kinetic energy. Which of the following is true?

A. Trajectory of electron is less curved
B. Trajectory of proton is less curved
C. Both trajectory are equally curved
D. Both move on straight line path

*Explanation:* $r = \frac{mv}{Bq} \Rightarrow r = \frac{\sqrt{2m K_E}}{Bq} \Rightarrow r \sim \sqrt{m}$

Hence, larger mass has larger radius of trajectory **Answer: B**

☞ Larger curvature means lesser radius of curvature.

34. The photoelectric effect is based on the conservation of

A. mass B. energy
C. momentum D. angular momentum

**Answer: B**

Generated by CamScanner from intsig.com

35. In Einstein's photoelectric equation $K_E = hf - \emptyset_o$. $K_E$ refers to
A. kinetic energy of all the emitted electrons
B. maximum kinetic energy of the emitted electrons
C. means kinetic energy of the emitted electrons
D. minimum kinetic energy of the emitted electrons

**Answer: B**

36. The electrons are emitted in the photoelectric effect from a metal surface
A. only if the frequency of the incident radiation is above a certain threshold value
B. only if the temperature of the surface is high
C. at a rate that is independent of the nature of the metal
D. With a maximum velocity proportional to the frequency of the incident radiation.

**Answer: A**

37. If the wavelength of light incident on the surface of a metal is decreased, the maximum velocity of emitted photo – electrons will
A. decrease
B. not change
C. increase
D. first decrease then increase

***Explanation***: $E = mc^2 = hf$, since $c = \lambda f \Rightarrow \lambda = \frac{h}{p}$

**Answer: B**

38. The equation E = PC is valid
A. for an electron as well as for a photon
B. for an electron but not for a photon
C. for a photon but not for an electron
D. neither for an electron nor for a photon.

**Answer: C**

39. Light of frequency 1.5 times the threshold frequency is incident on photosensitive material. If the frequency is halved and intensity is doubled, the photo current becomes
A. quadrupled
B. doubled
C. halved
D. zero

***Explanation***: When the given frequency is halved, the frequency becomes less than threshold frequency. So, no photo- emission.

**Answer: D**

40. The best metal to be used for photoemission is
A. potassium
B. sodium
C. cession
D. lithium

**Answer: C**

Generated by CamScanner from intsig.com

35. In Einstein's photoelectric equation $K_E = hf - \emptyset_o$. $K_E$ refers to
   A. kinetic energy of all the emitted electrons
   B. maximum kinetic energy of the emitted electrons
   C. means kinetic energy of the emitted electrons
   D. minimum kinetic energy of the emitted electrons

**Answer: B**

36. The electrons are emitted in the photoelectric effect from a metal surface
   A. only if the frequency of the incident radiation is above a certain threshold value
   B. only if the temperature of the surface is high
   C. at a rate that is independent of the nature of the metal
   D. With a maximum velocity proportional to the frequency of the incident radiation.

**Answer: A**

37. If the wavelength of light incident on the surface of a metal is decreased, the maximum velocity of emitted photo – electrons will
   A. decrease
   B. not change
   C. increase
   D. first decrease then increase

***Explanation***: $E = mc^2 = hf$, since $c = \lambda f \Rightarrow \lambda = \frac{h}{p}$

**Answer: B**

38. The equation E = PC is valid
   A. for an electron as well as for a photon
   B. for an electron but not for a photon
   C. for a photon but not for an electron
   D. neither for an electron nor for a photon.

**Answer: C**

39. Light of frequency 1.5 times the threshold frequency is incident on photosensitive material. If the frequency is halved and intensity is doubled, the photo current becomes
   A. quadrupled
   B. doubled
   C. halved
   D. zero

***Explanation***: When the given frequency is halved, the frequency becomes less than threshold frequency. So, no photo- emission.

**Answer: D**

40. The best metal to be used for photoemission is
   A. potassium
   B. sodium
   C. cesium
   D. lithium

**Answer: C**

Generated by CamScanner from intsig.com

41. If the frequency of light in a photo electric experiment is doubled, stopping potential will
A. be doubled
B. be halved
C. become more than double
D. become less than double

*Explanation*: $K_E = hf - \emptyset_o = ev$

$K_E^1 = 2hf - \emptyset_o = ev'$

Taking ratio $v^1 > 2v$

**Answer: C**

42. Light of frequency f is incident on a substance of threshold frequency $f_o$ ($f_o < f$). The energy of the emitted photo – electron will be

A. $h(f - f_o)$
B. $\frac{h}{f}$
C. $he(f - f_o)$
D. $\frac{h}{f_o}$

*Explanation*: $\frac{1}{2}mV_{max}^2 = hf - hf_o = h(f - f_o)$

**Answer: A**

43. A photon of energy hf is absorbed by a free electron of a metal having work function $\emptyset < hf$
A. The electron is sure to come out
B. The electron is sure to come out with a kinetic energy $hf - \emptyset$
C. Either the electron does not come out or it come out with a kinetic energy $hf - \emptyset$
D. It may come out with a kinetic energy less than $hf - \emptyset$

*Explanation*: maximum kinetic energy is $hf - \emptyset$.

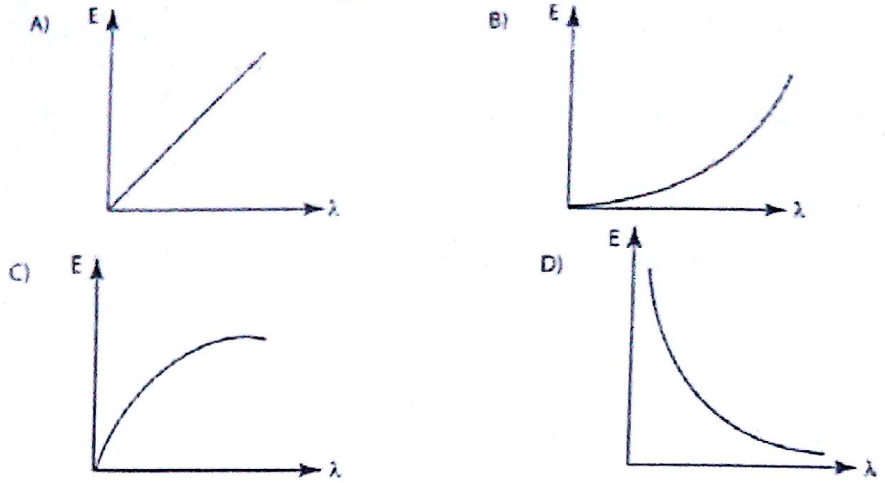
**Answer: D**

44. When light falls on a metal surface, the maximum kinetic energy of the emitted photo – electrons depends upon
A. the time for which light falls on the metal
B. frequency of the incident light
C. intensity of the incident light
D. velocity of the incident light

**Answer: B**

Generated by CamScanner from intsig.com

45. Which graph shows how the energy E of a photon of light is related to its wavelength λ?




***Explanation:*** $E = hf = \frac{hc}{\lambda} \Rightarrow E \sim \frac{1}{\lambda}$ **Answer: D**

46. Of the following having the same kinetic energy, the one which has the largest wavelength is

A. An alpha particle B. A neutron
C. A proton D. An electron

***Explanation:*** $E = mv^2 = hf$ and $f = \frac{v}{h} \Rightarrow \lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mE}} \Rightarrow \lambda \sim \frac{1}{\sqrt{m}}$

electron has least mass, maximum wavelength **Answer: D**

47. A photon of energy 3.4 ev is incident on a metal having work function 2 ev. The maximum $K_E$ of photo electrons is equal to

A. 1.4 ev B. 1.7 ev C. 5.4 ev D. 6.8 ev

***Explanation:*** $KE_{max} = hf - \emptyset = 3.4 \text{ ev} - 2\text{ev} = 1.4 \text{ ev}$

**Answer: A**

48. The ratio of momenta of an electron and an α–particle which are accelerated from rest by a potential difference of 100v is

A. 1 B. $\sqrt{\frac{2\,me}{m\,\alpha}}$ C. $\sqrt{\frac{me}{m\alpha}}$ D. $\sqrt{\frac{me}{2m\alpha}}$

***Explanation:*** $\lambda = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2mev}}$ and

$P = \sqrt{2mE} = \sqrt{2mev}$

$\frac{Pe}{P\alpha} = \frac{\sqrt{2meev}}{\sqrt{2m_\alpha (2e)^v}} = \sqrt{\frac{me}{2m_\alpha}}$ **Answer: D**

Generated by CamScanner from intsig.com

49. A metal surface of work function 1.07 ev is irradiate with light of wavelength 332 nm. The retarding potential required to stop the escape of photo electrons is

A. 4.81 ev B. 3.7 ev C. 2.66 ev D. 10.07 ev

*Solution*: The energy in ev corresponding to 332 nm is = 3.73 ev

$$\frac{1}{2}mu_{max}^2 = (3.73 - 1.07)\text{ev} = 2.66 \text{ ev}$$

**Answer: C**

50. The mass of a particle is 400 time that of an electron and the charge is double. The particle is accelerated by 5v. Initially the particle is at rest. Then its final kinetic energy will be

A. 5ev B. 10ev C. 100ev D. 2000ev

*Solution*: $w = qv = (2e)\,5v = 10ev$

**Answer: B**

51. The radio transmitter operates on a wavelength of 1500m. Then the frequency corresponding to this wavelength is

A. 500 KHZ B. 200 KHZ

C. 100 KHZ D. 400 KT

*Solution*: $c = \lambda f \Rightarrow f = \frac{c}{\lambda} = \frac{3\times10^8}{1500} = 200\times10^3 HZ$

$f = 200$ KHZ

**Answer: B**

52. The $K_E$ of the photoelectrons is E when the incident wavelength is $\frac{\lambda}{2}$. The $K_E$ becomes 2E when the incident wavelength is $\frac{\lambda}{3}$. The work function of the metal is

A. $\frac{hc}{\lambda}$ B. $\frac{2hc}{\lambda}$ C. $\frac{3hc}{\lambda}$ D. $\frac{hc}{3\lambda}$

*Explanation:* $E = hf - \varnothing_o = \frac{hc}{\frac{\lambda}{2}} - \varnothing_o = \frac{4hc}{\lambda} - 2\varnothing_o$ .....*

$$2E = \frac{hc}{\frac{\lambda}{3}} - \varnothing_o = \frac{3hc}{\lambda} - \varnothing_o \text{ .....(**)}$$

From equation * and **

$$\frac{4hc}{\lambda} - 2\varnothing_o = \frac{3hc}{\lambda} - \varnothing_o \Rightarrow \varnothing_o = \frac{hc}{\lambda}$$

**Answer: A**

DIGITAL LIBRARY FOR ETHIOPIAN STUDENTS
Digital Library for Ethiopian students
Link: https://telegram.me//digitallibrary4ethiopianstudents or digitallibrary4ethiopianstudents
YouTube Digital[0,1] Ethiopia
Generated by CamScanner from intsig.com

53. Ultraviolet radiation of 6.2 ev falls on an aluminium surface of 4.2 ev work function. The kinetic energy in joule of the fastest electron emitted is nearly

A. $3.2 \times 10^{-21}$ B. $3.2\times10^{-19}$ C. $3.2\times10^{-17}$ D. $3.2 \times 10^{-15}$

***Solution***: $\frac{1}{2}mV_{max}^{\ 2} = 6.2 \text{ ev} - 4.2 \text{ ev} = 2 \text{ ev} = 2\times1.6\times10^{-19}$

$= 3.2 \times 10^{-19}$ J **Answer: B**

54. In a photoelectric cell, the wavelength of incident light is changed from $4000 \times 10^{-10}$m to $3600 \times 10^{10}$m. The change in stopping potential will be

A. 0.14 v B. 0.24 v C. 0.35 v D. 0.44 v

***Solution:*** $ev = hf - \emptyset_o = ev^1 - ev = hf^1 - hf$

$Ev^1 = hf^1 - \emptyset_o$

$\Delta ev = \left(\frac{12375}{3600} - \frac{12375}{4000}\right) ev = 0.35v$ **Answer: C**

55. The threshold frequency for a certain metal is $f_o$. When frequency of incident radiation is $2f_o$, the maximum velocity of photoelectrons is $3\times 10^6$ m/s. If the frequency of incident radiation is increased to 10 of, then the maximum velocity of photoelectrons will be

A. $3 \times 10^6$ m/s B. $6\times10^6$ m/s

C. $9 \times 10^6$m/s D. $12\times10^6$ m/s

***Solution:*** $\frac{1}{2}mV_{max}^{\ 2} = 2hf_o - hf_o$

Again, $\frac{1}{2}mV_{max}^{\ 2} = 10hf_o - hf_o = 9f_o$

$\Rightarrow V_{max}$ is increased by a factor of 3 and stopping potential By a factor of 9. **Answer: C**

56. The work function for aluminium is 4.125ev. The cut – off wavelength for photoelectric effect for aluminium is

A. 300 nm B. 200 nm

C. 420 nm D. 150 nm

***Solution***: $\emptyset_o = hf_o = \frac{hc}{\lambda_o} \Rightarrow \lambda_o = \frac{hc}{\emptyset_o} = \frac{12375}{4.125}\times10^{-10}\times10^9 nm$

$\lambda_o = \frac{12375}{41.25} nm = 300 \text{ nm}$ **Answer: A**

Generated by CamScanner from intsig.com

57. Two radiations containing photons of energy twice and five times the work function of a metal are incident successively on the metal surface. The ratio of the maximum velocities of the emitted electrons in the two cases will be

A. 1: 3 B. 1: 4 C. 1:2 D. 1:1

*Explanation:* $2\varnothing_o = \varnothing_o + \frac{1}{2}mv_1^2 \Rightarrow \varnothing_o = \frac{1}{2}mv_1^2$ .....*

And, $5\varnothing_o = \varnothing_o + \frac{1}{2}mv_2^2 \Rightarrow 4\varnothing_o = \frac{1}{2}mv_2^2$ .....**

From equation * and ** $\frac{u_1 2}{v_2^2} = \frac{\varnothing_o}{4\varnothing_o} = \frac{1}{4} \Rightarrow \frac{v_1}{v_2} = \frac{1}{2}$ **Answer: C**

58. The ratio of the velocities of a proton and α – particle is 4: 1. The ratio of their De Broglie wavelength is

A. 1: 1 B. 1: 2 C. 2: 1 D. 4:1

*Explanation:* $\frac{\lambda_p}{\lambda_\alpha} = \frac{m_\alpha v_\alpha}{m_p v_p} = \frac{4}{1} \times \frac{1}{4} = 1:1$ **Answer: A**

59. The de Broglie wavelength of an electron is same as that of a photon. If the velocity of electron is 50% of the speed of light, then ratio of kinetic energy of photon to the kinetic energy of electron is

A. 0.5 B. 1.5 C. 2.5 D. 4.0

*Explanation:* $\frac{E_{ph}}{E_e} = \frac{2c}{v} = \frac{2c}{\frac{c}{2}} = 4$ **Answer: D**

60. The frequency of the incident light falling on a photosensitive metal plate is doubled. The kinetic energy of the emitted photo electron is

A. unchanged B. double
C. less than doubled D. more than doubled

*Explanation:* $K_E = hf - \varnothing_o$ and $KE^1 = 2hf - \varnothing_o$

$\frac{K_E^1}{K_E} = \frac{2hf - \varnothing_o}{hf - \varnothing_o}$ **Answer: D**

61. An important spectral emission line has a wavelength of 21cm. The corresponding photon energy is

A. $5.9 \times 10^{-11}$ ev B. $5.9 \times 10^{-8}$ ev
C. $5.9 \times 10^{-6}$ ev D. $5.9 \times 10^{-4}$ ev

Generated by CamScanner from intsig.com

***Solution:*** $E = hf = \frac{hc}{\lambda}$ **Answer: C**

62. The momentum of a photon is P. The frequency associated with it is given by

A. $\frac{pc}{h}$ B. $\frac{ph}{c}$ C. $\frac{hc}{p}$ D. $\frac{h}{pc}$

***Explanation:*** $p = mc = \frac{mc^2}{c} = \frac{E}{c} = \frac{hf}{c} \Rightarrow f = \frac{pc}{h}$ **Answer: A**

63. Sodium surface is illuminated by ultraviolet and visible radiation successively and the stopping potential determined. This stopping potential is

A. equal in both cases B. more with uv light
C. more with visible light D. varies randomly

***Explanation:*** The frequency of uv is more than the frequency of visible radiation. **Answer: B**

64. x– ray are used to irradiate sodium and copper surfaces in two separated experiments and stopping potential determined. The stopping potential is

A. equal in both cases B. greater for sodium
C. greater for copper D. infinite in both cases

***Explanation:*** Sodium has low work function so, maximum kinetic energy is more in sodium. The stopping potential is more for sodium **Answer: B**

65. Stopping potential depends

A. only upon the energy of incident photon
B. only on the work function of the metal
C. on the difference in energy of incident photon and work function of metal
D. on the sum of energy of incident photon and work function of metal

**Answer: C**

66. If the wavelength of incident radiation in a photoelectric experiment is decreased from $6000\times10^{-10}$m to $4500\times10^{-10}$m

A. The photoelectric current may stop
B. The photoelectric current will increase
C. The stopping potential will decrease
D. The stopping potential will increase

**Answer: D**

Generated by CamScanner from intsig.com

67. In order to increase the kinetic energy of ejected photoelectrons, there should be an increase in
A. intensity of radiation
B. wavelength of radiation
C. frequency of radiation
D. wavelength and intensity
**Answer: C**

68. An electron of charge 'e' coulomb passes through a potential difference of 'v' volt. Its energy in joule will be
A. $\frac{v}{e}$ B. ev C. $\frac{e}{v}$ D. v
**Answer: B**

69. Electron volt is a unit of
A. potential
B. charge
C. power
D. energy
**Answer: D**

70. Light of frequency f is incident on a certain photoelectric substance with threshold frequency $f_o$. The work function for the substance is
A. hf B. $hf_o$ C. $h(f - f_o)$ D. $h(f + f_o)$
**Answer: B**

71. In photoelectric effect
A. Light energy is converted into heat energy
B. Light energy is converted into electric energy
C. Light energy is converted into sound energy
D. Electric energy is converted into light energy
**Answer: B**

72. If the energy of a photon is 25ev and the work function of the material is 7ev, then the value of stopping potential is
A. 3v B. 9v C. 18v D. 27v
***Solution***: $eV_s = 25ev - 7ev = 18v$
**Answer: C**

73. A photo – sensitive material would emit electrons if excited by photons beyond a threshold. To cross the threshold you would increase
A. intensity of light
B. wavelength of light
C. frequency of light
D. voltage applied
**Answer: C**

74. The strength of a photo – electric current depends upon
A. frequency of incident radiation
B. intensity of incident radiation
C. angle if incident radiation
D. distance between anode and cathode
**Answer: C**

Generated by CamScanner from intsig.com

75. Photo – electrons are being obtained by irradiating zinc by a radiation of $3100 \times 10^{-10}$m. In order to increase the kinetic energy of ejected photo – electrons
A. The intensity of radiation should be increased
B. The wavelength of radiation should be increased
C. The wavelength of radiation should be decreased
D. Both wavelength and intensity of radiation should be increased

**Answer: C**

76. Photoelectric effect is the phenomenon in which
A. photons come out f a metal when it is hit by a beam of electrons
B. Photons come out of the nucleus of an atom under the action of an electric field
C. Electrons come out of a metal with a constant velocity which depends on the frequency and intensity of incident light wave
D. Electrons come out of a metal with different velocity not greater than certain value which depends only on the frequency of the incident light wave and not on its intensity.

**Answer: D**

77. The maximum kinetic energy of photoelectrons emitted from a surface when photons of energy 6ev fall on it is 4ev. The stopping potential in volts is
A. 2 B. 4 C. 6 D. 10
*Explanation*: $K_E = eV_s \Rightarrow 4ev = eV_s \Rightarrow V_s = 4$ volt

**Answer: B**

78. According to Einstein's photoelectric equation, the plot of the kinetic energy of the emitted photoelectrons from a metal versus frequency of incident radiation gives a straight line whose slop:
A. depends on the nature of the metal
B. depends on the intensity of radiation
C. depends both on the intensity of radiation and the metal used
D. is the same for all metals and independent of the intensity of radiation.

*Explanation*: $K_E = hf - \phi$
This is equation of a straight line of slope h

**Answer: D**

Generated by CamScanner from intsig.com

79. In a mean life of a radioactive sample
A. about $\frac{1}{3}$ of substance disintegrates
B. about $\frac{2}{3}$ of the substance disintegrates
C. about 90% of the substance disintegrated
D. almost all the substance disintegrates

*Explanation:* $N = N_o e^{-\lambda t}$ , at mean life, $t = \frac{1}{\lambda}$

$$\Rightarrow N = N_o e^{-\lambda\left(\frac{1}{\lambda}\right)} = \frac{1}{e} N_o \Rightarrow \frac{N}{N_o} = \frac{1}{2.718} \simeq 0.37$$

Fraction of the element that disappears $= (1 - 0.37) = 0.63 = \frac{2}{3}$

**Answer: B**

80. A nuclear reaction given by ${}_zX^A \rightarrow {}_{z+1}Y^A + {}_{-1}e^o + \bar{\nu}$ represents
A. $\beta$ – decay
B. $\gamma$ – decay
C. fusion
D. fission

**Answer: A**

81. What was the fissionable material used in bomb dropped at Negasaki of Japan in the year 1945
A. Uranium
B. Neptunium
C. Beryllium
D. Plutonium

**Answer: D**

82. Pick out the statement which is true
A. The energy released per unit mass is more in fission than in fusion
B. The energy released per atom is more in fusion than in fission
C. The energy released per unit mass is more in fusion and that per atom is more in fission
D. Both fission and fusion produce same amount of energy per atom as well as per unit mass.

**Answer: C**

83. If $E_1$ is the energy released per unit mass in nuclear fussion and $E_2$ that in nuclear fission, then
A. $E_1 = E_2$
B. $E_1 = -E_2$
C. $E_1 > E_2$
D. $E_1 < E_2$

**Answer: C**

Generated by CamScanner from intsig.com

84. Which of the following is not a mode of radioactive decay?
A. positron emission B. electron capture
C. fusion D. alpha decay

**Answer: C**

85. Half-life of radioactive element depends upon
A. amount of element B. temperature
C. pressure D. nature of element

**Answer: D**

86. The half- life of a certain radioactive element is such that $7/8$ of a given quantity decays in 12 days. What fraction remains undecayed after 24 days?

A. 0 B. $\frac{1}{128}$ C. $\frac{1}{64}$ D. $\frac{1}{32}$

***Explanation***: The radioactive element has $\frac{1}{8}$ of a given quantity that remains after 12 days. After 24 days, that is in additional 12 days, the fraction that remains undecayed is,

$$\left(\frac{1}{8}\right)\left(\frac{1}{8}\right)=\frac{1}{64}$$

**Answer: C**

87. The radioactive source is found to fall by 0.875 of its initial activity in 210 sec. what is the half – life of the source?
A. 30 sec B. 60 sec
C. 70 sec D. 120 sec

***Explanation***: The activity has fallen to $1 - 0.875 = 0.125$ of its initial activity. This corresponds to $\left(\frac{1}{2}\right)^3$ of its initial activity or passage of 3 half – lives. The half–life is then

$$\frac{1}{3}(210)=70\,\text{sec}$$

**Answer: C**

88. Fussion reaction takes place at high temperature because
A. atoms are ionized at high temperature
B. molecules break – up at high temperature
C. nuclei break – up at high temperature
D. kinetic energy is high enough to overcome repulsion between nuclei

**Answer: D**

Generated by CamScanner from intsig.com

In beta decay
A. the daughter nucleus has one proton more than the parent nucleus
B. the parent and the daughter nuclei have the same number of protons
C. the daughter nucleus has one neutron more than the parent nucleus
D. the daughter nucleus has one proton less than the parent nucleus

**Answer: A**

90. During a negative beta decay
A. an electron which is already present within the nucleus ejected
B. a neutron in the nucleus decays and emit an electron
C. part of binding energy of the nucleus is converted into an electron
D. an atomic electron is ejected

**Answer: B**

91. Two radioactive isotopes P and Q have half – lives of 10 minutes and 15 minutes, respectively. After 30 minutes the ratio of number of atoms of P to Q will be
A. 0.5 B. 2.0 C. 1.0 D. 3.0
*Explanation*: 30 minutes is 3 half-lives for P and 2 half-life for Q

$$\frac{3(10)}{2(15)} = \frac{30}{30} = 1$$

**Answer: C**

92. In the disintegration series $^{238}_{92}U \xrightarrow{\alpha} X \xrightarrow{B-} {}^{A}_{Z}Y$
The value of Z and A respectively will be
A. 92, 236 B. 88, 230
C. 90,234 D. 91,234
*Explanation*: α decreases mass number by 4 and reduces charge number by 2. And, β – decay keeps mass number unchanged and increases charge by 1.

**Answer: D**

93. A sample contains 10 mg of radio – active material of half – life 270 days. After 540 days the mass of radioactive material left will be
A. 5mg B. 2.5 mg C. 1.25 mg D. zero

Generated by CamScanner from intsig.com

***Explanation***: Time = 2 × half – life. So, half of half left. That is 2.5 mg **Answer: B**

94. If $N_o$ is the original mass of the substance of half-life 5 years, then the amount of substance left after 15 years is

A. $\frac{N_o}{8}$ B. $\frac{N_o}{16}$ C. $\frac{N_o}{2}$ D. $\frac{N_o}{4}$

***Explanation***: $N = \frac{N_o}{2^{15/5}} = \frac{N_o}{2^3} = \frac{N_o}{8}$ **Answer: A**

95. In a sample of radioactive material, what percentage of the initial number of active nuclei will decay during one mean life?

A. 69.3% B. 63 %
C. 50% D. 37%

***Solution***: $N = N_o e^{-\lambda t}$, at mean life, $t = \frac{1}{\lambda}$

$$N = N_o e^{-\lambda\left(\frac{1}{\lambda}\right)} = \frac{N_o}{e}$$

$$\Rightarrow \frac{N}{N_o} = \frac{1}{e} \text{ and the decay percentage is,}$$

$$\left(1 - \frac{1}{e}\right) \times 100 = 63.2\%, e \simeq 2.718$$ **Answer: B**

96. After an interval of one day, $\frac{1}{16}$ the initial amount of a radioactive material remains in a sample. Then its half-life is

A. 6 hours B. 12 hours
C. 1.5 hours D. 3 hours

***Solution***: $N = \frac{N_o}{2^{t/T}}$ and $\frac{N_o}{16} = \frac{N_o}{2^{t/T}}$

$$\Rightarrow 2^{t/T} = 16 = 2^4 \text{ or } \frac{t}{T} = 4 \Rightarrow T = \frac{t}{4} = \frac{24}{4}$$

$\Rightarrow$ T= 6 hours **Answer: A**

97. Which of the following processes represents gamma decay?

A. ${}^{A}X_Z + \gamma \rightarrow {}^{A}X_{z-1} + a + b$ C. ${}^{A}X_Z \rightarrow {}^{A}X_Z + f$

B. ${}^{A}X_Z + {}^{1}n_o \rightarrow {}^{A-3}X_{z-2} + c$ D. ${}^{A}X_Z + e_{-1} \rightarrow {}^{A}X_{z-1} + g$

***Explanation***: In gamma decay, neither A nor Z change **Answer: C**

Generated by CamScanner from intsig.com